The New Work of Generative AI

Evidence from 18 Million Duty Statements in US Job Postings

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Abstract

Whether generative AI raises or lowers the demand for labour depends in part on the new work it creates. I measure the new tasks that employers advertise in 18.4 million duty statements from US job postings, 2019–2026, drawn from a new dataset of applicant-tracking-system pages archived by Common Crawl. Duty statements with no close counterpart in early-2019 postings are flagged, and a language model separates new work—a new activity or a new object of work—from familiar tasks done with new tools and from variants of old duties. In the third of occupations that use AI most, new work rose about fivefold, from 0.5 to 2.6 per 100 duty lines between 2022 and 2026, and most of the increase is work on AI itself: building and governing AI, directing it and driving its adoption. Working with AI, rather than building it, rose from 4% of AI duties in 2019–23 to 22% in 2026, but about two-fifths of it asks workers only to use AI tools in their own work, which changes how familiar tasks are done rather than what they are. Beyond new work, existing tasks are crossing occupational lines: postings that ask for technical and data tools, such as dashboards, in occupations whose duties did not name them in 2019–20 rose about threefold between 2022 and 2026. Postings with AI duties advertise higher pay, though most of the premium is accounted for by the firms and job titles that post them. Most evidence on generative AI concerns the tasks it can perform; reading duty statements measures the work it creates as employers advertise it, before job titles and official task lists record it.

JEL codes: J23, J24, O33. **Keywords:** generative AI, new work, job postings, tasks, reinstatement.

1 Introduction

New technologies displace workers from some tasks and create new tasks for them (Autor et al., 2003). Whether a technology raises the demand for labour on net depends on how the displacement compares with the gains in productivity and with the creation of new tasks, which Acemoglu and Restrepo (2018, 2019) call reinstatement. Job postings record an early margin of this second effect: the new tasks that employers ask new hires to do. Historically, new work has appeared where innovations complement the tasks that workers already perform (Lin, 2011; Autor et al., 2024). For generative AI, most of the evidence so far is on the other side of the ledger: which occupations and tasks the technology can perform (Felten et al., 2021; Webb, 2020; Eloundou

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et al., 2024; Tomlinson et al., 2025), which ones people use it for (Handa et al., 2025; Massenkoff and McCrory, 2026; Chatterji et al., 2026), and whether employment in exposed occupations has fallen (Brynjolfsson et al., 2025a; Hampole et al., 2025; Humlum and Vestergaard, 2025). The studies that look for new work count skills new to a firm and occupation in postings (Chen et al., 2024), ask workers whether chatbots have created new tasks (Humlum and Vestergaard, 2025) or build task lists from interviews with workers (Wang et al., 2026b), rather than reading the duties that employers advertise.

Technology has the potential to change the content of jobs in four ways (Section 2). It can create new work, tasks whose activity or object did not exist before; give workers new instruments for familiar tasks; move existing tasks between occupations; and make some existing tasks more common. Each of the four can raise the demand for labour, but only the first adds to the tasks that workers do: it is the reinstatement that offsets automation in the task framework (Acemoglu and Restrepo, 2019), and the new work that Autor et al. (2026) define by the novel expertise it demands. The other three act through productivity, the division of labour and the scale of existing work. All four appear in job postings as new wording, and the evidence on generative AI has not told them apart.

This paper uses job postings to tell them apart. A job posting lists what an employer asks a new hire to do, one duty at a time. Using 1,629,317 US postings captured from 14 applicant tracking systems (ATS) archived by Common Crawl between 2019 and 2026, I compare each of their 18.4 million duty statements with the duty statements of early 2019, and flag a statement when nothing in the early-2019 postings is close to it. A language model then reads samples of the flagged statements and separates new work from familiar tasks done with new tools and from variants and re-wordings of familiar duties. I also classify every duty about AI by what the worker does with it—building it, governing it or working with it—to track the growth of AI work whether or not it is new, and use the technical tools named in duties to see where occupations are taking on tasks that other occupations already did. Each kind of change is then measured in the same unit, duty lines per 100 duty lines.

Table 4 sets the four side by side. In the third of occupations with the highest AI use according to the Anthropic Economic Index (Massenkoff and McCrory, 2026), between 2022 and 2026, new work rose from 0.5 to 2.6 per 100 duty lines; duties that ask workers only to use AI tools, a new instrument, from 0.02 to 0.7; AI work with a close counterpart among the duties of 2019 from 0.6 to 3.1; and duty lines that name a data, code or automation tool new to the occupation from 0.06 to 0.2. In the third with the least AI use, none of the four exceeds about 0.2 in 2026.

Four findings follow, one for each kind of change. First, most new work is work on AI itself. Of the new work in the high-use third in 2026, 2.0 per 100 duty lines is about AI, mostly building, deploying, selling and governing AI systems; the rest asks workers to drive the adoption of AI—finding uses for it, rolling it out and training colleagues to use it—or, still rarely, to direct AI systems and review their output, the strictest kind of new work. New work outside AI grew much less, on every reading, and five of the six largest new or fastest-rising job titles are AI titles. Second, working with AI is the fastest-growing side of AI work, but much of it is a new instrument rather than new work: duties that ask workers to use AI tools, direct or review AI output, or drive AI adoption grew from 4% of AI duties in 2019–23 to 22% in 2026, and two-fifths of them ask workers only to use AI tools in their own work. Third, much of the growth of AI work is more of work that already existed: most AI duty lines with a close counterpart in 2019 build, deploy or sell AI systems, as postings already asked for machine learning in 2019. Fourth, occupations are taking on technical and data tasks that other occupations already did, a change in the division of labour rather than new work: production managers now build dashboards and financial analysts write code

to automate their analysis, and the share of postings naming a data, code or workflow-automation tool that the occupation rarely used in 2019–20 more than tripled between 2022 and 2026, rising most in occupations that use AI more.

Two further results concern how AI work is priced and how well official task inventories capture it. Postings with AI duties advertise higher pay, mostly because higher-paying firms and dedicated AI job titles post them: the premium for building AI falls from 0.17 log points within the same occupation, state and year to 0.07 within the same firm, occupation and year, and to 0.01 within the same job title at the same firm. Within working with AI, duties that drive adoption or direct AI pay more than those that only use AI tools, as the scarcity of new expertise would predict (Autor et al., 2026). Official task inventories have not kept pace: even in its August 2026 release, O*NET, the task inventory on which most AI exposure measures are built, has no task statement about language models or working with AI.

The paper also contributes data. Research on job postings relies mostly on commercial datasets that are costly and cannot be redistributed (Hershbein and Kahn, 2018; Deming and Kahn, 2018; Atalay et al., 2020; Acemoglu et al., 2022). The postings used here come from Common Crawl's public web archive, and I will release the build code together with the posting-level dataset: for each posting, its metadata, occupation, industry and seniority codes, weights, and posting-level measures of new work and AI duties, alongside aggregate series by occupation, state and quarter. The sample captures a selective share of US hiring tilted towards professional and technical jobs, but its state, occupational and pay distributions closely match official statistics and commercial job-posting data, and it tracks major shifts such as the rise of remote work and state pay-transparency laws (Section 3 and Appendix A).

The closest work uses job postings to study generative AI. Chen et al. (2024) count the skills in Lightcast's taxonomy that are new to a firm and occupation, and find more of them where AI augments work; Koll et al. (2026) count and price skills new since 2010–11, AI developer and user skills among them, through 2024. Skill taxonomies record mostly tools, credentials and subjects; duty statements describe what the worker will do, which lets this paper test whether newly worded duties are new tasks and follow them through 2026. Wang et al. (2026a) and Shukla (2026) measure the AI exposure of the tasks that postings list, whereas this paper measures the duties that are new. Hosseini and Lichtinger (2026) identify firms that adopt generative AI by their first posting for workers who integrate it and find that junior employment falls at adopters, as Chandar and Klein Teeselink (2026) do across 41 countries; other studies of the job ladder ask the same question (Brynjolfsson et al., 2025a; Iscenko and Curto Millet, 2026). A broader literature measures new work by the job titles added to official classifications (Lin, 2011; Autor et al., 2024; Marguerit, 2025). Autor et al. (2026) define new work by the novel expertise it demands and find that its pay premium fades as the expertise spreads, and Skoglund et al. (2026), classifying Swedish occupations by function since 1880, find that occupations born of specialisation outlast those tied to a new technology. Both see new work only once a classification records it; duty statements capture it as employers advertise it, within existing job titles as well as new ones, and separate it from new tools for familiar tasks. Beyond job titles, Humlum and Vestergaard (2025) use worker surveys in Denmark to document new tasks from chatbots, and Acemoglu et al. (2022) study earlier AI in job vacancies. On the side of use, Chin et al. (2026) show that many ChatGPT conversations involve another occupation's tasks, and OpenAI's Signals data, which assign consumers' work messages to ChatGPT to O*NET's work activities each quarter, show that use has become less concentrated in the most common activities (OpenAI Economic Research, 2026; Chatterji et al., 2026). Wang et al. (2026b) build task lists for six occupations from interviews with workers and find that two-fifths

of their tasks have no counterpart in O*NET: over half of these describe oversight, coordination and teaching, work that keeps an occupation running but that formal descriptions seldom record, and a fifth depend on a recent tool or practice. Comparing duties with earlier postings rather than with a task inventory separates work that is new from work that was never written down. Whereas Meisenbacher et al. (2025) release aggregate O*NET features from postings, the dataset released here is at the posting level.

Section 2 sets out what counts as new work, Section 3 describes the data and Section 4 the measurement. Section 5 presents the results: the four kinds of change side by side; then new work and new job titles, new instruments and more of existing AI work, and tasks crossing occupational lines; and finally where new work grows across occupations, the official task lists and pay. Section 6 discusses the limitations, and Section 7 concludes.

2 What counts as new work

A job posting describes a job one duty at a time. I take each duty to describe a task: an activity applied to an object of work, in the manner of O*NET's task statements, such as preparing financial reports or building fraud-detection models. A duty can also name the instrument the worker uses, such as a spreadsheet, a design package or an AI assistant. Technology can change the content of jobs in four ways, and they bear differently on the demand for labour (Table 1).

The first is new work: tasks whose activity or object did not exist before. New work is the reinstatement that offsets automation in the task framework of Acemoglu and Restrepo (2018, 2019), adding tasks in which labour has a comparative advantage, and what Autor et al. (2026) define by the novel expertise it demands. The other three can also raise the demand for labour, through productivity or the scale of existing work, but they do not add to the tasks that workers do. The second is a new instrument for a familiar task. When the carriage driver becomes a truck driver, the function persists and the tool changes (Skoglund et al., 2026); the technology changes how productive the worker is at the task, not what the task is. The third is a change in the division of labour: a task that existed elsewhere in the economy moves into an occupation that did not perform it, as when production managers build the dashboards that analysts built before. Occupations born of specialisation have held more employment, and lasted longer, than those tied to a new technology (Skoglund et al., 2026), and generative AI may reverse the trend by letting workers do tasks that used to need a specialist. The fourth is more of an existing task, the counterpart in tasks of what Autor et al. (2026) call more work. Variants of familiar duties, done with a tool, object or setting that is not itself new, and re-wordings of them are none of these.

This paper counts the first as new work and measures the others separately. For generative AI, I classify duties by what the worker does with AI (Section 4.2): building AI, which covers developing, deploying and selling AI systems; governing it; and working with it, which covers using AI tools in one's own work, directing or reviewing AI systems and their output, and driving the adoption of AI by finding uses for it, rolling it out and training colleagues. A new duty about AI is new work unless it only asks the worker to use AI tools in their own work, which is a familiar task with a new instrument. New work outside AI follows the same rule, as with a new product, technology or regulation.

Applying this classification requires six judgement calls, and I apply each one in the same way throughout. The first three set where new work ends, and each is a step of the ladder that Figure 1 draws:

1. A new instrument alone does not make a task new: drafting with a language model is still drafting. Where using AI involves directing a system or checking its output, however, the duty describes a new activity and counts as new work; it is the strictest kind of new work and the first step of the ladder. The same rule is applied both to AI duties, using their activity labels, and to non-AI duties, using a separate reading that asks whether the activity, the object of work or only the instrument is new (Section 4.3).
2. A familiar activity applied to a new object counts as new work, such as building language-model systems or reporting under a new regulation. A technology counts as new if it was not present in early-2019 postings, even when it succeeds an older technology, as language models succeed earlier machine learning and AI agents succeed robotic process automation. Much of building AI is therefore a new vintage of work that existed in 2019, and the count is generous on this margin; AI work with a close counterpart among the duties of 2019 is measured separately (Section 5.3).
3. Driving the adoption of AI counts on the same basis, with the organisation's use of AI as its object, even though the activity resembles the rollout of earlier enterprise software. It is the last step of the ladder about AI; new work outside AI completes it.

The other three fix how duty lines are compared and classified:

4. New work is defined relative to all postings in the first half of 2019, before COVID-19 and generative AI. A task that other occupations already performed is not counted as new work when it moves into a new occupation; it is measured separately as a change in the division of labour (Section 5.4).
5. AI work is classified by the worker's activity rather than the employer's product: a bank's data scientist training a fraud model and an AI vendor's account manager selling its tools both count as building AI, which includes selling and supporting AI products, whereas a paralegal who trains colleagues to use an AI assistant is working with AI.
6. Categories blur at the margin. Workers who rate model responses to train models are grouped under reviewing AI output rather than building AI, though both count as new work; and some duties that mix using AI tools with helping colleagues adopt them are grouped under tool use and excluded from new work, so the count is if anything understated there.

Table 1: Kinds of change in job content, with duty lines from the postings

Kind of change	Examples (job title, year)	New relative to	New work?
A new activity or a new object of work, from the strictest kind up			
Working with AI: directing or reviewing AI	"Perform error and quality reviews of AI-patient communications" (Nurse Reviewer, RN, 2025); "Monitor alumni conversations with the AI Bot and follow-up with alumni who could benefit from additional 1:1 advising" (Postsecondary Success Advisor, 2024); "Orchestrate LLM-powered agents for research, summarization, outreach, and follow-ups" (AI-First Orchestration Engineer, 2026)	Economy	Yes
Building AI	"Architect and deploy scalable LLM + retrieval + agent systems in production environments" (Senior AI Engineer, 2026); "Design motion and animation for AI product interfaces, including loading states, streaming responses, task progress, and system transitions" (Motion Designer, 2026); "Deliver compelling demos and business cases that tie Serval's AI automation to measurable ROI and efficiency gains" (Account Executive, 2026)	Economy	Yes
Governing AI	"Lead efforts to establish and maintain effective governance frameworks for AI applications" (Internal Audit Manager, 2024); "Review AI shadow-IT assessments and surface compliance concerns to legal and IT leadership" (Paralegal, LegalOps, 2026)	Economy	Yes
Working with AI: driving adoption	"Build and maintain prompt libraries, workflow templates, and SOPs for AI-assisted legal tasks" (Paralegal, 2026); "Proactively explore and pilot AI tools to eliminate low-value manual work and build more scalable accounting operations" (Accountant, 2026)	Economy	Yes
New work outside AI	"Will assist with COVID-19 surveillance, screenings, and vaccination" (Licensed Practical Nurse, 2021); "Create NFT programs for brands designed to drive engagement and sales" (Product Manager, 2022); "Generate, analyze, and distribute monthly NIL transaction reports" (Associate Director of Athletics, 2025)	Economy	Yes
Other change in job content			
Working with AI: using AI tools in one's own work	"Leverage AI tools and platforms to optimize administrative workflows, including calendar management, data organization, and travel planning" (Executive Assistant, 2025); "Leverage AI tools (e.g., Claude) to analyze vendor proposals, summarize contracts, and extract key terms and risks" (Director of Procurement, 2026); "Uses AI tools (Copilot, Claude, ChatGPT) daily for code reviews, design drafts, and debugging" (Technical Lead Manager, 2026)	The tool, not the task	No
Existing task, new to the occupation	"Leverage SQL and Python to automate reporting" (Senior Financial Analyst, 2025); "Use SQL, dashboards, and automation tools to establish real-time monitoring and alerting systems that detect anomalies or potential compliance issues" (Compliance Analyst, 2026)	Occupation	No
More of an existing task	"Collaborate with litigation teams to design and implement Technology Assisted Review (TAR) and other AI-driven review workflows" (Litigation Support Project Manager, 2026), beside 2019's "Develop and execute approaches to technology assisted review tools in litigation and investigation response"; "Take end-to-end ownership of machine learning systems" (Machine Learning Tech Lead, 2026), beside 2019's "Take end to end ownership of Machine Learning systems"	—	No
Variant or re-wording	"Directing newscasts in studio using Grass Valley Ignite Production System and ENPS Newsroom software" beside 2019's "Floor directs newscasts"; "You will prepare and lead presentations of the monthly S&OP readout" beside "Lead the monthly S&OP process"	—	No

Notes: Duty lines quoted from postings (some shortened), classified by the paper's measures: the reading of flagged lines, the AI activity labels and the three-question reading for the first six rows (Section 4); the technical-tool families for tasks new to the occupation (Section 5.4); and embedding similarity to a 2019H1 duty for the last two rows. Appendix Tables C2 and C3 report three further examples of each kind drawn at random.

3 Data

3.1 Postings from applicant tracking systems

Large employers publish their job adverts through applicant tracking systems: software that hosts the advert and collects applications, such as Greenhouse, iCIMS, Workday or Lever. An ATS can be detected for 97% of the Fortune 500 (Purcell, 2026). ATS job pages sit at predictable web addresses, and Common Crawl, a public archive of the web, has captured many of them since 2017 (Common Crawl Foundation, 2026). I query the archive's index for the job pages of the 14 ATS platforms whose archived pages reliably give at least a title and a location, in all 96 crawls from January 2017 to September 2026, parse each page for its structured data and with a template for the platform, merge reposted adverts and keep the postings in the United States: 3,341,872 postings from 52,473 employers (Appendix A). Each posting has its title, employer, location, pay where stated, text and dates. I date each posting by its stated posting date when that falls within a year before the crawl that first captured it (87% of postings), and by the date of first capture otherwise. A text classifier trained on 63,000 postings coded by a language model (Claude Opus 5.5) gives each posting a detailed code in the 2018 Standard Occupational Classification; on held-out postings it gives the same code as the language model for 74% of them, about as often as two language models agree with each other (73%; Appendix A.3).

The postings capture a small, selective share of US hiring (0.5 postings in 2024 for every 100 hires in the Job Openings and Labor Turnover Survey): Common Crawl archives only a sample of each platform's job pages, these platforms cover part of the ATS market, and many employers, notably school districts, public employers and health systems, post elsewhere. Like all online vacancies, these postings over-represent professional work: to match the occupation mix of employment, 34% of them would have to change major group, against 21% of Lightcast's commercial postings (see also Hershbein and Kahn, 2018). Where the archive captures postings, though, their state, occupational and pay distributions match external benchmarks, and the shares of postings that state pay or offer remote work respond to known shocks, such as state pay-transparency laws and the pandemic (Figure A1 and Appendix A). As the sample over-represents computer and business occupations, I report results by occupation group and check the aggregates with weights that match employment and hiring.

3.2 Duty statements and the analysis sample

The unit of measurement is a duty statement: a sentence in which the employer lists something the new hire will do. A text layer restores the line structure of descriptions published as flat text, types each section header, splits the text into sentences and classifies each sentence as a duty, a requirement, a statement about the role, the company, pay and conditions, or process and legal text, dropping generic duties such as "other duties as assigned" (Appendix A). Trained on 45,000 sentences that Claude Sonnet 5.5 labelled while reading the whole advert, the classifier finds Sonnet's duties with precision 0.92 and recall 0.94 in cross-validation that holds out whole postings (Table A8). About one line in twelve is therefore not a duty, most often a general description of the role, and the reading of flagged lines sets such lines aside (Section 4.3). Throughout the paper, a duty (or duty line) is a single duty sentence in a posting, whereas a task is the activity and object of work it describes (Section 2).

The analysis sample covers postings dated from January 2019 to September 2026; the crawls of

2017–18 captured some platforms at several times their later rate (Appendix A). I keep English-language postings with a description of at least 200 characters and at least one duty line, drop bulk posters (employer-years with at least 1,000 postings across 50 or fewer titles), and keep only the earliest posting when an identical title and description is advertised across multiple locations (Appendix Table A4). The final sample comprises 1,629,317 postings from 48,304 employer accounts, containing 18.4 million duty lines and 10.1 million distinct duty sentences (Table 2). Each posting also has a seniority level, from its job title; an industry, a two-digit NAICS sector for its firm; and a firm identifier that links employer accounts across platforms, all from language-model labels extended by classifiers (Appendix B.8).

Table 2: Analysis sample by year

Year	Postings	Employers	Duty lines (M)	Lines per posting	Flagged (%)	Flagged AI (%)
2019	186,588	9,838	1.97	10.5	10.0[b]	0.11[b]
2020	133,435	9,577	1.39	10.4	11.8	0.15
2021	225,174	12,485	2.44	10.8	12.9	0.19
2022	179,575	13,227	2.00	11.1	14.2	0.15
2023	133,581	12,579	1.52	11.4	14.9	0.21
2024	263,295	17,474	3.02	11.5	14.4	0.36
2025	287,258	18,521	3.32	11.6	15.0	0.90
2026[a]	220,411	14,648	2.72	12.3	17.4	1.69
All	1,629,317		18.37	11.3		

Notes: Postings in English with a description of at least 200 characters and at least one duty line, excluding bulk posters, one posting per multi-location advert. Employers: employer accounts (platform and employer identifier). Flagged: duty lines with no close counterpart among the duty sentences of 2019H1 postings (Section 4.1). Flagged AI: flagged duty lines that describe AI work. [a]January to September. [b]2019H2 only: no line of the reference half-year is flagged.

3.3 Release

With a later version of this paper, I will release a posting-level dataset covering all 3,341,872 US postings, together with aggregate series by quarter, occupation, state and platform, and the code for the build. For each posting, the release will provide its dates, location, pay fields, occupation code with its confidence, seniority, industry, crawl-intensity and representativeness weights, flags for agencies, bulk posters, remote work and closed adverts, and posting-level summary measures of duty lines, novelty, AI work by side and tool use. The raw text of the adverts, which is the employers' own writing, will not be redistributed; the data will be released under CC BY 4.0 and the code under the MIT licence.

4 Measuring new work

Measuring new work across 18.4 million duty lines proceeds in four steps (Table 3). First, an embedding screen compares every duty line with the duties posted in the first half of 2019 and flags lines with no close counterpart. Second, a language model classifies every duty sentence containing an AI keyword by the worker's activity, and a classifier trained on those labels extends them to the full corpus. Third, the language model reads stratified random samples of the flagged lines alongside their closest 2019 counterparts and judges which describe new tasks. Fourth, the rule

of Section 2 separates new work from new instruments among the new tasks. Three terms recur: a *flagged line* has no close counterpart among the duties of 2019H1; a *new task* is a flagged line that the reading judges to describe a task uncommon in 2019; and *new work* is a new task whose activity or object of work is new, not only its instrument. A second language model independently re-reads subsamples of each labelling task (Appendix B.6).

Table 3: How new work is measured

Step	Question	Method	Applied to
1. Screen (Section 4.1)	Does the duty line have a close counterpart among the duty sentences of 2019H1?	Sentence embeddings; a similarity threshold set on 2019H2	Every duty line
2. AI labels (Section 4.2)	Is the line about AI, and what does the worker do with it?	Keyword rules; Sonnet 5.5 reads every keyword sentence; a classifier for the rest	Every duty line
3. Reading (Section 4.3)	Does a flagged line describe a new task, a familiar task with a different tool, object or setting, or a familiar task in new words, or is it not a duty?	Sonnet 5.5, beside the closest duty of 2019H1	Random samples of flagged lines
4. Rule (Section 4.3)	Is the new task's activity or object of work new, or only its instrument?	AI lines: the activity labels; other lines: Sonnet 5.5 asks three questions	Lines read as new tasks

4.1 Duty lines with no counterpart in 2019

A duty line is flagged if no duty sentence in the first half of 2019 is close to it in meaning. I embed all distinct duty sentences using a sentence-embedding model (all-MiniLM-L6-v2; Reimers and Gurevych, 2019; Wang et al., 2020) and compute each sentence's maximum cosine similarity to the 679,314 distinct duty sentences from the first half of 2019 (2019H1). Using the first half of 2019 as the baseline provides a pre-pandemic, pre-generative-AI reference while leaving the second half of 2019 (2019H2) as an immediate pre-shock comparison period. The reference is earlier postings rather than a task inventory such as O*NET, whose lists omit much long-standing work (Wang et al., 2026b), so that a duty is flagged only if employers did not ask for anything close to it before.

Even over six months with little technological change, some duty lines lack a close match in the reference pool because employers phrase familiar duties in new ways and the archive captures a sample of postings. I use 2019H2 to calibrate the similarity threshold, setting $\tau = 0.66$ so that 10% of 2019H2 duty lines fall below it. This cutoff is intentionally loose: it flags a tenth of the lines of a quiet period—only 1.6% of which the language model judges to be genuine new tasks (Section 5.1)—so that the screen misses as few true new tasks as possible before the language-model reading filters out the rest (Section 4.3). Evaluating pairs across the full similarity distribution confirms that new tasks are rare above $\tau = 0.66$, while most pairs below it are variations on familiar tasks (Appendix Figure B3 and Appendix B). Varying the threshold so that between 5% and 20% of 2019H2 lines are flagged shifts the overall level of new work but leaves cross-occupation comparisons unchanged (Appendix Table C4 and Appendix D).

As time passes from the 2019H1 baseline, the share of flagged lines drifts upward, reaching about 14% by 2022 across high- and low-AI-use occupations alike. Most of this drift reflects turnover in the phrasing, tools and settings of familiar duties: of the 4.2 percentage-point increase in flagged

non-AI lines between 2019H2 and 2022, Sonnet judges only 4% to be new tasks (Section 5.1). When novelty is measured against a rolling two-year reference window of fixed size instead of 2019H1, flagged non-AI lines remain flat over time while flagged AI lines still rise sharply (Appendix B). Given this wording drift, I rely on the language-model reading rather than the raw screen count to measure new work.

Duties from the COVID-19 pandemic provide a check on whether the pipeline detects a known wave of new tasks. Of the duty lines of 2020–21 that mention COVID-19, 67.7% are flagged, and Sonnet reads 38% of these as new tasks, against a few per cent of other flagged non-AI lines; clustered into themes, the flagged lines of 2020–21 recover COVID-19 duties, from screening and safety protocols to vaccination programmes (Appendix B).

4.2 AI work: building it or working with it

I find duty lines about AI in two steps. Keyword rules flag every duty sentence that mentions AI, machine learning, language models, generative AI or named AI tools. Claude Sonnet 5.5 then reads every one of the 134,656 keyword sentences, with the job title and occupation group but not the date, and records whether it describes AI work, which side of the technology it is on and what the worker does. Building AI covers developing, training and evaluating models and AI products; deploying and integrating them; and selling, implementing and supporting them for clients. Working with AI covers using AI tools in one's own work; prompting, directing or reviewing AI systems and their output; and finding uses for AI, redesigning workflows around it and training colleagues in it. Governing AI (policy, risk and compliance) is a third side. These sides classify the worker's activity rather than the employer: a bank's data scientist training a fraud model and an AI vendor's customer-success staff implementing its software for client firms are building AI, whereas a firm's own staff who find uses for AI or train their colleagues to use it are working with AI (a sentence that mentions both takes the side it is mainly about).

A second model (Claude Opus 5.5) independently reads 2,000 of these sentences and agrees with Sonnet on the side of AI work in 90% of cases (Cohen's κ 0.82). Duties about building AI often name specific methods rather than "AI" itself, so the keyword rules catch about half of building-AI work compared with most working-with-AI duties from 2024 on. I account for this in two ways (Appendix B.3): a classifier trained on all labelled sentences extends the AI labels to every duty line in the corpus, and the overall split between building AI and working with it is reported after dividing each side's keyword lines by its keyword catch rate, estimated on random samples of keyword-free lines.

4.3 Reading the flagged lines

Every language-model label in the paper comes from either a census or a stratified random sample (Appendix Table B7). Sonnet 5.5 reads random samples of flagged AI and non-AI lines, each paired with its closest duty from 2019H1 and stripped of its year and occupation cell, and classifies each line as a new task, a familiar task with a different tool, object or setting, a re-wording of a familiar task, or not a duty. The prompt defines a new task as an activity, responsibility or object of work that was uncommon in 2019 postings for that job, including a familiar activity performed with a genuinely new technology or under a new product or regulatory regime. The sample comprises 500 flagged non-AI lines per year (2019H2–2026) in each third of occupations by AI use (plus 250 per year in occupations without an AI-use score), together with 400 flagged AI lines on each side

of AI work in 2023–24, 2025 and 2026. Lines are presented in mixed batches of 40 drawn across years, thirds and AI status (Appendix B.6 and Appendix D report unpooled strata and alternative batching). Opus 5.5 independently re-reads a fifth of the batches, agreeing on most classifications while judging fewer non-AI lines to be new tasks (Appendix B.6).

The reading prompt counts a familiar task performed with a new technology as a new task, so a second step applies the definition of new work from Section 2, separating new activities and new objects of work from new instruments for familiar tasks. For AI lines, the activity labels of Section 4.2 make this distinction directly: duties that only ask workers to use AI tools in their own work (or fit no specific activity) are reported separately and excluded from new work. For non-AI lines that the first reading classifies as new tasks, a follow-up reading asks three separate questions—whether the activity is new, whether the object of work is new, and whether only the instrument is new—and counts a line as new work only if its activity or object is new. About three in five pass this filter, while the remainder involve only a new tool or, on re-reading, nothing new. Applied to the AI new tasks as a check, the same three questions closely match the activity labels, classifying most tool-use lines as only a new instrument and nearly all directing-or-reviewing lines as a new activity (Appendix B.6). Finally, although the reading prompt evaluates novelty relative to 2019 postings for that kind of job whereas the embedding screen compares against all occupations, the two criteria rarely conflict: fewer than 1% of lines read as new tasks involve a technical tool that was merely new to that occupation (Section 5.4).

To compare occupations by their uptake of AI, I group them into posting-weighted thirds of observed AI use from the Anthropic Economic Index (Massenkoff and McCrory, 2026), which measures the share of an occupation's task time spent on tasks rated feasible for language models by Eloundou et al. (2024) and observed in work-related conversations with Claude in 2025. The high-use third spans 72 detailed occupations and the low-use third 459. Observed AI use is measured in 2025, so Appendix D repeats the main comparisons using thirds of the pre-existing exposure score of Eloundou et al. (2024).

5 Results

Table 4 sets the four kinds of change in job content side by side, each in duty lines per 100 duty lines, for 2022 and 2026 and for each third of occupations by AI use. All four grew most where AI is used most, and two of them dominate: new work, and AI work that has a close counterpart among the duties of 2019. In the high-use third, new work rose from 0.5 to 2.6 per 100 duty lines and that existing AI work from 0.6 to 3.1. New instruments, almost all of them duties that ask workers to use AI tools, rose from 0.04 to 0.8, and duty lines that name a data, code or automation tool new to the occupation from 0.06 to 0.2. In the third that uses AI least, each stayed at or below about 0.2. The rest of this section takes them in turn—new work and new job titles, then new instruments and more of existing AI work, then tasks crossing occupational lines—before where new work grows across occupations, the official task lists and pay.

5.1 New work

New work has grown sharply in the occupations that use AI most (Figure 1). Across all occupations, new work rose from 0.3 to 1.2 per 100 duty lines between 2022 and 2026 (panel a). In the high-AI-use third of occupations it rose about fivefold, from 0.5 to 2.6 per 100 duty lines, while in the

Table 4: Four kinds of change in job content, per 100 duty lines

	Most AI use		Middle third		Least AI use		All	
	2022	2026	2022	2026	2022	2026	2022	2026
1. New work: a new activity or a new object of work								
About AI	0.13	2.01	0.03	0.74	0.01	0.14	0.05	0.92
Outside AI	0.39	0.57	0.22	0.32	0.17	0.08	0.26	0.32
2. A new instrument for a familiar task								
Using AI tools in one's own work	0.02	0.71	0.01	0.28	0.00	0.08	0.01	0.35
Outside AI: only the tool is new	0.02	0.13	0.01	0.07	0.01	0.02	0.01	0.07
3. An existing task, new to the occupation								
Data, code and automation tools[a]	0.06	0.19	0.04	0.10	0.01	0.05	0.04	0.12
4. More of an existing task								
AI work with a 2019H1 counterpart[b]	0.61	3.06	0.12	1.05	0.01	0.16	0.22	1.35
All AI duty lines	1.04	7.38	0.23	2.63	0.03	0.49	0.38	3.34

Notes: Duty lines of each kind per 100 duty lines, by third of occupations by observed AI use; 2026: January to September. Row 1: new work as in Figure 1. Row 2: duty lines that ask the worker only to use AI tools in their own work, every line counted, flagged or not (Section 4.2); outside AI, lines read as new tasks that the three-question reading finds to change only the instrument (Section 4.3). [a]Duty lines that name a tool from a family of data, code or automation tools that the posting's occupation did not use in 2019–20, each weighted by the share of such matches that Opus 5.5 confirms (Section 5.4); in the occupations with confident codes and at least 500 postings in 2019–21, which hold 70% of the duty lines of the high-use third and 54% of all duty lines, per 100 of their duty lines. [b]Duty lines about building, governing, directing or adopting AI that have a close counterpart among the duties of 2019H1 and so are not flagged; reading random duty lines without the screen finds about 24% (Sonnet) to 41% (Opus) of these lines in the high-use third in 2026 to be new work (Appendix D). All AI duty lines also include general statements about AI and flagged AI lines read as variants, re-wordings or not duties.

low-use third it remained flat at about 0.2 (panels b and c). Figure 1 stacks new work from the strictest kind up, following the judgement calls of Section 2. Directing AI systems and reviewing their output, the clearest new activity, is still rare: 0.06 per 100 duty lines in the high-use third in 2026. Building and governing AI, familiar activities applied to a new object, bring new work to 1.80; driving the adoption of AI to 2.01; and new work outside AI to 2.58. Duties that ask workers only to use AI tools are new instruments rather than new work and are not counted (Appendix Table C1 adds them). Computer occupations hold 74% of the flagged AI lines of the high-use third in 2026, but the contrast between the thirds holds without them and under other weights (Appendix Table C4).

Outside AI, new work is small and has grown much less. Most flagged non-AI lines are variants or re-wordings of familiar duties (Appendix Figure B2): only a few per cent are new tasks, and the three-question reading finds about three in five of those to be new work. On the paper's reading, new work outside AI in the high-use third rose by 0.2 per 100 duty lines between 2022 and 2026, against 1.9 for new work about AI. The screen misses some new work phrased in familiar terms (Appendix Table B1), so I also read random duty lines of 2022 and 2026 without the screen. On that reading, new work outside AI rose by 1.1 per 100 duty lines (Sonnet) or 0.6 (Opus), part of it AI work that the AI labels miss, and new work about AI by 2.8 (Appendix D). The reading by occupation and half-year of Section 5.5, with far more lines, puts the two rises at 0.6 and 2.4. On every reading, new work about AI grew at least 2.6 times as fast as new work outside it. Clustering the flagged lines of 2025–26 into themes gives the same picture: of the 60 fastest-rising themes, 49 concern AI, and new work outside AI, from blockchain software and crypto assets to new compliance regimes, is too dispersed across topics to form rising themes (Appendix C.1).

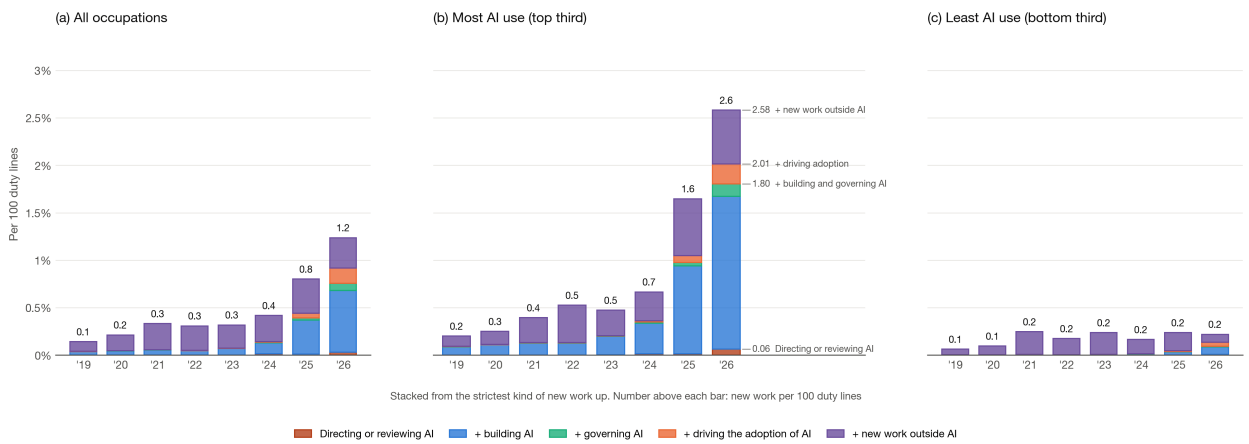


Figure 1: New work in duty statements, stacked from the strictest kind up. New work per 100 duty lines, by year (2019: second half; 2026: January to September): flagged duty lines (no close counterpart among the duty sentences of 2019H1 postings) that Sonnet 5.5 reads as new tasks and whose activity or object of work is new (Section 4.3). Bottom to top, the steps of the judgement calls of Section 2: directing or reviewing AI, a new activity; building and governing AI, familiar activities applied to a new object; driving the adoption of AI, with the organisation's use of AI as its object; and new work outside AI. Duties that only ask the worker to use AI tools are new instruments and are not drawn (Appendix Table C1 gives them). The shares come from Sonnet's reading of 500 flagged non-AI lines in every year and third of occupations by AI use and of flagged AI lines in 2023–24, 2025 and 2026, by activity, read in batches that mix the samples (2019–22: the shares of 2023–24; governing AI lines: the pooled share of building and working with AI); the non-AI new tasks are scaled by the share that a second reading finds to be new work. (a) All occupations. (b, c) The thirds of occupations with the most and least observed AI use; beside the last bar of (b), the running total at each step. Numbers above the bars: new work per 100 duty lines.

5.2 New job titles

New job titles carry the most distinctive duties (Figure 2). Five of the six largest titles that are new since 2022 or have grown fastest are AI titles, and they rose steeply from late 2024 (Appendix Table C9 gives the counts): forward deployed engineers, posted by 9 employers in 2019–22, were posted by 247 in 2024–26, and GTM engineers, a title absent from these postings before 2024, by 118. The new titles also come with duties unlike their neighbours': more of their duty lines have no close counterpart among the duties of their three nearest established titles than is true of established titles (panel b), a measure of distinctiveness from similar titles rather than of new work in the sense of Figure 1. New titles are rare, however: titles that at least three employers used in a year and that no posting of 2019–22 used, in any spelling, are under 1% of 2026 postings (Appendix C). Measures of new work built on official title lists (Lin, 2011; Autor et al., 2024, 2026) record such titles only once a classification adds them; the Census index of titles that Autor et al. (2026) use was last revised for 2018.

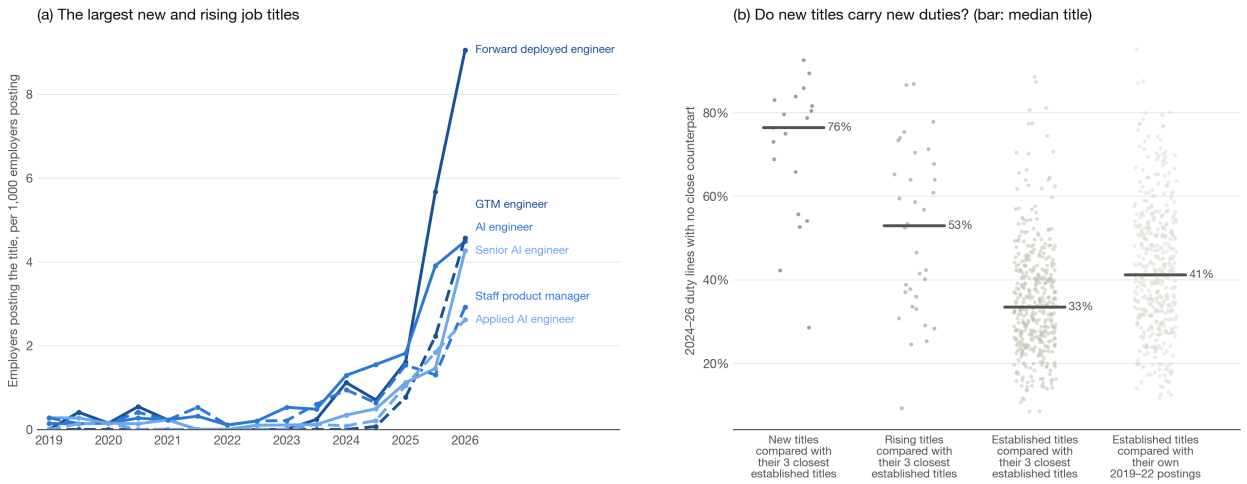


Figure 2: New job titles. (a) Employers posting the largest new and rising titles, per 1,000 employers posting in the half-year, 2019H1–2026H1 (Appendix Table C9 gives the counts). New titles: at least 10 employers in 2024–26 and none in 2019–22, in any spelling (ignoring word order, seniority words and initialisms) and in any posting of those years; rising titles: at least 20 employers in 2024–26 and five times their 2019–22 count. All but staff product manager are AI titles, with an AI duty in at least half of their postings. (b) Share of 2024–26 duty lines with no close counterpart among the duties of the three most similar established titles (new and rising titles, and established titles against each other) or of the title's own 2019–22 postings; bar: median title.

5.3 New instruments and more of existing AI work

New work is only part of the growth of AI work. Counting every AI duty line, new or not, AI work grew from 0.4 to 3.3 per 100 duty lines between 2019 and 2026 across all occupations (Figure 3a), and its mix shifted from building AI towards working with it. The sides classify activities rather than employers (Section 4.2), following the distinction between AI developer and AI user skills in Maslej et al. (2025) and Koll et al. (2026) but measured in duties rather than skill tags. Building AI grew first and most in computer occupations, where about a third of postings had a building-AI duty by mid-2026 (Appendix Figure C3). Working with AI barely existed before 2024; by mid-2026 it appears in about one in seven postings in computer occupations and one in nine in management and business occupations. Allowing for the share of each side's lines that the keywords catch (Appendix B.3), working with AI rose from 4% of AI work in 2019–23 to 22% in 2026, though building AI is still the larger side. AI duty lines rose in every professional and technical group, most in computer occupations and, second, in legal occupations, which had almost no AI duties in 2022 (Figure 4); the groups with little AI work are those whose work is done by hand or in person, such as health-care support, production, transportation, construction and food preparation.

Working with AI straddles the line between new work and new instruments. Driving the adoption of AI—finding uses for it, rolling it out and training colleagues—and directing AI systems or reviewing their output are new work, whereas using AI tools in one's own job applies a new instrument to a familiar task. In 2026, about half of working-with-AI duty lines drive adoption, two-fifths ask workers to use AI tools in their own work and only a small share ask them to direct AI systems or review their output (counting all duty lines across occupations; Figure 3b shows the breakdown among keyword lines). Duties that ask workers to use AI tools were virtually absent in 2022 but appear in 3.2% of all postings by 2026 and in 6.1% of postings in the high-use third, where they are 0.7 per 100 duty lines (Table 4). They describe familiar work—drafting, research, coding—done

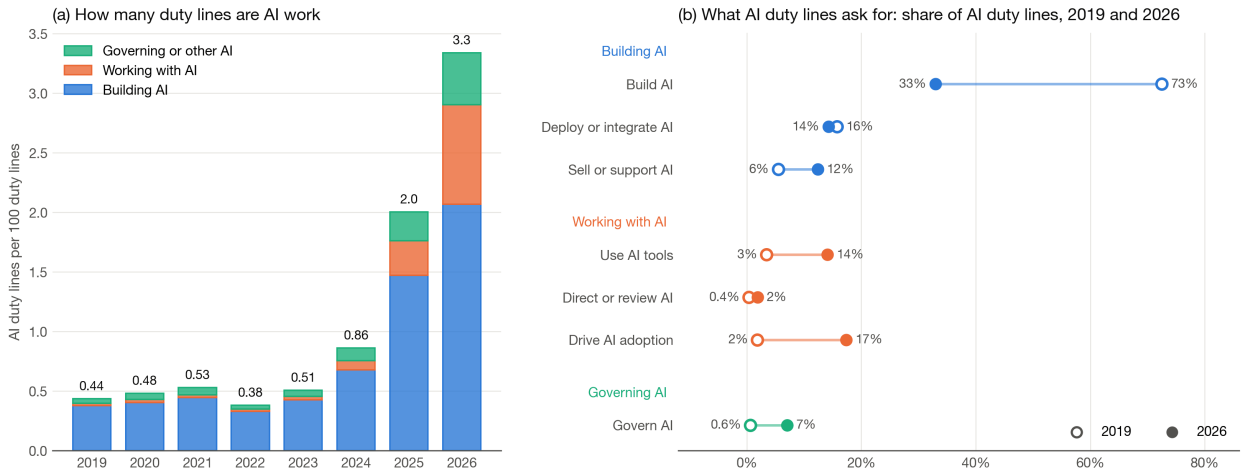


Figure 3: What AI duties ask for. (a) AI duty lines per 100 duty lines, by year and side; every duty line is classified (Sonnet 5.5 labels of sentences with an AI keyword, a classifier for the rest). (b) The activities of AI duty lines as shares of all AI duty lines, 2019 (open) and 2026 (filled): Sonnet 5.5 labels of every duty sentence with an AI keyword; the shares exclude AI work that fits no activity and are not corrected for the keywords' catch rates (Appendix B.3). Appendix Table C5 gives every year.

with a new tool (Table 1): AI as an instrument that may raise productivity rather than a source of new tasks. Surveys find generative AI adopted at work faster than earlier technologies (Bick et al., 2024), and experiments find that it raises the productivity of the people who use it (Brynjolfsson et al., 2025b; Noy and Zhang, 2023); these duties are the demand side of that adoption. So far, employers ask workers to use AI tools much more often than they ask them to direct or check AI systems.

Much of the rest of the growth of AI work is more of work that existed before generative AI. In the high-use third, duty lines about building, governing, directing or adopting AI that have a close counterpart among the duties of 2019H1 rose from 0.6 to 3.1 per 100 duty lines between 2022 and 2026 (Table 4), more than new work about AI. Most of them build, deploy or sell AI systems, as in "Take end-to-end ownership of machine learning systems" (Table 1), work that postings of 2019 already described for machine learning. A direct reading finds some new work among them (Table 4), but a large part of the growth of AI work is the growth of AI work that already existed, the counterpart in tasks of what Autor et al. (2026) call more work.

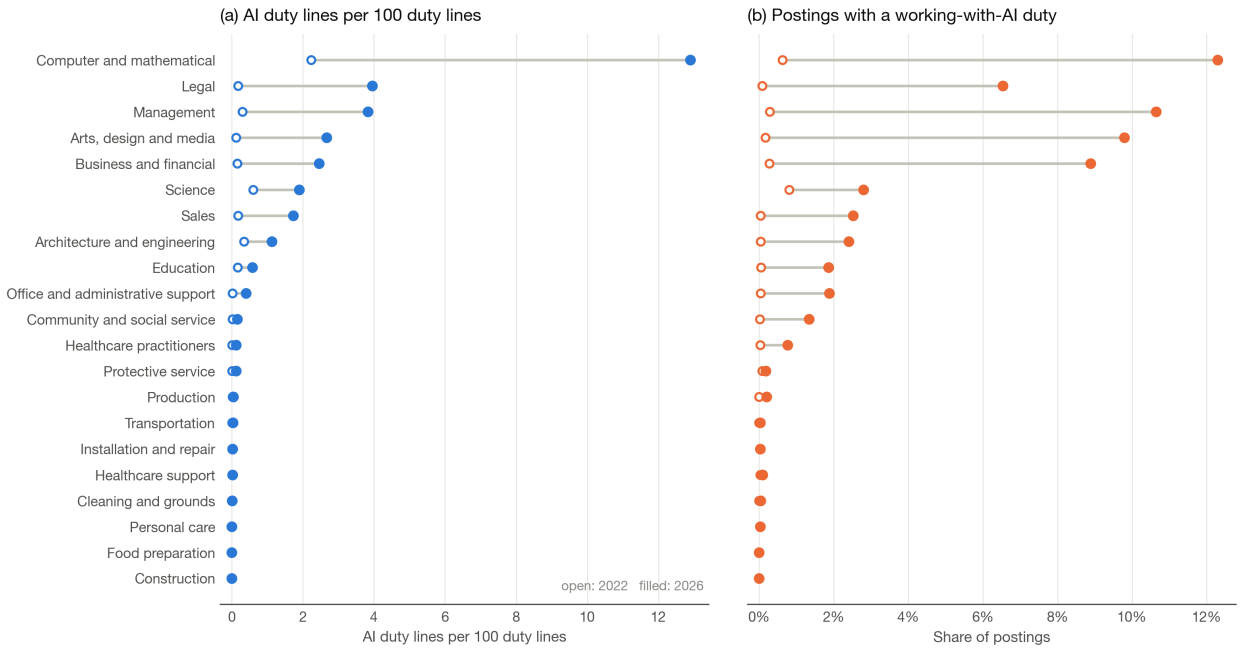


Figure 4: AI duties by occupation group, 2022 (open) and 2026 (filled). (a) AI duty lines per 100 duty lines, every duty line counted (Sonnet 5.5 labels for keyword sentences, a classifier for the rest). (b) Share of postings with a working-with-AI duty. SOC major groups with at least 1,000 postings in both years, sorted by panel (a) in 2026.

5.4 Existing tasks crossing occupational lines

The new work of the previous sections is new to the economy's postings: it has no close counterpart among the duties of early 2019 in any occupation. A task can also be new to a job without being new to the economy, when an occupation takes on a task that other occupations already did, such as a production manager building the dashboards that analysts built before. That is a change in the division of labour rather than new work (Section 2), and possibly a reversal of the specialisation that has shaped occupations in the past (Skoglund et al., 2026): tools, AI among them, may let workers do tasks that used to need a specialist, and ChatGPT users already ask for help with the tasks of other occupations (Chin et al., 2026). I measure one well-defined kind of task crossing: tasks done with technical and data tools. I define five families of technical and data tools in advance: four drawn from O*NET's November 2022 software taxonomy (dashboards and business intelligence, databases, programming, and statistical software, combining O*NET's "hot technologies" with newer tools not yet in O*NET) plus workflow automation, which O*NET lacks. I classify an occupation as not having used a tool family prior to generative AI if fewer than 1% of its 2019–20 postings named a tool from that family in a duty statement, and then track how often subsequent postings in those occupations name one. Opus 5.5 evaluates stratified samples of matched lines to verify that the worker is actually asked to use the tool, and the series are adjusted for the resulting precision rates (Appendix B.2).

The share of postings asking for a technical tool family that the occupation rarely used in 2019–20 more than tripled between 2022 and 2026, rising from 0.36% to 1.24%, with most of the increase occurring after 2024 and concentrated in dashboards and workflow automation (Figure 5a). In percentage-point terms the increase is largest in the high-AI-use third of occupations, though from its lower baseline the low-use third grew at a similar relative pace, and postings that adopt these

tools also ask workers to use AI more often than other postings in the same firm and occupation (Appendix B.2). Concrete examples include production managers building dashboards to monitor quality and output, financial analysts writing Python, SQL and VBA to automate reporting, and management analysts building low-code workflows (panel b). A tool named for the first time may also be a new instrument for a task the occupation already did, such as tracking output in a dashboard rather than a spreadsheet, so the series may include some new instruments as well. Counted in duty lines rather than postings, this crossing remains small beside new work: 0.2 per 100 duty lines in the high-use third in 2026, against 2.6 for new work (Table 4). Use of these tools also deepened within the occupations that already used them in 2019–20 (and business-intelligence tools had already begun spreading into managerial roles earlier; Koll et al., 2026). Virtually none of this crossover is double-counted as new work in Section 5.1: fewer than 1% of the lines read as new tasks name a tool new to their occupation. Appendix B.2 and Appendix D (Figure D1) show that the post-2022 rise is not an artefact of regression to the mean from the 2019–20 baseline, platform shifts or the 1% cutoff, and that dashboards and workflow automation spread several times faster after 2022 than the remaining eleven O*NET software families.

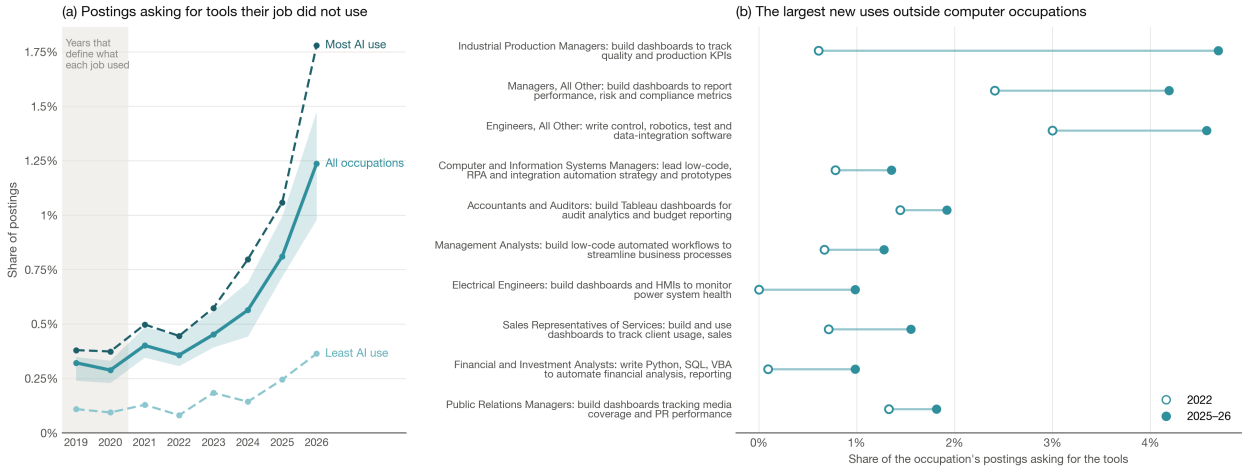


Figure 5: Technical and data tools new to the job. (a) Share of postings with a duty that names a tool from a family that the occupation did not use in 2019–20 (fewer than 1% of its postings), each weighted by the share of such matches that Opus 5.5 confirms (Appendix B.2): all occupations and the thirds with the most and least observed AI use; 145 detailed occupations with confident codes, weighted by their 2019–21 postings. Shaded: 95% interval from resampling employers and the validation sample. Grey: the years that define each occupation's earlier use. (b) The largest such occupation and tool pairs outside computer occupations whose recent lines Opus confirms, with Opus's summary of what the workers do: share of the occupation's postings in 2022 (open) and 2025–26 (filled).

5.5 Where new work grows across occupations

The paper's readings are drawn by year and third of occupations, too thin to follow new work occupation by occupation. A further reading covers every occupation and half-year: up to 50 flagged AI duty lines of each, all of them where there are fewer, and seven flagged non-AI lines for each AI line, read in the paper's batches and weighted to the flagged lines of their cell (126,696 lines in 3,602 cells; Appendix B.7). Added up by third, it reproduces Figure 1: new work about AI in 2026 is 2.46 per 100 duty lines in the high-use third and 0.14 in the low-use third, against 2.01

and 0.14. I estimate, for 222 detailed occupations over half-years from 2019 to 2026,

$$y_{ot} = \alpha_o + \gamma_t + \sum_{s \neq 2022\text{H2}} \beta_s \, \mathbf{1}[t = s] \, U_o + \varepsilon_{ot},$$

where U_o is the occupation's observed AI use, standardised, and y_{ot} is new work per 100 duty lines; occupations are weighted by their 2019–22 postings, and standard errors are clustered by occupation. New work about AI is unrelated to occupations' AI use before 2023 and rises with it from 2023 (Figure 6a): averaged over 2024–26 against 2019–22, an occupation one standard deviation higher in AI use gains 0.29 more lines of new work about AI per 100 duty lines (standard error 0.14). New work outside AI rises less (0.11, standard error 0.06; panel b), and duties that only ask workers to use AI tools rise from 2025 (panel c). The gradient runs between broad occupation groups rather than within them: new work about AI is concentrated in computer occupations (4.5 per 100 duty lines in 2026), management (1.2), legal occupations (1.1) and arts and media (0.6), and occupations compared only with others in their major group gain no more of it when they use AI more (Appendix C.2). The comparison is descriptive: observed AI use is measured in 2025 and overlaps closely with whether an occupation's work is done at a screen.

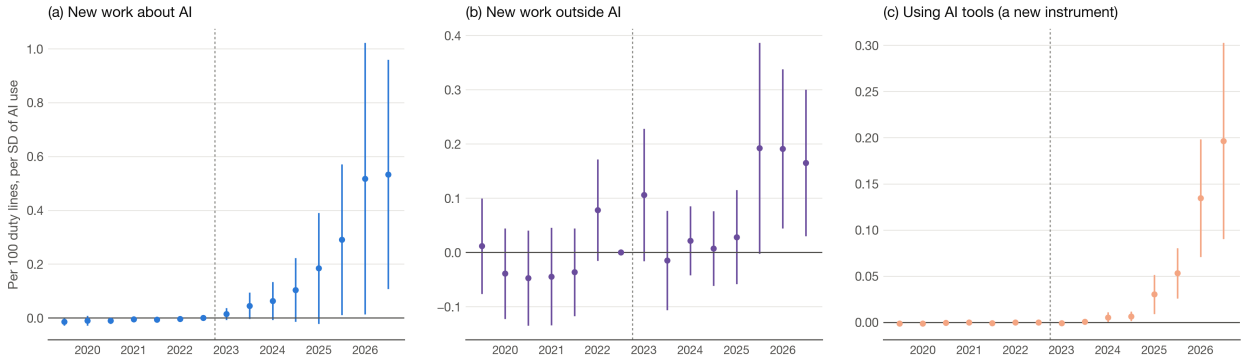


Figure 6: New work by observed AI use of the occupation: coefficients β_s on standardised AI use (Anthropic Economic Index) by half-year, relative to 2022H2, with 95% intervals; occupation and half-year effects, occupations weighted by their 2019–22 postings, standard errors clustered by occupation. 222 occupations with confident codes, 3,254 occupation-half-years. Outcomes per 100 duty lines, from the reading by occupation and half-year (Appendix B.7): (a) new work about AI; (b) new work outside AI; (c) duty lines that only ask the worker to use AI tools, read as new tasks. Appendix Figure C5 adds major group × half-year effects.

5.6 New work and the official task lists

Most research on the labour-market effects of AI relies on O*NET's occupational task lists to define what each occupation does and which tasks AI can perform (Webb, 2020; Felten et al., 2021; Eloundou et al., 2024). These lists lag behind the duties that employers advertise. Of the 19,265 task statements in O*NET 27.1 (2022), only a handful mention artificial intelligence or machine learning, mostly as one technology in a list in engineering and science occupations, and none concerns language models or working with AI; O*NET 31.0, released in August 2026, nearly four years after ChatGPT, has 103 task statements that 27.1 lacked, none of them about AI. Interviews with workers point the same way: the software developers interviewed for TaskNet in 2026 describe designing prompts and retrieval pipelines, evaluating model outputs and integrating language models into their systems, tasks with no counterpart in O*NET 31.0's list for the occupation (Wang et al., 2026b). Matched to the task list of their own occupation (Appendix B.5), 89% of continuing duty

lines have a closest O*NET task that describes related work, against 49% of flagged lines about building AI and 58% of those about working with AI, whose closest tasks are older analogues such as programming robots or building business-intelligence tools; part of this gap reflects the screen's selection of lines unlike those of 2019, since flagged non-AI lines also match less often (67%). Exposure measures built on these lists still point to the right occupations: across detailed occupations, the share of duty lines read as new tasks has a rank correlation of 0.71 with the exposure score of Eloundou et al. (2024), because employers are adding AI work in the same occupations whose existing tasks are exposed to language models. Tracking how tasks change inside those occupations requires reading job postings directly.

5.7 Pay

Postings with AI duties advertise higher pay, and most of the premium is accounted for by which firms and job titles post them. Figure 7 decomposes the posted-pay premium, on 428,836 postings that state pay, across successively tighter comparisons. Against postings in the same occupation, state and year, building-AI postings advertise 0.17 log points more and working-with-AI postings 0.05. Within the same firm, occupation and year the premia fall to 0.07 and 0.02, and within the same job title at the same firm and year neither is distinguishable from zero. That last comparison rests on the few title cells in which some postings have the duty and others do not, and its interval for working with AI allows a premium of up to 5 per cent. Firms often introduce AI work through dedicated, high-paying job titles (Appendix Table C9), so conditioning on the exact title absorbs that channel rather than ruling out a premium. These patterns hold when AI duties are identified from keyword matches alone, when the sample is restricted to states with pay-transparency mandates or to a balanced panel of firms, and when pay is measured at the top or bottom of the posted range; correcting for classification error increases the estimated premium for building AI (Appendix D). They are consistent with the premia estimated for AI developer and AI user skills in Lightcast postings (Koll et al., 2026), and Appendix Table C6 reports estimates by major occupation group.

The line between new work and new instruments carries a prediction for pay. New work demands expertise that is scarce at first, so it should command a premium that fades as the expertise spreads: Autor et al. (2026) find such a premium within detailed occupations, largest for new work linked to a technology, and Skoglund et al. (2026) find none for occupations created in the twenty-first century. A new instrument for a familiar task demands no comparably scarce expertise. The posted pay of working-with-AI duties fits this prediction (Figure 7). Within the same firm, occupation and year, postings with duties that drive the adoption of AI or direct it advertise about five per cent more than other postings, and postings that ask workers only to use AI tools about two per cent less, a gap that remains, somewhat smaller, when seniority and state are held fixed too (Appendix D reports every comparison).

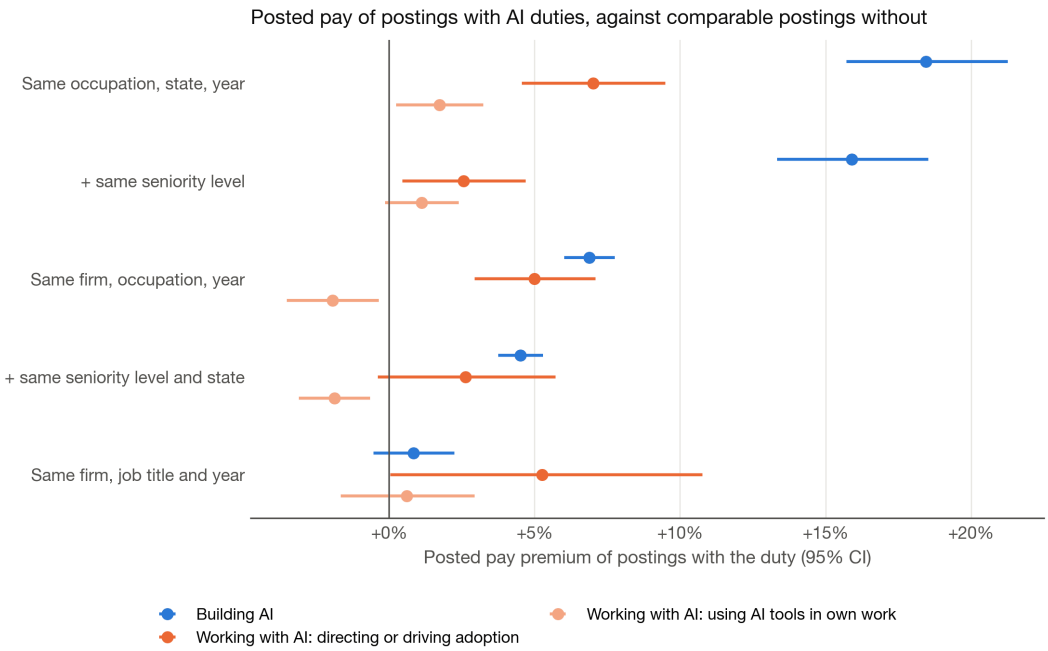


Figure 7: Posted-pay premium of postings with an AI duty, 2023–26, as the comparison tightens: log annual posted pay on indicators for a duty that builds AI, a working-with-AI duty that is new work (directing or reviewing AI, or driving its adoption) and a duty that only asks the worker to use AI tools in their own work (Section 2), with fixed effects for the cells named on the left (rows marked + add to the row above); every duty line counted; premia in per cent, $100(e^{\beta} - 1)$, with 95% intervals (the text reports log points). Standard errors clustered by firm.

6 Limitations

Several limitations qualify these findings. First, job postings record what employers ask of new hires rather than what existing employees do day to day; interviews with workers find oversight, coordination and teaching work that formal descriptions of jobs seldom record (Wang et al., 2026b), and postings may understate such work. The Common Crawl ATS sample captures a selective segment of US hiring tilted towards large, professional and technical employers. The mix of platforms and firms in the archive also shifts over time—notably as Workday descriptions enter in 2022 and Lever leaves in late 2025—and potential changes in how platforms or employers restrict web crawlers after 2023 cannot be observed directly. Reweighting the sample to match employment or hiring distributions alters the aggregate level of new work, though the cross-occupation contrasts hold when Workday and Lever are dropped, when platform shares are held constant, and when estimation is restricted to a balanced panel of firms or to continuing firm-by-occupation cells (Appendix Table C4 and Appendix D).

Second, all classifications rely on language models and classifiers trained on their outputs, validated against a second language model rather than a human-coded benchmark. Inter-model agreement is high for occupation codes and AI activity labels, and blinding the model to the job title barely alters the AI classifications. Uncertainty is greater for new work outside AI: non-AI new tasks are rare, more sensitive to prompt wording and batch composition, more often missed by the embedding screen when written in familiar language, and judged more conservatively by Opus than by Sonnet. Section 5.1 therefore reports the range across both models and a reading of random duty lines

without the screen, and Appendix D the bounds across prompts and recall rates.

Third, job descriptions have lengthened since 2023, and some employers now use language models to draft them. Longer and more standardised adverts increase the raw rate at which the embedding screen flags text: benefits and legal sentences, for example, are flagged 34% of the time in 2026, compared with 17.4% of duty lines. The language-model reading, however, filters out non-duty boilerplate—classifying 99% of flagged benefits and legal lines as non-duties and none as new tasks—and flagged AI duty lines are no more likely to be duplicated across unrelated firms than continuing duties are (Appendix D).

Fourth, the level of new work depends on the judgement calls of Section 2, which Figure 1 and Appendix Table C1 trace out step by step, and the screen counts as continuing some AI work that a direct reading finds to be new (Section 5.3). Tasks that cross occupational boundaries are measured only for technical and data tools, where a tool named for the first time may also be a new instrument (Section 5.4). Measuring task crossover across all domains would require assigning every duty line to a fine-grained task category, which embedding similarity to O*NET work activities does not do reliably (Appendix B.2); a coarser check using a classifier trained on 14,664 Opus-labelled lines across O*NET intermediate work activities detects no broad shift of duties outside occupations' 2019 activity mixes between 2022 and 2026 (Appendix D).

Finally, several patterns sharpen in 2026, for which the data cover only January to September, and the occupation-level design of Section 5.5 is descriptive: observed AI use is measured in 2025, shares task ratings with pre-existing exposure indices, and correlates with pre-2022 occupational traits such as remote-work prevalence.

7 Conclusion

Whether a new technology raises the demand for labour depends in part on the new work it creates (Acemoglu and Restrepo, 2018, 2019), and in the past most new work appeared where technologies complemented what workers already did (Lin, 2011; Autor et al., 2024). For generative AI, the evidence has so far come mostly from what the technology could do and what people use it for. Judged from the duties in US job postings, the content of jobs is changing most in the occupations that use AI most, and in all four ways at once. New work has grown, and most of its growth since 2022 is work on or with AI itself: building and governing AI systems and, increasingly, driving their adoption across the organisation. Directing AI systems and reviewing their output remains rare, and new work outside AI has grown much less than new work about AI. Much of the growth of AI work is more of work that existed before generative AI, and duties asking workers to use AI tools in their own work are spreading rapidly, but they represent a new instrument for familiar tasks rather than new work. While job postings alone cannot establish whether these new duties raise labour's share of value added, as task reinstatement requires, they show where and how quickly the content of jobs is changing. Beyond new work, existing technical tasks such as building dashboards and automating workflows are spreading into occupations that rarely performed them in 2019–20, particularly in high-AI-use occupations—a shift in the division of labour that may begin to reverse the specialisation associated with earlier technologies (Skoglund et al., 2026). And whereas new technologies have historically taken decades to spread in job postings from their pioneers to a wider set of jobs (Kalyani et al., 2025), the shift from building AI to working with it is already visible within four years.

These findings carry two broader lessons. First, official occupational task inventories lag behind

technological change: nearly four years after ChatGPT, O*NET still lists no task about language models or working with AI. Pre-existing AI exposure measures built on O*NET (Webb, 2020; Eloundou et al., 2024) still point to the occupations where new AI work is appearing, because new AI tasks are emerging in the same occupations whose existing tasks are exposed to AI, but tracking how work changes within an occupation requires reading what employers actually ask for. Second, history counsels caution about how long this new work will last. Occupations tied to a new technology have proved less durable than those born of specialisation (Skoglund et al., 2026), and the pay premium of new work, highest for work tied to a technology, fades as the expertise spreads (Autor et al., 2026). The new work of generative AI so far is largely of that kind, work on the technology itself, and whether it remains a durable part of jobs or proves to be a passing stage before further automation will depend on how far model capabilities advance (Kwa et al., 2025). The posting-level data from this study will be released publicly so that researchers can follow that evolution as generative AI diffuses.

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A The dataset

This appendix summarises the construction and quality of the postings. The build covers January 2017 to September 2026; this paper uses postings dated from 2019. Fuller documentation, including the extraction rules and the occupation classifier's training labels (posting identifiers and codes), will accompany the release.

A.1 Construction

Table A1 lists the platforms and Table A2 accounts for every captured page. Table A3 gives counts and field completeness by year. Counts follow Common Crawl's crawling, not the labour market: from September 2017 to January 2019 the archive captured Workday and iCIMS pages at several times their later rate, iCIMS pages before September 2017 carry no text, and 2023 had fewer crawls than the years around it. I therefore start the analysis in 2019 and do not interpret posting counts as vacancies. The crawl-intensity weight of Table A3 is the inverse of a crawl's capture intensity for the platform, relative to the platform's 2019–20 rate, where that intensity exceeds one; it differs from one only in the crawls of 2017 to January 2019, so no result in this paper uses it, and Appendix B shows the measure without the January 2019 crawl instead. Table A4 follows the postings dated 2019–26 to the analysis sample.

I start from 20 widely used ATS platforms. Eighteen of them have enough archived job pages to test: for each, I fetched 20 random archived job pages in each of four years (2018, 2021, 2024 and 2026) and checked what could be extracted, and I keep the 14 platforms whose pages reliably give at least a title and a location. They hold 84–95% of the US job pages that the archive holds for the 20 platforms (Table A1). Each page is parsed twice, once for structured data (a schema.org `JobPosting` object) and once with a template for the platform, and each field comes from the structured data where it exists. The 10.5 million captured pages collapse to 5.2 million jobs by platform, employer and job identifier. Employers often repost an advert under a new identifier, so jobs with the same title, location and text are merged; the merge removes 7.9% of US jobs. A posting is classified as US from its location or, failing that, from the page metadata, the text, its other captures or the employer's other postings. First capture occurs a median of 37 to 67 days after the stated date of posting, depending on the year, and more than a year later for 4–12% of postings; dating every posting by its first capture instead leaves the results unchanged (Appendix D). The analysis sample retains 28,835 US postings whose state could not be resolved and 17,338 postings by recruitment agencies, which list real duties; dropping the former changes no result (Appendix D). The last crawl of the period of unusual capture rates, in January 2019, falls in the reference half-year, and 23% of the reference's distinct duty sentences appear only in it; Appendix B shows that omitting it changes little.

Table A1: Platforms

Platform	US ATS pages (%)	US postings	Years	Description (%)	Salary (%)	State (%)	Verdict
iCIMS	36.2	1,393,482	2017–2026	88	17	84	yes
Workday	20.2	634,446	2017–2026	26	11	66	caveats
Greenhouse	10.3	446,489	2017–2026	100	44	83	yes
Jobvite	4.1	171,110	2017–2026	100	21	85	yes
JazzHR	3.2	96,274	2017–2026	84	30	78	caveats
Breezy HR	2.6	99,834	2017–2026	95	30	87	yes
Lever	2.5	117,637	2017–2026	100	31	71	caveats
UKG (UltiPro)	2.3	69,272	2017–2026	76	16	74	caveats
SmartRecruiters	2.2	83,018	2017–2026	78	12	83	caveats
Dayforce	1.9	76,437	2019–2026	97	23	93	yes
Paylocity	1.3	56,490	2017–2026	100	33	93	yes
Ashby	1.2	51,563	2020–2026	79	50	55	caveats
Paycom	1.0	38,143	2019–2026	93	37	86	caveats
Recruitee	0.3	7,677	2018–2026	94	36	91	caveats

Notes: US ATS pages: the platform's share of the US job pages captured on the 20 platforms first queried, in one crawl per quarter from 2017 to 2026. US postings: postings in the release. Description, salary, state: share of the platform's US postings with a description of at least 200 characters, salary information and a resolvable state.

Table A2: From captured pages to postings, 2017–2026

Step	All	US	Non-US	Unresolved
Captured job pages on the 14 platforms	10,505,800			
not fetched or not parsed (kept)	3			
Jobs (platform, employer and job ID)	5,205,625	3,630,815	1,348,062	226,748
repost merge (same title, location and text)	−350,263	−285,346	−63,262	−1,655
Postings	4,855,362	3,345,469	1,284,800	225,093
non-US postings dropped	−1,284,800		−1,284,800	
Postings kept (US and unresolved)	3,570,562	3,345,469		225,093
iCIMS portal merge	−4,163	−3,597		−566
Release	3,566,399	3,341,872		224,527
of which: closed-ad notice (flagged)	34,995			
of which: no title	24,232			
of which: no description of 200+ characters	888,023			

Notes: Counts at each step of the build, which covers 2017–2026. Jobs are called US, non-US or unresolved before the repost merge, postings after it.

Table A3: US postings by first-seen year

		Weighted		Field present (%)						Median
Year	Postings	postings	Employers	Description	Duties	Date	Employer	State	Salary	length
2019	274,531	225,538	10,739	87	69	80	75	91	6	3,142
2020	197,526	197,526	10,311	85	64	84	75	88	7	3,217
2021	308,332	308,332	12,408	88	70	87	84	88	13	3,456
2022	287,957	287,957	13,161	87	69	85	78	81	25	3,594
2023	192,502	192,502	12,606	91	70	88	82	83	35	3,691
2024	326,187	326,187	16,624	90	68	87	84	82	42	4,045
2025	371,142	371,142	18,314	88	69	85	79	78	48	4,551
2026	327,003	327,003	16,065	87	69	84	79	75	51	4,967

Notes: Weighted postings use the crawl-intensity weight, which scales crawls that captured a platform at an unusual rate to the platform's 2019–20 capture rate. Description: at least 200 characters. Duties: a duties section was found by header (the text layer of Section 3 replaces this rule). 2026: January to September.

Table A4: From the release to the analysis sample, 2019–26

Step	Postings
release, dated 2019–26 (US or unresolved location)	2,390,292
US location	2,256,069
(the steps below keep unresolved locations)	
English, description of at least 200 characters	2,009,281
not a bulk poster	1,929,848
one posting per location group	1,712,526
at least one duty line (analysis sample)	1,629,317
of which: location unresolved	28,835
of which: recruitment agency	17,338
of which: closed-advert notice	50

Notes: Postings of 2019–26, dated as in Section 3, at each step of the filters of Section 3. The last three rows are flags inside the analysis sample, which keeps those postings.

A.2 Coverage and representativeness

Figure A1 summarises the benchmarks. Where and when states required pay ranges in postings, the share of postings with pay information jumped (panel a); the share offering remote or hybrid work rose with the pandemic, as in Lightcast's and Indeed's series, if to a higher level (panel b); posting shares by detailed occupation track Lightcast's (panel c); and median posted pay tracks the occupation's median wage (panel d). Figure A2 compares the postings with hires, Figure A3 with the occupation mix of employment and of other posting data, and Table A5 summarises the differences as dissimilarity indices: the share of postings that would have to change category for two distributions to match. Figure A4 compares states, and Table A6 the shares of postings that state pay or offer remote work against Indeed's and Lightcast's series. Live postings per crawl fell by 39% between February and May 2020 and track the Indeed postings index until 2023 but not after (Figure A5), as the share of each platform that the archive captured changed. Table A7 compares extracted fields with Web Data Commons' independent parse of the same pages.

State shares of postings correlate with employment shares at 0.92–0.98 a year; across detailed occupations, log posting shares correlate with Lightcast's at 0.74, and median posted pay with the occupation's median wage at 0.89; titles agree with Web Data Commons' parse of the same pages for 99.9% of postings. The shares of postings that state pay or offer remote work track the levels of Indeed's and Lightcast's series (correlations of 0.85–0.98), though not their quarter-to-quarter changes, with the remote-work share running higher than both from 2021 onwards. When Colorado mandated pay ranges in 2021, the share of its postings listing pay jumped from 10% to 55%, and the national share offering remote work rose from 3% to 13% between early 2020 and late 2021. Even a complete census of online adverts would lie well below 100 postings per 100 hires, as employers often fill several vacancies from a single advert and many hires occur without an online posting.

Table A5: Dissimilarity indices

Comparison	Level	ATS postings weighted by		
		crawl intensity	OEWS	hires
ATS vs OEWS employment	major group	0.34	0.00	0.14
	detailed	0.45	0.32	0.35
ATS vs Lightcast postings	major group	0.22	0.21	0.28
	detailed	0.38	0.38	0.42
Lightcast vs OEWS	major group	0.21		
	detailed	0.35		
ATS vs CES employment, 2024	state	0.12		
ATS vs JOLTS openings, 2024	state	0.13		

Notes: Share of postings that would have to change category for the two distributions to match. Occupations: postings first seen in 2023–26; OEWS employment in May 2025. Columns: crawl-intensity weight, and the representativeness weights that give each major group its share of OEWS employment or of new hires in the CPS (the OEWS weights set the major-group index against OEWS to zero by construction). States: 2024.

Table A6: Fit of the pay-information and remote-work series

Series	Period	Quarters	Correlation			Gap (points)		
			levels	changes	changes[a]	2021	2023	2025
Pay information vs Indeed	2020–26	27	0.98	0.24	0.30	−11.9	−6.0	−1.7
postings with a description	2020–26	27	0.98	0.19	0.32	−10.0	−2.1	5.0
Remote work vs Lightcast	2019–26	31	0.88	0.14	0.33	3.3	1.8	7.7
Remote work vs Indeed	2019–26	31	0.85	0.48	0.33	3.8	3.2	9.5

Notes: Quarterly shares of postings against Indeed's pay-transparency and remote-work trackers and Lightcast's remote-work share. Changes: first differences. [a]Postings dated by their date posted where available.

Table A7: Agreement with Web Data Commons

Field	Route	Matched pairs	Compared	Agree (%)
title	URL (Workday)	2,628	7	
	page template	7,577	7,543	99.4
	structured data	102,617	102,579	100.0
	all	112,822	110,129	99.9
US call	metadata or description	1,954	3	
	page template	9,918	7,486	100.0
	structured data	100,950	100,855	100.0
	all	112,822	108,344	100.0
location	URL (Workday)	2,628	1,104	99.7
	page template	7,577	7,497	97.4
	structured data	102,617	97,036	97.2
	all	112,822	105,637	97.2
description	page template	8,489	6,525	63.8
	structured data	104,333	104,321	100.0
	all	112,822	110,846	97.8

Notes: Web Data Commons postings on the fourteen platforms in the October 2022 and October 2024 crawls, matched by platform, employer and job identifier. Route: where the field came from in this build.

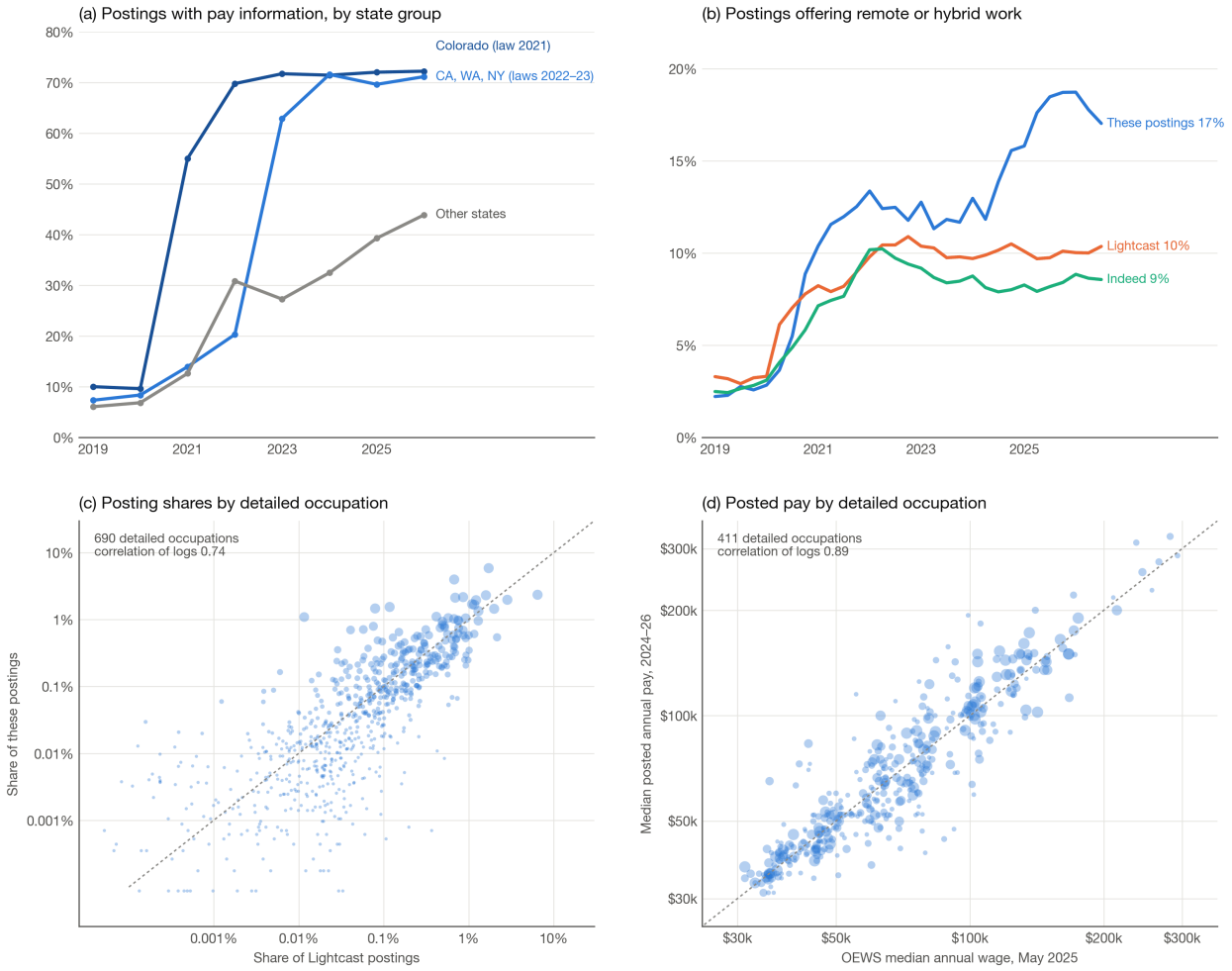


Figure A1: The postings against known facts. (a) Share of postings with pay information by state group; Colorado's pay-transparency law took effect in 2021, California's, Washington's and New York's in 2022–23. (b) Share of postings offering remote or hybrid work, by quarter, against Lightcast's and Indeed's series. (c) Shares of postings by detailed occupation against Lightcast's, 2023–26 (log scales). (d) Median posted annual pay, 2024–26, against the OEWS median wage by detailed occupation (log scales). Dotted lines: equality.

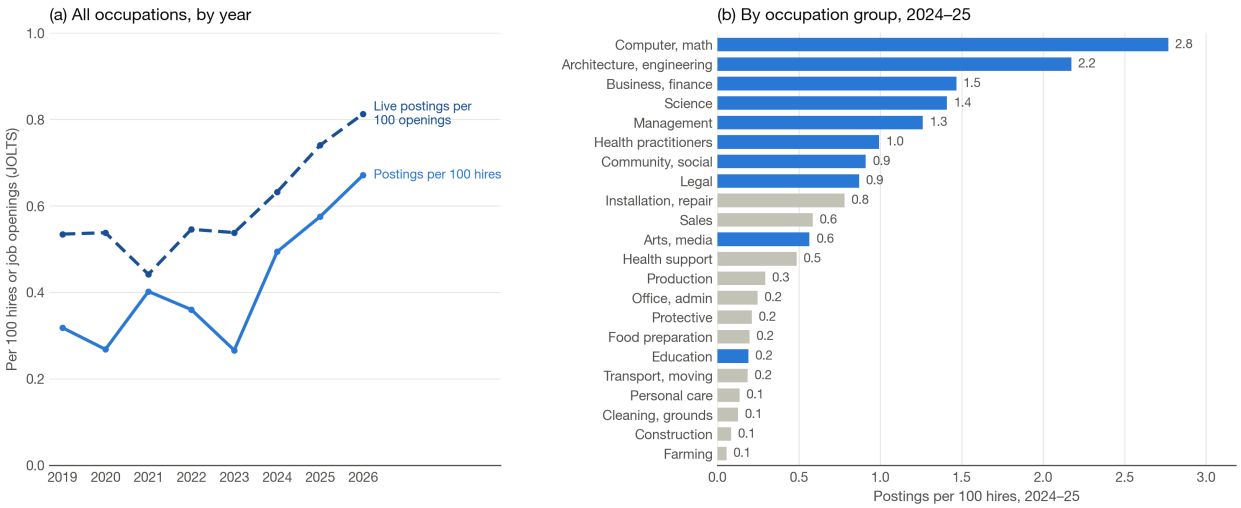


Figure A2: Coverage of US hiring. (a) Postings per 100 hires in JOLTS, and live postings per crawl per 100 job openings. (b) Postings per 100 hires by occupation group, 2024–25; blue: professional and technical groups.

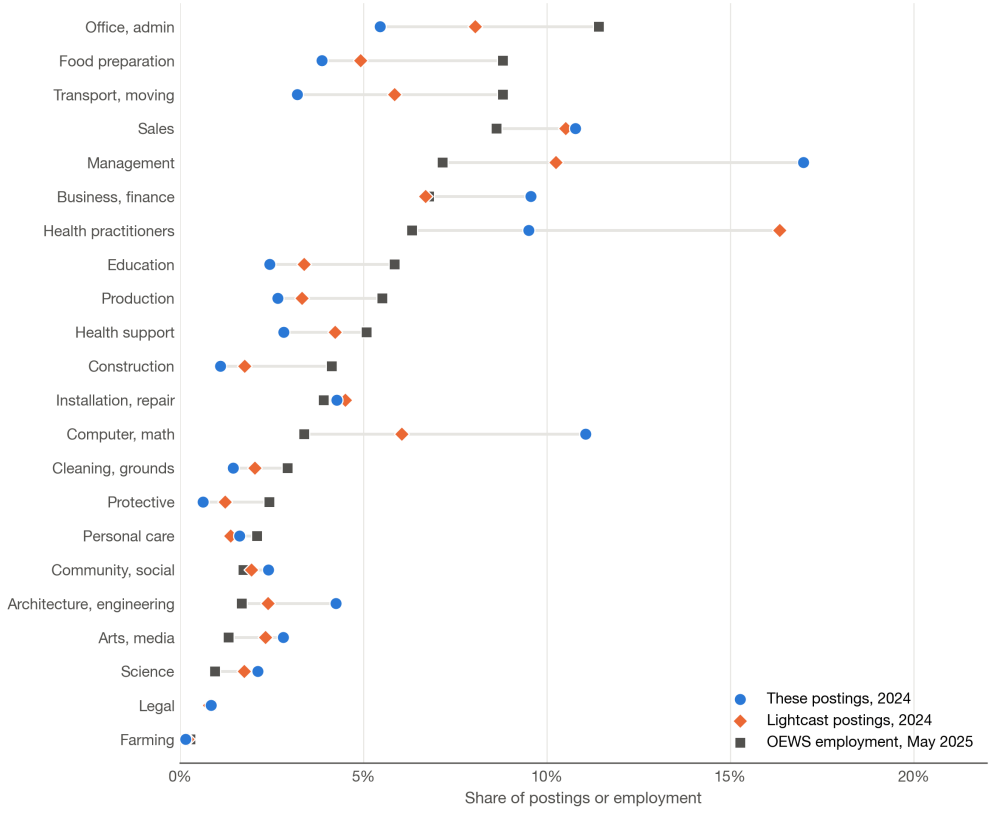


Figure A3: Occupation mix by SOC major group: these postings and Lightcast postings in 2024, and OEWS employment in May 2025.

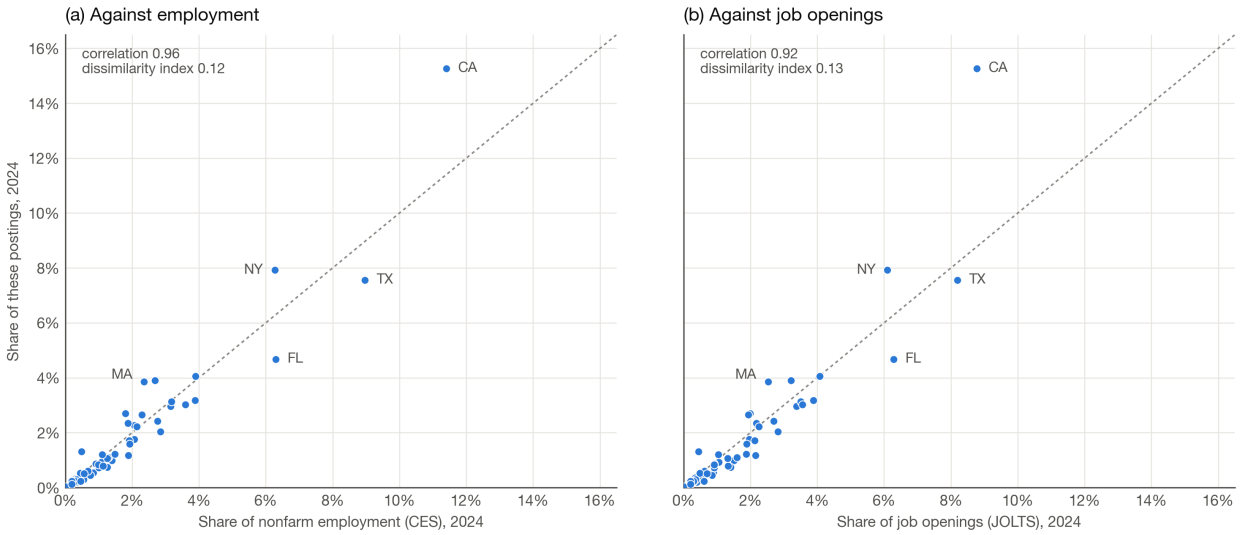


Figure A4: State shares of postings against state shares of nonfarm employment (a) and of job openings (b), 2024. Dotted lines: equality.

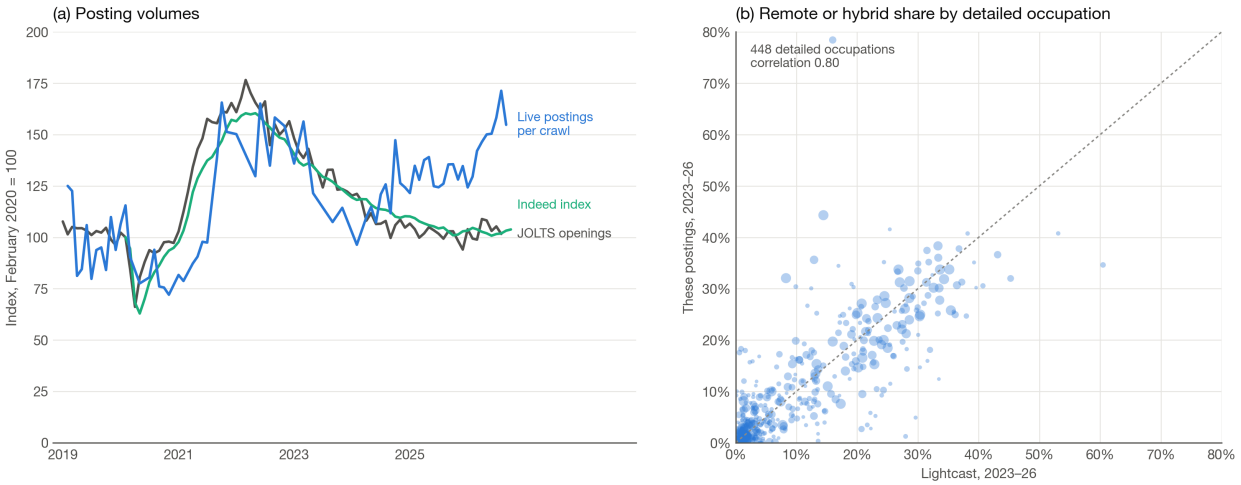


Figure A5: (a) Live US postings per crawl, JOLTS job openings and the Indeed postings index (February 2020 = 100). (b) Share of postings offering remote or hybrid work by detailed occupation, against Lightcast, 2023–26. Dotted line: equality.

A.3 Occupation codes

The classifier blends a linear model of the title's words and characters and the posting's duty and role sentences with a small language model (MiniLM) fine-tuned on the same text. It is trained on 63,000 postings coded by Opus 5.5, 60,000 drawn by platform and year and 3,000 with AI or new titles, and on postings whose titles match an O*NET title. On 5,179 held-out postings drawn by platform and year it agrees with Opus 5.5 on 74% of detailed codes and 87% of major groups; Sonnet 5.5 and Opus 5.5 agree with each other on 73%, and Opus with itself on a second reading on 89%. On 800 postings with AI or new titles it agrees on 78%. A code is confident when the classifier's probability is at least 0.55, where its codes agree with Opus as often as Opus agrees with itself: 64% of all postings (62% of the held-out postings of Figure A6, which are drawn by platform and year), at 90% agreement, and 89% on AI and new titles. The right code is among the three likeliest for 89% of postings. Two parts of the pipeline use title-based codes from a simpler classifier that reads only the title and the first 400 characters of the description: the strata from which the reading samples are drawn (Appendix B.6) and the themes (Table C7); because these codes rest on the title rather than the duties, they also serve as a check (Appendix D). Figure A6 shows that the classifier is conservative, with agreement at or above its confidence, and how agreement rises with the threshold. Every posting's detailed code places it in a third of occupations by AI use and gives it an O*NET task list, and the occupation-level design of Appendix 5.5 uses only confident codes. Confident codes cover 64–68% of postings in every year, and 79% of computer postings, 85% of health-care postings, 43% of production postings and 47% of science postings.

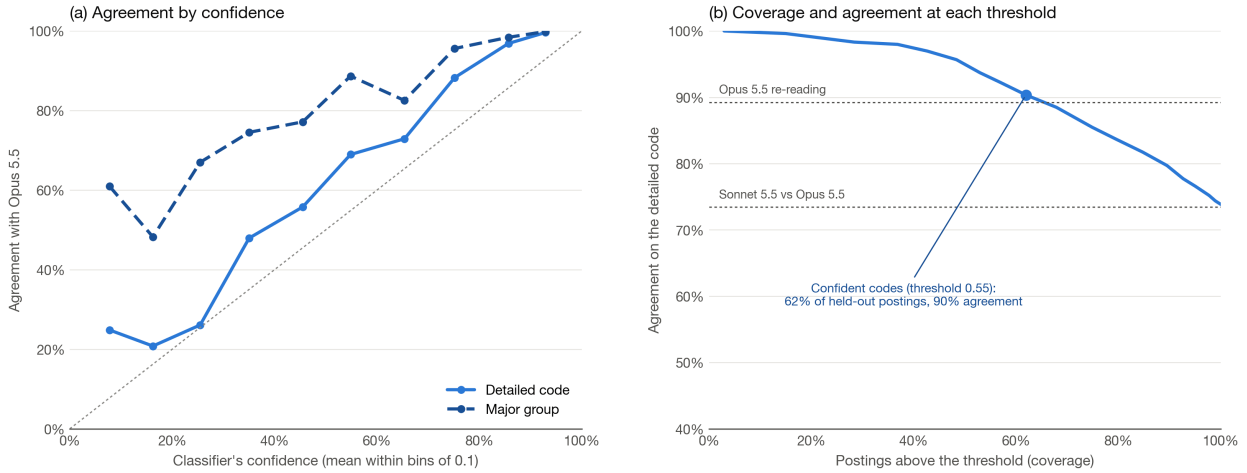


Figure A6: (a) Agreement with Opus 5.5 by bin of 0.1 of the classifier's confidence. (b) Share of held-out postings above each confidence threshold and agreement on them; dot: the threshold for a confident code (0.55). Dotted lines: Sonnet 5.5 against Opus 5.5, and Opus against its own second reading.

A.4 The text layer

The duty classifier is a regularised logistic regression on the sentence's words and character sequences, its section header, the type of section, its position in the advert and, in a second stage, the predicted classes of the sentences around it. Classifying a sentence is a narrow task that depends mostly on its wording, section header and position in the advert, which a linear model learns well from the 45,000 labelled sentences of 1,200 postings (1,000 drawn by year and 200 with flat text); an independent re-labelling of 100 postings by Sonnet agrees on 99% of duty labels. Table A8

reports the classifier's agreement with the labels by year, in five-fold cross-validation that holds out whole postings, so each score is on adverts the classifier was not trained on. The weakest class is statements about the role, a fifth of which the classifier calls duties. Flat text is restored to lines by rules before the sentences are classified. Flattening 4,061 duty sentences of pages that have native line breaks and restoring them recovers 56% exactly, and most of the rest merged with a neighbour; merged lines sit farther from any single 2019 duty and are flagged more often (19% against 12% of the original lines), which is why Appendix D repeats the comparisons without the flat-text platforms. In 2026, 52% of AI duty lines and 62% of other duty lines sit under a duties header, and 22% of AI lines under a header about the role, where statements about the role pass the classifier; the reading of the flagged lines sets such statements aside.

Table A8: Duty classifier against language-model labels

Year	Postings	Units	Precision	Recall	Duty share, Sonnet (%)	Predicted (%)
2017	100	3,067	0.90	0.94	32.0	33.3
2018	100	3,359	0.92	0.95	30.9	31.9
2019	100	3,525	0.92	0.93	28.3	28.7
2020	100	3,669	0.92	0.95	30.9	32.0
2021	100	3,351	0.94	0.93	27.9	27.8
2022	100	3,927	0.92	0.92	20.0	20.0
2023	100	3,799	0.94	0.94	24.1	24.0
2024	100	3,694	0.92	0.96	26.7	27.7
2025	100	4,379	0.91	0.96	23.7	25.1
2026	100	4,695	0.89	0.94	23.2	24.3
All (weighted)	1,000	37,465	0.92	0.94	26.4	27.1
Flat text	283	10,148	0.92	0.93	21.3	21.5

Notes: Five-fold cross-validation grouped by posting. All: the 1,000 postings drawn by year, each year weighted by its postings in the frame. Flat text: the 283 postings with flat text in either draw (the 200 drawn for flat text and 83 of the 1,000), unweighted. Units: sentences and headers. Precision: share of predicted duties that the language model labels duties; recall: share of the language model's duties predicted. Duty share: share of units labelled or predicted duty.

A.5 Known gaps

Workday pages carry descriptions only where they include structured data, almost none before 2022 and about half from 2023, and their titles and locations come from the page address. Common Crawl holds almost no UKG pages in 2021 and almost no Lever or Paycom postings in 2026. Four tested platforms whose archived pages mostly carry no job data, Taleo, ADP, Workable and BambooHR, are not included, nor are ATS platforms outside the 20 first queried, and school districts, public employers and health systems mostly post elsewhere.

B Measurement

B.1 Novelty and the COVID-19 check

The embedding model is all-MiniLM-L6-v2 (mean pooling, 128 tokens, unit-normalised vectors). The search is exact: every distinct duty sentence of 2019H2–2026 is compared with every distinct

duty sentence of 2019H1. Figure B3 shows what the similarity means. Sonnet 5.5 read 100 duty lines of 2023–26 in each bin of 0.05 of similarity to their most similar duty of 2019H1, from 0.45 to 0.95, beside that duty, with the prompt of the flagged-line reading (Appendix B.6). Re-wordings dominate above 0.75, variants between 0.55 and 0.75, and new tasks and lines that are not duties are most common in the most distant pairs. To measure what the screen misses, Sonnet also read 400 non-AI lines in each of three bins just above the threshold (0.66–0.70, 0.70–0.75 and 0.75–0.80); with the non-AI lines of the 100-per-bin sample above 0.80, it read 1,498 lines above the threshold and called 2 of them new tasks, one in each of the two bins nearest the threshold. Weighting each bin's rate of new tasks by its lines in 2023–26, the screen passes 87% of the non-AI lines Sonnet would call new tasks. Most duty lines sit in the bins above the threshold, so even a sample rate of zero in 400 lines leaves substantial uncertainty: drawing each bin's rate from its sampling distribution gives a median recall of 73% and a 95% interval of 47–91%.

Figure B1 shows all flagged duty lines by what Sonnet reads them to be, and Figure B2 the flagged non-AI lines. Most are variants, re-wordings and lines that are not duties, a turnover of wording that is present in 2019H2 by construction and grows with time since the reference; new work is a small part of them in every year.

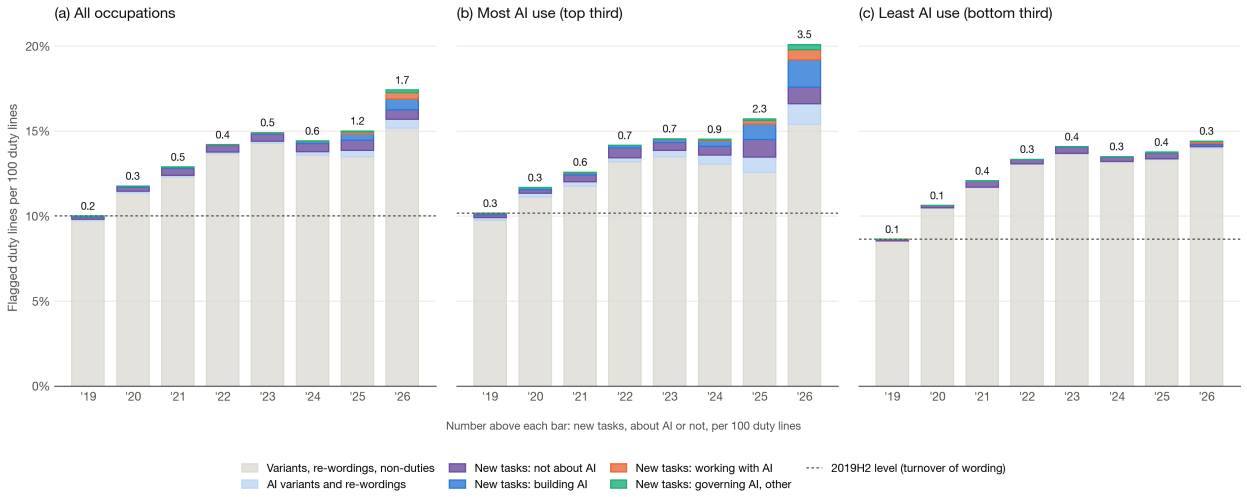


Figure B1: All flagged duty lines, by what Sonnet 5.5 reads them to be. Flagged duty lines (no close counterpart among the duty sentences of 2019H1 postings) per 100 duty lines, by year (2019: second half; 2026: January to September), split by what Sonnet 5.5 reads them to be: variants and re-wordings of familiar tasks and lines that are not duties, not about AI (grey); AI lines that vary or re-word AI work already described in 2019 (pale blue); new tasks that are not about AI; and new AI tasks by side. The split uses Sonnet's reading of 500 flagged non-AI lines in every year and third of occupations by AI use and of 400 flagged AI lines on each side in 2023–24, 2025 and 2026, read in batches that mix the samples (2019–22: the shares of 2023–24; governing and other AI lines: the pooled share of the two sides; Appendix D gives the series with those shares measured directly, and 95% intervals). (a) All occupations. (b, c) The thirds of occupations with the most and least observed AI use. Dotted line: the panel's 2019H2 level of flagged lines, the turnover of wording (10% in panel a, by construction). Numbers above the bars: every flagged line read as a new task, about AI or not, including the AI lines that ask workers only to use AI tools, which are not new work (Section 2).

Table B1 applies the screen and the reading to new work that is known to have arrived. Of the COVID-19 duty lines, the screen flags 67.7% and Sonnet reads 38% of these as new tasks; another 32% are notices about COVID-19 policy that the text layer took for duties. For four kinds of non-AI work that are new, or much more common, since 2019, Sonnet reads 20–72% of the flagged

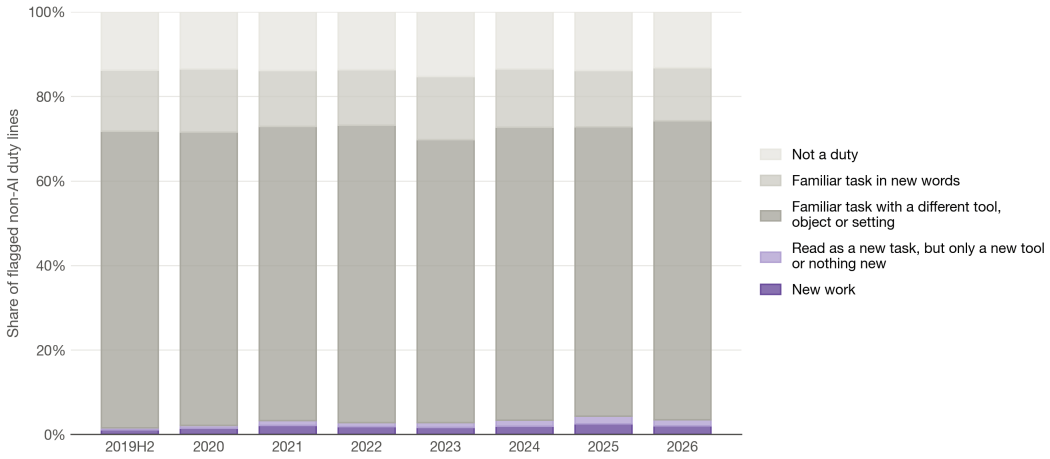


Figure B2: What the flagged non-AI duty lines are. Sonnet 5.5 compares each of 500 flagged non-AI lines in every year and third of occupations by AI use with the most similar duty sentence of 2019H1, without seeing the year or the cell (Appendix B.6). Shares of the flagged non-AI lines, all occupations: new work (read as a new task whose activity or object of work is new); read as a new task but only a new tool for a familiar task, or nothing new on a second reading; a familiar task with a different tool, object or setting that is not itself new (such as scanning inventory with a smartphone app, beside 2019's handheld scanner); a familiar task in new words (such as taking care of permits, beside securing them); and not a duty (examples paraphrased). The thirds and the occupations without a measure of AI use are weighted by their flagged non-AI lines; Figure 1 gives new work outside AI by third. 2019: second half.

lines as new tasks, but the screen flags only 26–49% of the lines, and some of the rest are new tasks written in familiar words: by these readings the screen passes 48–69% of the new tasks.

Table B1: The screen and the reading on known new work

	Lines read	Flagged new (%)	New tasks, flagged (%)	New tasks, not flagged (%)	Recall (%)
COVID-19 duties (2020–22)	800	68	38	12	88
Climate disclosure (CSRD, ISSB, TCFD, SEC)	84	48	52	52	48
Electric-vehicle batteries and charging	400	49	26	11	69
New compliance regimes (CMMC, SBOMs, No Surprises Act, SOFR, T+1)	400	44	72	39	59
Telehealth	400	26	20	7	49

Notes: Duty lines of 2023–26 (COVID-19: 2020–22) that contain the named terms and are not about AI. Flagged: share below the novelty threshold (COVID-19: of 2020–21 duty lines). New tasks: share of the flagged and of the other lines that Sonnet 5.5 reads as new tasks, each beside its most similar 2019H1 duty (400 lines of each kind drawn at random, flagged or not, and all 84 lines of climate disclosure; COVID-19: 400 flagged lines and 400 others). Recall: share of the lines read as new tasks that the screen flags.

Dropping January 2019 from the reference changes little. The last crawl that captured certain platforms at several times their later rate falls in January 2019, and 23% of the reference sentences appear only in postings first captured in that crawl; excluding them shifts the 10% threshold on 2019H2 from 0.66 to 0.652, the levels of flagged AI lines fall slightly (Table C4), and new work in the high-use third goes from 0.53 to 2.41 per 100 duty lines, against 0.5 to 2.6 (Appendix D).

Measured against the duties of the two previous years instead, on 20,000 lines a year, the result depends on the size of the reference, since a larger reference finds closer matches: the full two-year windows grow from 2.1 to 3.7 million distinct sentences, and against them flagged non-AI lines fall from 5.5 per 100 duty lines in 2021 to 4.0 in 2026. Against windows subsampled to the size of the 2019H1 reference (679,314 sentences), flagged non-AI lines stay at 11.5 to 11.7 per 100 over 2021–26 while flagged AI lines rise from 0.13 to 0.62: non-AI novelty did not accelerate, and the moving reference counts each year's new wording rather than all change since 2019. A larger embedding model (BAAI bge-base), with its own threshold for the same 10% churn (0.77), gives the same series on samples of 20,000 lines per half-year: 1.5 flagged AI lines in every 100 duty lines in 2026, against 1.8 with MiniLM on the same lines, and the same contrast between the thirds (Table C4). The two models agree on the totals more than on single lines: each flags 58–71% of the lines the other flags.

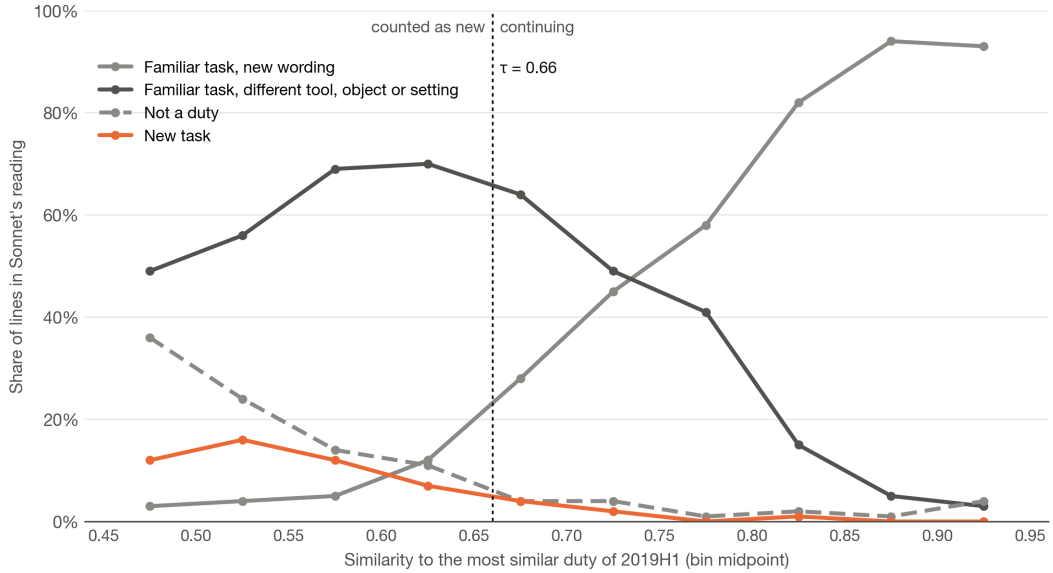


Figure B3: What similarity to 2019 means: Sonnet 5.5's reading of 1,000 duty lines of 2023–26, each beside its most similar duty of 2019H1, 100 per bin of similarity. Dotted line: the novelty threshold.

Figure B5 shows the COVID-19 check. Sentences of all kinds with a COVID-19 term rose in 2020 and peaked in 2022, but most were policy notices, legal text and statements about the company; duty lines with a COVID-19 term peaked earlier, at 0.21% of all duty lines in the first half of 2021 (panel a); in 2021, 11% of sentences with a COVID-19 term were duty lines. COVID-19 duty lines were most common in the occupations that use AI least: in 2021H1 they were 0.32 in every 100 duty lines of the third with the least AI use and 0.12 in the third with the most. The themes of the flagged lines of 2020–21, which include the policy notices, show both kinds (panel b): four rising themes are COVID-19 duties and eight are notices. Figure B4 sets the flagged AI lines of 2023–26 against the COVID-19 duty lines of 2020–22: across all occupations, flagged AI duty lines in 2026 are 8 times as common as COVID-19 duty lines were at their 2021H1 peak (21 times after reading).

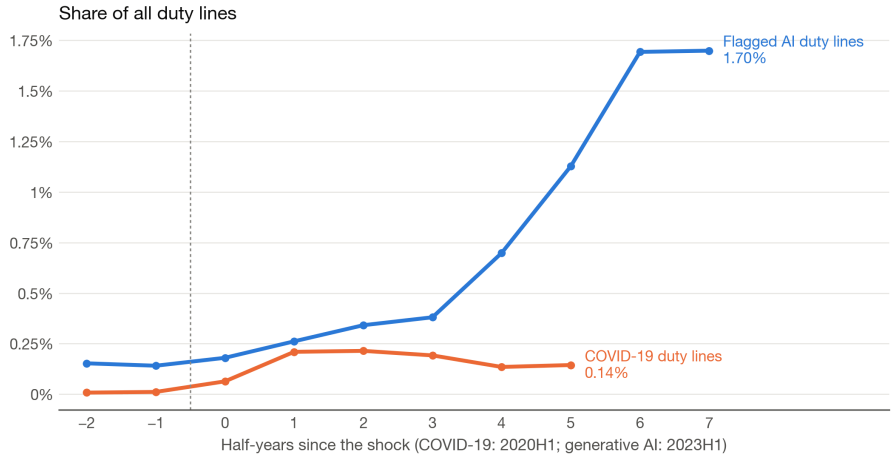


Figure B4: Flagged AI duty lines and COVID-19 duty lines as a share of all duty lines, in half-years since each shock (COVID-19: 2020H1; generative AI: 2023H1). COVID-19 duty lines: duty lines with a COVID-19 term; notices of COVID-19 policy, such as vaccination requirements, are not duty lines and are not counted. COVID-19 series to 2022H2.

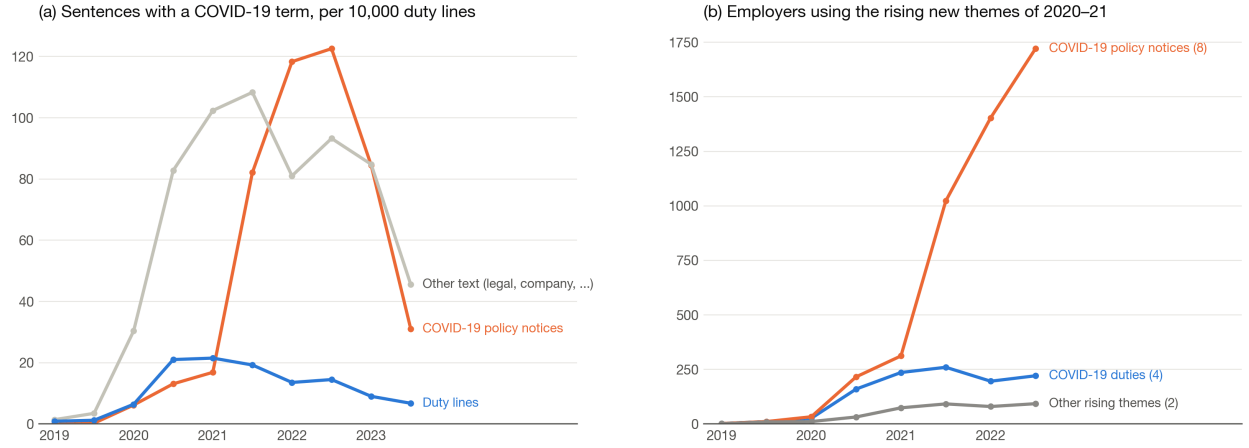


Figure B5: The novelty measure finds COVID-19. (a) Sentences with a COVID-19 term per 10,000 duty lines, by the text layer's class: duty lines, COVID-19 policy notices and other text. (b) Employers using the rising new themes of 2020–21, by whether Sonnet 5.5 judged the theme to be COVID-19 duties, COVID-19 notices or neither. Themes: clusters of the flagged lines and notices of 2020–21, rising when used by at least three times as many employers as in 2019H2.

B.2 Tools new to the job

Measuring tasks that are new to an occupation across all domains is difficult because duty lines cannot be mapped reliably to fine-grained O*NET work activities. Assigning each duty line to one of O*NET's 2,087 detailed work activities (or 332 intermediate work activities) via its closest task statement gives the same activity to only 21–27% of pairs that Sonnet judges to be the same familiar task in different words; as a result, against a 2019–20 baseline, 16–68% of 2021 postings—before generative AI—would already appear to contain a task new to their occupation. Section 5.4 therefore focuses on a narrower, well-defined domain: technical and data tools.

Table B2 lists the five families. Each follows O*NET 27.1's Technology Skills categories (November 2022). Its tools are the examples that O*NET flags as hot technologies, those most often requested

in postings (47 tools after 13 are dropped as not data or code tools, such as LinkedIn and cloud infrastructure, as office software, Microsoft Access, or as ambiguous words, such as C, R, Go and Swift), and 32 current tools that O*NET 27.1 lacks, such as Power BI, Looker, Snowflake and Zapier, with the words dashboard and data visualisation. Workflow automation has no O*NET category and is defined by its tools alone. Matching is by regular expression on each duty line and, separately, each requirement sentence. The list was fixed before counting and revised once, after a first check by Opus 5.5 of 1,440 matched lines found two systematic false matches: tool vendors describing their own products, and SQL meaning a sales-qualified lead in marketing postings. A tool does not count in a posting by its own vendor, and SQL does not count next to MQL, sales-qualified or SQLs.

A second check, on a fresh sample of 1,440 lines, 80 in each of 18 cells of family, period (2019–22 or 2025–26) and whether the occupation used the family before, measures how often a matched duty asks the worker to use the tool. Where the occupation used the family before, 91–100% do; where it did not, 56–94%, lowest for databases, where most of the remaining false matches are partner firms that sell alongside data platforms. The series weights each matched posting by its family's precision in its period (for 2023–24, the mean of the two periods; a posting matching multiple families receives a weight of one minus the product of their false-match rates), and its interval comes from 200 replicates that resample employers and draw each precision from its validation sample. Uncorrected, the share goes from 0.49% in 2022 to 1.46% in 2026; counted in requirement sentences instead of duties, from 0.78% to 1.54%. By family, uncorrected, dashboards and business intelligence rose from 0.33% to 1.21%, workflow automation from 0.16% to 0.50%, databases from 0.09% to 0.30% and programming from 0.07% to 0.20%; statistical software did not rise.

In the third of occupations that use AI most, the share of postings that ask for a tool their occupation did not use rose from 0.45% in 2022 to 1.78% in 2026, and in the third that uses AI least from 0.08% to 0.36%. In 2025–26, 16% of these postings also ask the worker to use AI, against 4% of other postings in the same occupations, a gap of 10 points within the same firm and occupation. Production managers' postings that ask them to build dashboards went from 0.6% of the total in 2022 to 4.7% in 2025–26, and financial analysts' postings that ask them to write code from 0.1% to 1.0%; in the occupations that already used these tools, the share rose from 10.7% to 17.0%. Of the flagged lines read as new tasks, 0.8% of non-AI lines and 0.4% of AI lines name such a tool.

The cutoff and the baseline matter little. Non-use defined at 0.5% or 2% of 2019–20 postings gives 0.89% and 1.83% in 2026 (0.22% and 0.60% in 2022); occupations with at least 1,000 or 2,000 postings in 2019–21 instead of 500 give 1.28% and 1.20%; without Workday the share goes from 0.35% to 1.30%, and in the firms that posted in every year from 2022 to 2026 it goes from 0.32% to 1.05%. Counting a tool named in the requirement sentences of 2019–20 as prior use lowers the level to 0.18% in 2022 and 0.79% in 2026, and reduces the increase by about a third, indicating that part of the rise reflects tools that occupations already listed in requirements moving into their duty statements. Defining non-use on 2019 alone and counting new tools in 2020–21, before generative AI, adds 0.11 points in two years, against 0.88 from 2022 to 2026. Of the matched duty lines of 2025–26 in occupations that did not use the family, 36% also pass the novelty screen. The precision of the matches is 73% in the third that uses AI most, 79% in the middle third and 78% in the third that uses it least, so the uniform weights overstate the gradient slightly; among the confirmed lines the work is data analysis (18%), automating workflows (17%), dashboards and reports (16%) and code (13%). The gap in working-with-AI duties between these postings and others in the same occupations is 12 points, and 10 within the same firm and occupation.

Table B2: Families of technical and data tools

Family	O*NET categories	O*NET tools	Added	New, 2019–22	New, 2025–26	Earlier, 2019–22	Earlier, 2025–26
Business intelligence and dashboards	Business intelligence and data analysis; data base reporting	Alteryx, MicroStrategy, SQL Server Reporting Services SSRS, QlikView, Tableau	Power BI, Looker / Looker Studio, Google Data Studio, Domo, Sisense, ThoughtSpot, dashboard, data visualisation	76	94	94	91
Databases and data platforms	Data base user interface and query; data base management system	DynamoDB, Redshift, Cassandra, Hadoop, Hive, Spark, Elasticsearch, GraphQL, DB2, SQL Server, MongoDB, NoSQL, Oracle Database, PL/SQL, SQL Developer, Redis, SQL, Teradata Database, Transact-SQL	Snowflake, BigQuery, Databricks, PostgreSQL, MySQL	60	56	91	91
Programming and scripting	Object or component oriented development; development environment	Kafka, Subversion SVN, C#, C++, Eclipse IDE, .NET Framework, PowerShell, Team Foundation Server, Visual Basic, Visual Basic for Applications VBA, Visual Studio, Objective C, Java, Java 2 Platform Enterprise Edition J2EE, Perl, Python, Scala, TypeScript, jQuery	JavaScript, Jupyter, pandas, Google Apps Script	74	85	92	91
Statistical and analytical software	Analytical or scientific	ArcGIS, SPSS Statistics, SAS, MATLAB	Stata, Minitab, JMP	76	84	92	100
Workflow automation	(none)	—	Zapier, Power Automate / Microsoft Flow, Power Apps, UiPath, Automation Anywhere, Blue Prism, robotic process automation, Make / Integromat, n8n, Workato, Airtable, low-code or no-code	70	84	–	–

Notes: O*NET categories: O*NET 27.1 Technology Skills commodity categories. O*NET tools: the hot-technology examples kept, vendor names shortened; dropped are EC2, AWS and Azure (cloud infrastructure), Microsoft Access (office software), LinkedIn, ServiceNow and Yardi (not data or code tools), and C, R, Go, Ruby, Swift and Maven (ambiguous words). Added: current tools that O*NET 27.1 lacks, and generic terms. Last four columns: precision (%), the share of matched duty lines that Opus 5.5 reads as asking the worker to use, build or work with the tool, in occupations that did not use the family in 2019–20 (new) and in those that did (earlier), in 2019–22 and 2025–26; 80 lines each; –: not sampled.

B.3 AI labels

On 2,000 AI sentences drawn at random and rated blind by both models, Opus 5.5 and Sonnet 5.5 agree on whether the sentence describes AI work in 98% of cases, on the side in 90% (Cohen's κ 0.82) and on the activity in 85%, and they give the same aggregate shares. AI work without an AI keyword is not rare: Sonnet finds it in 0.6% of a random sample of 139,346 duty sentences without a keyword (18,750 lines a year) and in 8.5% of sentences with near-miss words such as "agent", "automation" or "model". Table B5 scales the AI rate of the random sample to all keyword-free lines: the keywords catch 37–55% of building-AI work, depending on the year, but 87–88% of working-with-AI work in 2025–26. The classifier that extends the labels to every duty line agrees with Sonnet on the labelled sentences (precision 0.98, recall 0.95) but finds few of the AI sentences without keywords (recall 0.16). The paper therefore measures the mix of sides on the keyword lines, scaled by each side's catch rate in each year (Table B4); counting the keyword lines alone, or every duty line with the classifier's labels, gives a larger share of working with AI in 2026 (30% and 25%) and the same rise. Read without the job title and occupation group, 82–93% of AI sentences keep their side, and the share of working with AI in each occupation group moves by at most 1 point.

In symbols, with k_s the keyword lines of side s in a year and c_s the share of the side's work that the keywords catch, estimated as $k_s/(k_s+\hat{n}_s)$ where $\hat{n}_s$ is the side's AI work among keyword-free lines, scaled up from the random sample and from a second stratum of lines with near-miss words, the corrected share of working with AI, side u, is $(k_u/c_u)/\sum_s k_s/c_s$, summing over building, working with, and governing or other AI. The catch rate for working with AI in 2022 rests on 5 keyword-free hits (62%, interval 29–85%), so the 2022 share alone has an interval of 3–8%; pooled over 2019–23, when the share was flat, it is 4% (3.8–6.0%), the figure the text uses, and in 2026 it is 22% (20.7–22.8%), with catch rates of 51% for building AI and 86% for working with it. Table B3 lists the keyword and near-miss rules.

Table B3: The AI keyword and near-miss rules

Rule	Patterns (regular expressions, word-bounded)
Any case	artificial intelligence; machine[-]learning; deep[-]learning; neural networks?; large language models?; language models?; foundation models?; generative ai; gen[-]?ai; chatgpt; openai; chat ?bots?; agentic; prompt (engineering|design|writing|engineer); natural language processing; computer vision; retrieval[-]augmented generation; vector databases?; transformer models?; mlops; (github|microsoft|m365|microsoft 365) copilot; copilot studio; google gemini; claude ai
Case-sensitive	A\.?I; LLMs?; GPT(-?\d(\.\d)?o?)?; NLP; GenAI; ML/AI; ML (?:models?|engineer\w*| pipelines?|platforms?|systems?|algorithms?|infrastructure|research\w*|scientists?|ops| solutions?|techniques?|methods?|frameworks?|applications?|features?|products?)
Near-miss (any case)	automat; model(s|ing|ling)?; agent(s|ic)?; copilot; prompt(s|ing|ed)?; bots?; algorithm; intelligen; predictive; data scien; claude; gemini; perplexity; midjourney; dall-?e; synthetic data; llm

Notes: Each pattern is bounded by a word boundary on both sides, so A\.?I also matches "A.I." (the final full stop falls outside the match); it does not count in AI/AN (American Indian or Alaska Native).

Table B4: Working with AI as a share of AI duty lines (%)

Year	Keyword lines	Every duty line	Corrected for catch rates
2019	4.9	4.6	4.9 [2.9, 9.2]
2020	5.4	5.2	3.1 [2.3, 5.8]
2021	4.4	4.3	3.5 [2.2, 6.7]
2022	5.4	5.2	4.0 [2.8, 8.4]
2023	6.6	5.9	5.0 [3.6, 8.4]
2024	10.7	9.1	7.4 [6.3, 9.6]
2025	17.7	14.5	12.5 [11.5, 14.0]
2026	30.4	25.0	21.5 [20.4, 22.8]

Notes: Keyword lines: duty lines with an AI keyword, by Sonnet 5.5's side. Every duty line: Sonnet's labels for keyword lines and the classifier's for the rest. Corrected: keyword lines plus the AI lines without a keyword estimated from the random sample of keyword-free lines alone (Table B5; the text's estimates add the stratum of near-miss lines, Appendix B.3), with 95% intervals from the sample.

Table B5: Share of AI work caught by the keyword rules

Year	Building AI	Working with AI	Any AI work	Keyword AI lines
2019	39 [33, 46]	43 [20, 69]	42 [36, 48]	4,744
2020	37 [32, 43]	73 [32, 94]	42 [36, 48]	3,688
2021	39 [33, 45]	56 [26, 82]	44 [38, 49]	7,309
2022	50 [42, 58]	68 [28, 92]	51 [44, 59]	4,273
2023	47 [41, 54]	66 [35, 88]	49 [43, 56]	4,699
2024	50 [45, 56]	81 [59, 92]	56 [51, 61]	18,640
2025	55 [51, 59]	88 [78, 94]	62 [58, 65]	49,217
2026	51 [48, 54]	87 [82, 91]	62 [59, 64]	70,102

Notes: Percent of AI duty lines that contain an AI keyword, with 95% intervals from the binomial interval of the random sample's AI rate. Keyword AI lines: duty lines with an AI keyword that Sonnet labels AI work. Random sample: 18,750 keyword-free duty lines per year; the catch rates in the text add the stratum of near-miss lines (Appendix B.3).

B.4 A map of duty space

To show where duties sit and where they change, I draw 40,000 duty lines a year from 2019 to 2026 at random, embed their 287,234 distinct sentences with the same model, and place them in two dimensions with UMAP (cosine distance, 30 neighbours; McInnes et al., 2018). A map of all 10.1 million distinct sentences would be impractical to fit and unreadable as a scatter, and drawing the same number of lines each year gives every year equal weight. Nearby points describe similar work; the axes have no units. Figure B6 names 30 regions of the map (k-means clusters of the map, named by Sonnet 5.5 from random sentences in each, without dates). Figure B7 shows where the share of sampled duty lines grew and shrank between 2019–22 and 2025–26, with the occupation mix held fixed: each line of 2025–26 is weighted by its SOC major group's share of the 2019–22 lines over its share of the 2025–26 lines, so the map shows changes in what occupations ask for rather than in which occupations post. The labelled clusters are among 300 finer clusters, with changes of at least four standard errors and at least 15 employers on the side that is larger. One cluster of AI work gained 1.2 percentage points of the sampled lines, more than any other part of the map; recruiting and nursing-documentation duties lost most (0.22 and 0.21 points), while patient and family care coordination gained (0.16). Figure B8 places the AI duties of 2026 on the map: most sit at its top edge, next to data and engineering work, and some in sales, marketing, administrative

and patient-care regions.

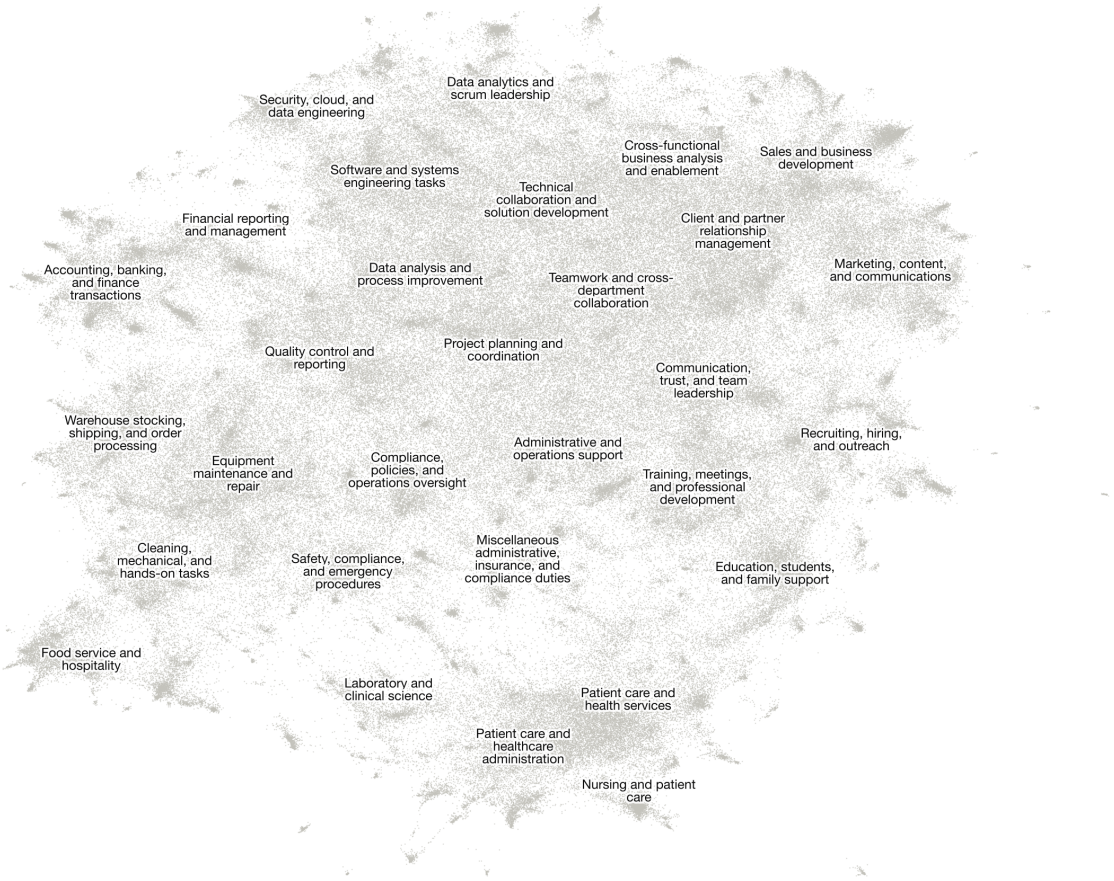


Figure B6: A map of duty space: 287,234 distinct duty sentences from a random sample of 320,000 duty lines (40,000 a year, 2019–2026), placed by meaning (sentence embeddings, UMAP). Region names by Sonnet 5.5 from random sentences in each of 30 regions.

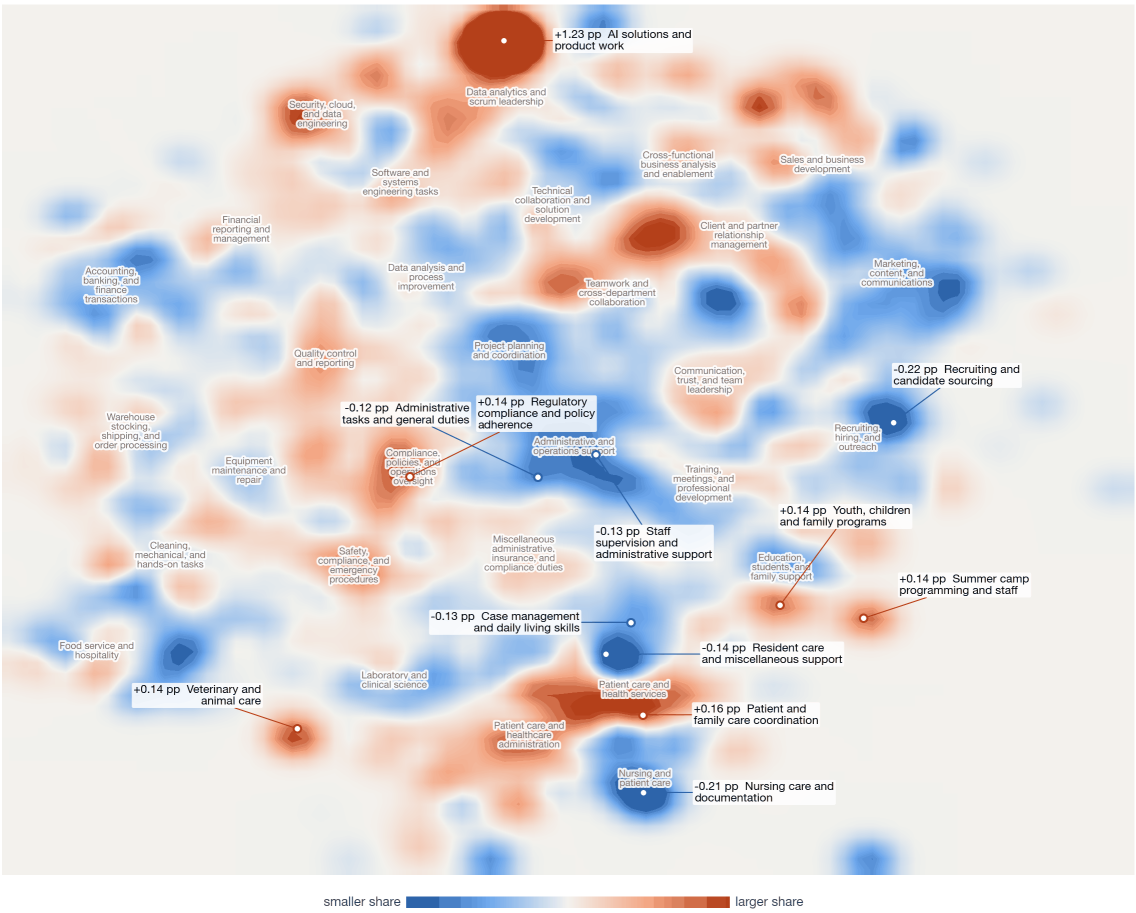


Figure B7: Where duties grew and shrank between 2019–22 and 2025–26, with the occupation mix held fixed: smoothed change in the share of sampled duty lines across the map of Figure B6. Labelled: the largest gains and losses among 300 finer clusters, with changes of at least four standard errors and at least 15 employers, in percentage points of all sampled lines; names by Sonnet 5.5.

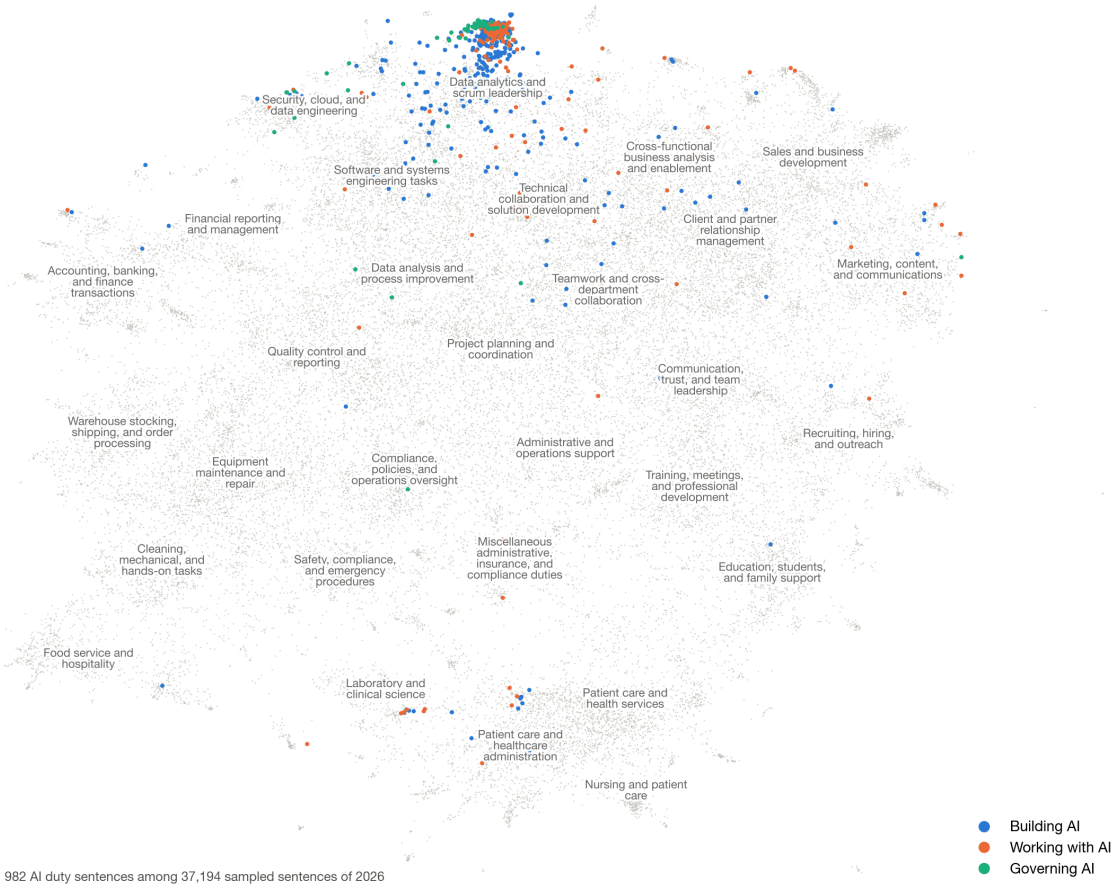


Figure B8: AI duties of 2026 on the map of Figure B6, by side (Sonnet 5.5 labels of sentences with an AI keyword). Grey: all sampled sentences of 2026.

B.5 Duty lines and O*NET tasks

Most measures of AI exposure rely on O*NET's occupational task statements to define what each occupation does, and can capture new work only insofar as O*NET lists it. To test how well O*NET covers the duties in job postings, I embed all 19,265 task statements in O*NET 27.1 (November 2022) and match every duty sentence to its closest O*NET task, both across all occupations and within the posting's own occupation. I consider a duty line covered by O*NET when its closest task is similar enough that a language model is more likely than not to judge the pair as describing at least related work; these similarity thresholds are calibrated from Sonnet 5.5 ratings of 1,530 pairs spanning the similarity range. The nearest embedding match need not be the most relevant task on the list, so Sonnet also evaluates the five closest O*NET tasks for random samples of lines on both sides of the threshold, alongside five tasks drawn at random from the occupation's list as a chance baseline. This comparison provides an external benchmark independent of the 2019H1 postings pool: a duty with no related O*NET task is absent from the official 2022 task inventory.

Figure B9 shows, for the duty lines of 2019–26, the share whose closest task statement in the posting's own O*NET 27.1 list describes related work, and Figure B10 shows how the matching thresholds were set. Sonnet 5.5 graded 85 duty lines in each bin of 0.05 of similarity to their closest O*NET task, separately for the closest task among all occupations and among the posting's own occupation, with the prompt of Appendix B.6 (2: the same work; 1: related work; 0: unrelated). The thresholds where a logistic fit of the share graded at least related work crosses one half are 0.44 among all tasks (95% bootstrap interval 0.42–0.46) and 0.35 within the occupation (0.32–0.37). The share graded the same work crosses one half only at 0.73 and 0.71, at the top of the graded range, so the paper uses no threshold for the same work. Fitted on 400 pairs of flagged AI lines alone, the threshold within the occupation is 0.36 (0.33–0.38), close to the pooled one. For the five closest tasks, Sonnet graded 160 lines of each class whose closest task falls below the threshold, each against its five closest tasks in its occupation's list; at the standard of the same work, it finds such a task for 24% of these building-AI lines and 9% of the working-with-AI lines. The threshold counts every line above it as having a related task. Grading 300 such lines of each class directly, the closest task is related for 65–75% of them and one of the five closest for 92–97% (Opus: 86–93%); five tasks drawn at random from the same list are related for 67–77% of lines, so the lists' coverage of the new duties is well above chance, and the standard is loose. At the stricter standard of the same work, the five closest tasks of these lines describe 50% of continuing lines, 35% of flagged building-AI lines and 27% of flagged working-with-AI lines, against 7–15% for five random tasks. Integrating the fitted probability over each class's similarities instead of cutting at the threshold gives 73% of continuing lines, 50% of flagged building-AI lines, 53% of flagged working-with-AI lines and 58% of flagged non-AI lines with a related closest task, against 89, 49, 58 and 67% at the cutoff. Of the non-AI lines read as new tasks, 20% have no related task among the five closest (405 lines). O*NET revises occupations on a rolling cycle, and 27.1 still carries many lists written before 2019: where the list was last revised before 2019, the closest task is related for 40% of flagged building-AI lines and 49% of flagged working-with-AI lines, against 52% and 64% where it was revised in 2019 or later (continuing lines: 88% and 90%). O*NET 31.0, the release of August 2026, has 103 task statements that 27.1 lacks, 0 of them about AI, and matching to it changes none of the shares by more than a point. Table B6 lists the O*NET task statements that the flagged duty lines of 2025–26 in high-use occupations are most often matched to, by the number of employers.

Exposure across occupations. Across 215 detailed occupations, the share of duty lines read as new tasks has a rank correlation of 0.71 with the published exposure measure of Eloundou et al.

(2024) and 0.77 with observed AI use in Massenkoff and McCrory (2026).

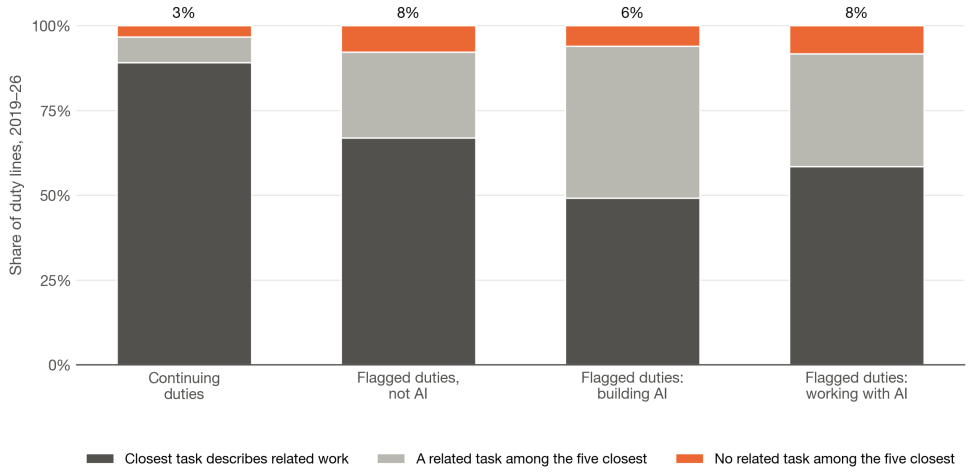


Figure B9: Duty lines of 2019–26 and the O*NET 27.1 task list of the posting's own occupation, by class of line: the closest task statement describes at least related work (dark; the threshold of Figure B10); otherwise, Sonnet 5.5 grades at least one of the five closest tasks as related work (light); or none of the five is related (orange, with its share). Random samples of continuing and flagged non-AI lines and every flagged AI line; the five closest tasks of 160 lines per class.

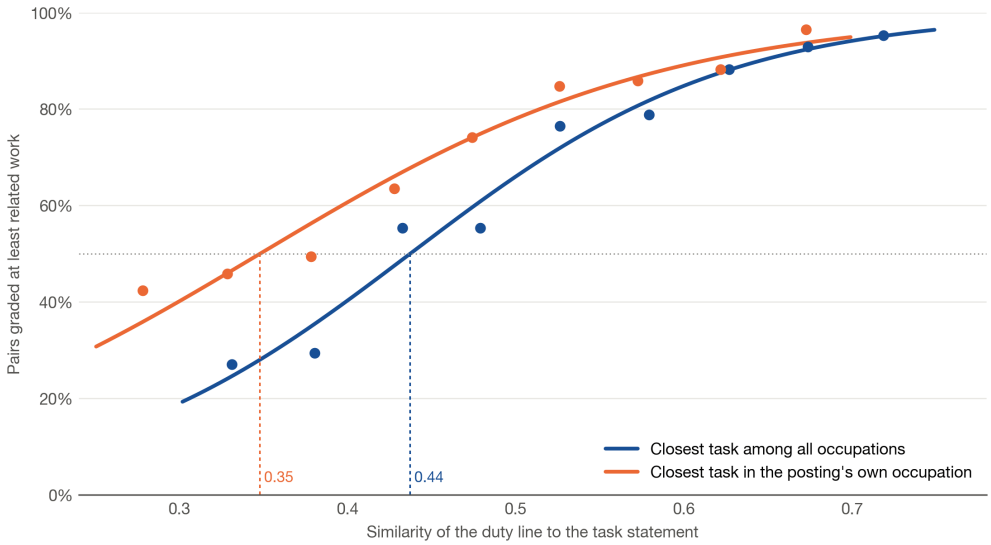


Figure B10: Setting the O*NET matching thresholds: share of duty-line and task pairs that Sonnet 5.5 grades at least related work, by similarity, with logistic fits, for the closest task among all occupations and in the posting's own occupation; 85 pairs per bin. Dotted lines: the thresholds, where the fits cross one half.

Table B6: O*NET tasks closest to flagged duty lines, 2025–26, occupations with high AI use

Closest O*NET 27.1 task	Employers	Lines
A. Flagged AI duty lines		
Develop, test, or program new robots.	687	1,332
Create back-ups of robot programs or parameters.	404	680
Train others to install, use, or maintain robots.	388	642
Create business intelligence tools or systems, including design of related datab...	353	487
Collaborate with system architects, software architects, design analysts, and ot...	283	414
Collaborate with product development teams to research, model, validate, or impl...	260	404
Help clients improve decision making, abstract reasoning, memory, sequencing, co...	235	353
Develop or apply data mining and machine learning algorithms.	226	357
Research computerized automotive applications, such as telemetrics, intelligent ...	202	277
Develop defense plans or tactics, using intelligence and other information.	202	353
B. Flagged duty lines not about AI		
Develop or implement marketing strategies.	530	696
Provide expertise to help financial institutions manage risks and maximize retur...	529	1,048
Collaborate with vendors to obtain or develop desired products.	508	881
Provide customers or installation teams guidelines for implementing secure syste...	482	917
Provide customers with ongoing technical support.	474	787
Collaborate with product development teams to research, model, validate, or impl...	451	729
Serve as liaisons between organizations, shareholders, and outside organizations...	429	601
Implement security practices to preserve assets, minimize liabilities, or ensure...	386	567
Develop or track quality metrics.	356	464
Provide training and support to system designers and users.	355	552

Notes: Flagged duty lines (no close counterpart in 2019H1 postings) whose closest O*NET 27.1 task statement, among all occupations, describes at least related work by the threshold of Figure B10; ranked by distinct employers.

B.6 Language-model labels

Table B7 lists every labelling task. All labels come from Claude Sonnet 5.5, with Opus 5.5 as the second rater, except the occupation codes, which come from Opus 5.5 with Sonnet 5.5 as the second coder, and the checks that use a new prompt, which Opus 5.5 reads alone. The models are called through Anthropic's command-line client at medium reasoning effort, with structured output and every response cached, and they never see the date of the posting being labelled (the theme prompt marks example sentences as recent or older); Table B8 gives the model identifiers and the other specifications.

The flagged-line readings are made in batches of 40 lines that mix the AI and non-AI samples and every year and third. The batch matters: the same lines are read as new tasks more often when mixed than beside similar lines (66% against 47% for AI lines and 10.0% against 6.7% for non-AI lines, on 530 lines read both ways), and one line per call falls in between (60%). Read in mixed batches, the 14,646 lines of the two samples that are new under the threshold give new-task shares of 2.9% for non-AI lines (3.1% in one-class batches), 50% for building AI (31%) and 84% for working with AI (63%); the paper uses the mixed-batch readings, and Appendix D gives the headline on the one-class readings. Opus 5.5 reads a random fifth of the same batches (2,929 lines): it agrees with Sonnet on 84% of non-AI lines, 75% of building-AI lines and 87% of working-with-AI lines, with κ for the new-task class of 0.43, 0.66 and 0.64, and calls 2.0%, 45% and 79% of them new tasks. On non-AI lines the positive class is rare, so agreement on it is the least certain part of the reading: on 1,414 lines drawn to include every non-AI line that Sonnet called a new task, κ for that class is 0.46 (0.39–0.54) and Opus calls 0.78 as many new tasks as Sonnet; for AI lines κ is 0.60 (F1 0.80). Reading a line beside its five most similar 2019 sentences and the most similar one from postings of its own occupation group, instead of the single closest, makes Opus stricter rather than looser: it calls 42% of 500 AI lines new tasks, against 51% beside the single closest.

The 500 flagged non-AI lines of each year and third were drawn on the thirds of the title-based occupation codes (Appendix A.3); with the final codes each line moves to its posting's third, and the share of each reading in a third is the average over the sampling cells it draws from, weighted by their flagged lines, with cells of fewer than 20 lines (9 of 128, holding 1.4% of flagged non-AI lines) taking their sampling third's share. Opus also re-reads random samples of the remaining tasks: on the near-miss and keyword-free samples it agrees with Sonnet on whether a sentence is AI work and on its side in 98% and 100% of cases (κ 0.85 and 0.78); on the threshold reading in 77% (κ 0.65); on the O*NET grades in 79% (κ 0.66); and on the themes' categories in 91% (κ 0.88). Two variants of the prompts check the reading: running the AI prompt without the job title and occupation group, and rewording the four kinds of the flagged-line prompt, which changes the share of new tasks from 4.0% to 5.5% for non-AI lines and from 50% to 55% for AI lines (κ 0.63–0.67). A separate reading by Opus of the lines classified as new tasks asks two distinct questions—whether the underlying activity is new and whether the tool or object of work is new. Among the new AI tasks of 2026, 8% (building AI) and 16% (working with AI) describe a new activity, such as directing agents or evaluating model output, while 76% and 79% describe a familiar activity done with or on AI, such as drafting or coding with an assistant. Among the non-AI lines classified as new tasks by Sonnet and coded by Opus (those also read as new tasks in batches of one class), 12% describe a new activity and 45% describe neither a new activity nor a new tool or object. Within working-with-AI new tasks, the share of new activities fell from 42% in 2023–24 to 16% in 2026 as AI tools spread into everyday workflows. The paper's classes of AI work come from the activity labels rather than from this reading (Section 4.3), and the two agree: Opus calls 81% of the new tasks that direct or review AI new activities and 83% of those that use AI tools familiar activities with a new tool. It calls 21% of the new tasks that build AI and 20% of those that drive its adoption neither, the lines most open to the question of vintage in Section 2. Applied to every flagged line read as a new task (Appendix Figure B1; using the 2023–24 shares for 2022 and the pooled shares for non-AI tasks), new activities account for 0.37 of the 2.6 such lines per 100 duty lines in the high-use third in 2026, familiar activities with a new tool or object for 2.35, and lines rated as neither for 0.76 (mostly non-AI lines). Counted on their own, new activities in the high-use third rose from 0.09 per 100 duty lines in 2022 to 0.37 in 2026, compared with 0.04 in the low-use third and 0.19 across all occupations in 2026.

A third reading asks three separate questions of every line classified as a new task in the main and unscreened readings (2,221 lines): whether the activity is new, whether the object of work is new, and whether the instrument is new, each judged relative to all US postings of early 2019. Sonnet 5.5 reads all 2,221 lines and Opus 5.5 reads a random fifth (444 lines) in mixed batches of 40. This reading applies the rule of Section 2 to the non-AI new tasks: 62% have a new activity or object of work, 9% involve only a new instrument, and 29% have neither on re-reading; the resulting period-specific new-work shares (66% in 2019–22 and 59% in 2023–26) scale the non-AI new-task counts. On the AI lines, the three questions cross-check the activity labels: 71% of lines labelled as using AI tools involve only a new instrument, 95% of those labelled as directing or reviewing AI involve a new activity, and 89% of building-AI lines and 75% of adoption lines involve a new activity or object. Opus agrees with Sonnet on whether a line is new work for 80% of lines but is stricter with non-AI lines, calling 39% of them new work against 66% for Sonnet on the same lines. Opus also re-reads the named themes of Appendix C.1: judged on the same four categories as the flagged lines, 64% of the AI themes are new tasks (93% of the working-with-AI themes and 52% of the building themes). The system prompts follow; line breaks are added for printing.

Table B7: Language-model labelling tasks

Task	Unit	Design	Labelled	Model	Second rating
Side and activity of AI work	Duty sentence with an AI keyword	Census	134,656	Sonnet 5.5	Opus 5.5, random 2,000: side 90%
AI work without a keyword	Duty line without an AI keyword	Random, 18,750 lines a year[d]	139,346	Sonnet 5.5	Opus 5.5, random 1,372: 100%
AI work, near-miss words	Duty line with a near-miss word	Random, 5,000 lines a year[d]	35,166	Sonnet 5.5	Opus 5.5, random 1,000: 98%
What flagged non-AI lines are	Flagged non-AI line and nearest 2019H1 duty	Stratified: year × third of AI use, 500 a cell[a,c]	12,386	Sonnet 5.5	Opus 5.5, random 2,457: 84%
What flagged AI lines are	Flagged AI line and nearest 2019H1 duty	Stratified: side × period, 400 a cell[c]	2,260	Sonnet 5.5	Opus 5.5, random 472: 81%
Checks of the reading	Line and nearest 2019H1 duty	COVID-19 lines; lines just above the threshold; known new work; reworded prompt	3,684	Sonnet 5.5	Opus 5.5, random 20%: 71–86%
AI work without context	Duty sentence with an AI keyword	Stratified: occupation group, 400 a cell	1,200	Sonnet 5.5	Same side as with context: 87%
O*NET grades for AI lines and five closest tasks	Duty line and O*NET task	Stratified by similarity; lines without a related closest task	4,400	Sonnet 5.5	Opus 5.5, random 20%: 85–88%
Novelty threshold	Duty line and nearest 2019H1 duty	Stratified: 100 per 0.05 of similarity	1,000	Sonnet 5.5	Opus 5.5, random 500: 77%
O*NET matching thresholds	Duty line and closest O*NET task	Stratified: 85 per 0.05 of similarity, two searches	1,530	Sonnet 5.5	Opus 5.5, random 500: 79%
Theme names and kinds	Rising theme of flagged lines	The 60 that rose most and the 40 largest not about AI	90	Sonnet 5.5	Opus 5.5, all: 91%
Region names	Region or cluster of the map	Census of the 30 regions and the labelled clusters	51	Sonnet 5.5	
Seniority	Job title	Half in proportion to postings, half uniform	30,000	Sonnet 5.5	Title rules: 87%
Industry	Firm	Census of firms with at least three postings	30,539	Sonnet 5.5	Second pass on low or medium confidence
Duty lines (text layer)[b]	Every sentence of a posting	100 postings a year and 200 with flat text	1,200	Sonnet 5.5	Retest of 100: 99%
Occupation codes[b]	Posting	Stratified by platform and year, plus AI and new titles	63,000	Opus 5.5	Sonnet 5.5, random 20%: detailed 73%
Technical and data tools	Duty line naming a tool, with the job title	Stratified: family × period × earlier use, 80 a cell	1,440	Opus 5.5	Also read the examples of Figure 5
What is new about a new task	Line read as a new task and nearest 2019H1 duty	Census: every line read as a new task, main and unscreened readings	2,221	Sonnet 5.5	Opus 5.5, random 444: 80% on new work

Notes: Labelled: items labelled (sentences, lines, pairs, themes, titles, firms or postings). [a]Eight years (2019: second half) by the three thirds of observed AI use, and 250 a year in occupations without a measure. [b]From the dataset (Section 3). [c]Drawn as designed (14,000 non-AI and 2,400 AI lines) on postings dated by first capture; the counts are the lines still new under the threshold on stated dates. [d]Lines drawn: 150,000 and 40,000; the counts are distinct sentences.

Table B8: Models and specifications

Component	Specification
Language models	claude-sonnet-5-5 (reader) and claude-opus-5-5 (second reader; new prompts), through the Claude Code command line 2.1.283 in print mode with JSON-schema output, reasoning effort medium; calls made 2026-10-11 to 2026-10-17
Prompts	printed in Appendix B.6; every reading is released with its model and prompt
Sentence embedding	sentence-transformers/all-MiniLM-L6-v2, mean pooling, 128 tokens, L2-normalised; larger-model check: BAAI/bge-base-en-v1.5
Novelty screen	maximum cosine to the 679,314 distinct duty sentences of postings dated to 2019H1; threshold 0.66 (10% of 2019H2 lines flagged)
AI-activity classifier	logistic regression (stochastic gradient descent, log loss, alpha 1e-6) on word 1–2-gram TF-IDF (300,000 features) and the MiniLM embedding (weight 3); trained on all Sonnet-labelled sentences (keyword, near-miss and random samples); Sonnet labels override its predictions where they exist
Themes	spherical k-means on MiniLM vectors, 1,500 clusters, 20 iterations, seed 7, fitted on the distinct flagged sentences of 2025–26; every flagged sentence of 2019–26 assigned to its nearest centroid; rising: employer share in 2025–26 at least 3 times 2019–22 and at least 20 employers
O*NET matching	O*NET 27.1 task statements, MiniLM; related-work thresholds 0.348 (own occupation) and 0.437 (all), each fitted on 765 Sonnet-graded pairs of its own lookup

AI work: side and activity. Each call sends 60 items as "id | job title | occupation group | sentence".

```
You code duty sentences from US job postings for research on how AI changes work. Each item gives a
  job title, an occupation group and one duty sentence from that posting. Code every item.

ai: true if the sentence describes work that involves AI or machine-learning systems: generative AI,
  LLMs, chatbots, copilots, AI agents, machine-learning or deep-learning models, computer vision,
  speech or NLP systems, AI-driven automation. false for automation without AI (scripts, RPA,
  macros, workflow rules), statistics or analytics without ML, "models" meaning financial, data,
  business or 3D models, "agents" meaning people (insurance, customer service, real estate), and
  "AI" in other senses.

activity (if ai is true, the main one; if ai is false, "none"):
  build: research, design, develop, train, fine-tune or evaluate AI models, agents or AI-powered
    products and features that the organisation builds
  deploy: integrate AI into products, systems or infrastructure; MLOps; data pipelines for AI;
    operate, maintain or monitor AI systems in production
  sell_support: sell, market, consult on, implement for clients, or support AI products or services
  govern: AI policy, risk, compliance, ethics, security, legal or responsible-AI oversight
  use_tool: use AI tools to do the person's own work (drafting, coding with an assistant, analysis,
    research, design, answering customers)
  direct_review: direct, prompt or orchestrate AI agents or tools, or review, verify, correct or
    approve AI-generated output, as part of the person's own work
  adopt_enable: find AI use cases, lead or drive adoption of AI tools in a team or function,
    redesign workflows around AI, or train and coach colleagues to use AI
  other_ai: anything else involving AI (e.g. "stay current on AI trends")

gen (if ai is true): "generative" if it clearly involves generative AI (LLMs, chatbots, copilots,
  generative agents, text, image or code generation); "other_ml" if it is clearly other machine
  learning; "unclear" otherwise. If ai is false: "na".
```

Flagged lines and the novelty threshold. Each call sends 40 items as "id | job title | recent sentence | most similar 2019 sentence".

```
You compare a duty sentence from a recent US job posting with the most similar duty sentence found
  in US job postings from early 2019, to judge whether the recent sentence describes new work. For
  each item give:
kind:
  new_task: the recent sentence describes an activity, responsibility or object of work that the
    2019 sentence does not, and that would not have been common in 2019 job postings for this kind
    of job
  variant: the same kind of activity as the 2019 sentence, with a different object, tool, setting or
    level of detail that is not new in itself
  reworded: the same activity as the 2019 sentence, in different words
  not_a_duty: the recent sentence is not a description of work (company, benefits, role context,
    requirements, or a fragment)
source (only for new_task; otherwise "na"): what makes it new: technology (non-AI tools, platforms,
  data systems), regulation_compliance, health_safety, product_market (new products, services,
  markets or business models), organisation_people (e.g. diversity, remote or hybrid work,
  wellbeing, culture), sustainability, other
```

Three questions about new tasks. Each call sends 40 items as "id | job title | recent sentence | most similar 2019 sentence".

```
You read a duty sentence from a recent US job posting, with its job title, next to the most similar
  duty sentence found in US job postings from early 2019. An earlier reading judged the recent
  sentence to describe a new task. Say what is new about it by answering three separate questions. A
  task is an activity applied to an object of work, and it may be done with an instrument.
activity_new: is the activity itself new? The activity is what the worker does: for example write,
```

```
  analyse, build, test, sell, teach, administer, review, decide, coordinate or comply. Answer "yes"
  only if the worker is asked to do something that US job postings rarely asked anyone to do in
  2019. Directing, prompting or orchestrating AI systems, and reviewing, correcting or signing off
  the work they produce, count as new activities.
object_new: is the object of work new? The object is what the activity is applied to or produces: a
  product, service, system, market, programme, rule or standard. Answer "yes" if that object did not
  exist, or was rarely mentioned in US job postings, in early 2019: for example an AI model, agent
  or AI-powered product that the worker builds, deploys, tests, governs or sells; the adoption of AI
  tools by colleagues that the worker leads; a new product, asset or market; a new law, reporting
  rule or programme. Answer "no" if the object is familiar, even when the sentence names a new tool
  used to work on it.
instrument_new: is the instrument new? The instrument is the tool, software, platform or device the
  worker uses to do the task. Answer "yes" if the sentence names or clearly refers to an instrument
  that did not exist, or was rarely mentioned in US job postings, in early 2019, such as an AI
  assistant used to draft, code or research, or a software package that has appeared since. Answer
  "no" otherwise.
Judge novelty against US job postings of early 2019 in general, not only those for this kind of job.
  Answer each question on its own: a sentence can be new in any combination of the three, or in
  none. The 2019 sentence shows the kind of duty that postings already described in 2019; it need
  not come from the same kind of job.
```

Technical and data tools. Each call sends 40 items as "id | job title | tool family | duty sentence".

```
You read duty sentences from US job postings that name a technical or data tool, each with the job
  title and the family of tools it was matched to. For each item give:
uses_tool: "yes" if the sentence asks the person in this job to use, build or work with a tool of
  that family (for example, building dashboards in Tableau, writing SQL queries, automating a
  process with Power Automate); "no" if the tool is named for another reason (the employer's own
  product, what another team uses, a qualification, or a word that is not the tool, such as a
  therapy called DBT or a dashboard of a vehicle)
activity: what the person does with the tool: dashboards_reports, data_analysis, code_scripts,
  automate_workflows, manage_data, or other
ai: "yes" if the sentence also involves AI tools or models, otherwise "no"
```

O*NET matching thresholds. Each call sends 20 duties, each with its job title and one candidate task statement.

```
You judge how well standard task descriptions from the O*NET database describe a duty from a US job
  posting. For each candidate, give a grade:
2 = it describes the same work as the duty (wording may differ; small differences in scope are fine)
1 = related work (same area or a closely connected step), but not the same work
0 = unrelated, or only superficially similar
Judge the work itself, using the job title for context. Grade every candidate independently.
```

Themes. Each call sends up to eight recent and four older example sentences per theme.

```
You classify recurring duties in US job postings. Each group is a set of similar duty sentences from
  different postings: "recent" sentences, and the "older" sentences closest to the same theme. For
  each group give a short name (at most 8 words, sentence case, describing the recent work) and one
  category for the recent sentences:
ai_building: building, deploying, integrating, selling or supporting AI systems (models, agents, LLM
  products)
ai_using: using, prompting, directing, reviewing or driving the adoption of AI tools in the worker's
  own work
ai_other: governing AI, or other work about AI
new_task: genuinely new work that is not about AI and would not have been common in 2019 postings
  for such jobs
variant: a familiar kind of work with a new object, tool, platform or setting that is not new in
```

```
  itself
reworded: familiar work in new wording, or a shared template
not_work: culture, values, benefits, notices or generic statements rather than work
source (only for new_task; otherwise "na"): technology (non-AI), regulation_compliance,
  health_safety, product_market, organisation_people, sustainability, other
```

B.7 Reading by occupation and half-year

Section 5.5 needs new work in every occupation and half-year. The cells are the detailed occupations with confident codes and at least 200 postings in 2019–22, by half-year from 2019H2, with at least 20 postings (3,602 cells). Sonnet 5.5 reads every flagged AI duty line of a cell where it has at most 50, and a random 50 otherwise (15,837 lines), with seven flagged non-AI lines for each, spread evenly over the same cells (126,696 lines in all), in 3,168 calls of 5 AI and 35 non-AI lines mixed across cells and years, with the prompt and comparison sentence of the main reading; each line is weighted by its cell's flagged lines of its kind over the lines of that kind read. The mix of a batch matters. In a pilot of 50 calls, AI lines that the main reading had already read were called new tasks 54% of the time in batches of half AI lines, against 66% before, while re-read in batches of the paper's mix they gave 67% against 68%; this reading therefore keeps the paper's mix of about one AI line in eight. The three questions of Appendix B.6 are asked of every non-AI line read as a new task, mixed with AI new tasks as before: 48% of these non-AI lines are new work, and on the AI lines the answers confirm the activity labels (33% of tool-use lines and 98% of directing lines are new work). Opus 5.5 re-reads a fifth of the calls, whole: it agrees with Sonnet on 79% of AI lines and 85% of non-AI lines, and calls 41% of the AI lines new tasks against Sonnet's 50%. Added up by third, the reading gives new work about AI of 0.08 and 2.46 per 100 duty lines in the high-use third in 2022 and 2026 (Figure 1: 0.13 and 2.01), 0.76 in the middle third and 0.14 in the low-use third in 2026 (0.74 and 0.14), and new work outside AI of 0.25 and 0.88 in the high-use third (0.39 and 0.57).

B.8 Seniority, industry and firms

Seniority comes from the job title: a model trained on 30,000 titles labelled by Sonnet 5.5 assigns one of eight levels from intern to executive. Industry is a two-digit NAICS sector for each firm with at least three postings, coded by Sonnet 5.5 from the firm's name, titles and occupation mix, and recoded from its company-description sentences where Sonnet was not confident; it covers 99% of postings. Employer accounts that switched ATS platforms are linked into firms, consolidating the 48,304 employer accounts into 45,678 firms, and the firm fixed effects of Section 5.7 use this identifier. Seniority levels are intern, junior, mid, senior, staff or principal, manager, director and executive. Title rules alone agree with Sonnet on 87% of 1,000 titles; the title model, trained on 30,000 Sonnet-labelled titles (half drawn in proportion to postings, half drawn uniformly from the remaining titles), agrees on 88% in cross-validation. Industry codes were checked and recoded where Sonnet's confidence was low or medium, using the employer's company-description sentences; 77% of firms (90% of postings) are coded with high confidence. Employer accounts are linked across platforms when a normalised name is unique to one account on each of at least two platforms; the links rest on the name alone, 5,040 accounts become 2,414 firms, and every other firm is a single account.

C Further results

Table C1 gives the ladder of Figure 1 and extends it: from directing and reviewing AI alone, the strictest reading, to every flagged AI line whether or not it is read as a new task. Tables C2 and C3 list three randomly drawn duty lines from each category alongside the illustrative examples in Table 1, showing the full range of lines that each category admits.

Table C1: New work under each judgement call (per 100 duty lines)

	Most AI use, 2022	Most AI use, 2026	All, 2026	Least AI use, 2026
Directing or reviewing AI	0.00	0.06	0.03	0.01
+ building and governing AI	0.13	1.80	0.76	0.09
+ driving the adoption of AI (new work about AI)	0.13	2.01	0.92	0.14
+ new work outside AI (new work, Figure 1)	0.53	2.58	1.24	0.22
+ using AI tools in one's own work, and other AI	0.74	3.48	1.72	0.33
Every flagged AI line, read or not, and new work outside AI	0.98	4.70	2.24	0.40

Notes: Each row adds a class of flagged lines read as new tasks to the row above, by what the worker does with AI (Section 2); the fourth row is new work as the paper counts it (Figure 1). The fifth adds flagged AI lines read as new tasks that ask the worker only to use AI tools in their own work or that fit no activity (Appendix Figure B1). The last counts every flagged AI line, read as a new task or not, with new work outside AI. Thirds of occupations by observed AI use; 2026: January to September.

Table C4 collects the headline comparisons under other choices. Two further checks bear on what the flagged AI lines are. They are not boilerplate copied across employers: 2% of the working-with-AI lines of 2025–26 recur word for word at a second firm, against 10% of continuing lines, and among distinct sentences, allowing for paraphrase, 2.1% of working-with-AI lines and 1.4% of building-AI lines have a near-duplicate (cosine similarity of at least 0.90) at another firm, against 1.9% of continuing lines and 1.1% of flagged non-AI lines, on 20,000 distinct sentences of each kind. And they do not come from new employers: in the firms that posted both in early 2019 and in 2025–26, flagged AI lines are 1.6 in every 100 duty lines in 2026, against 1.7 in all postings, with 3.6 in the high-use third and 0.29 in the low-use third.

Two other counts of new work are broader. Postings in titles that at least three employers used in a year and that no posting of 2019–22 used, in any spelling, rose from 0.1% of postings in 2023 to 0.7% in 2026, and to 1.2% in the third of occupations with the most AI use; 46% of them have an AI duty. Duty lines new to a firm and occupation, the analogue in text of the new skills counted by Chen et al. (2024), are 55–59% of the duty lines of each year from 2023; 77–83% of them are close to a duty of early 2019, and Sonnet reads 3% of the rest that are not about AI as new tasks.

Table C2: Duty lines counted as new work, drawn at random

Duty line	Job title, year
Building AI	
Identify edge cases where LLMs struggle and set up automated systems for catching hallucinations/unexpected behavior.	AI Software Engineer, 2025
Develop key methodologies, lab HW and system SW capabilitiesas well as system visibilities and debug tools necessary for successful system (HW/SW/FW)...	Server CPU Hardware Systems Lead - US, 2026
In this role, you will evaluate cloud AI platforms, orchestrate vector databases, and help build the Model Context Protocol (MCP) gateways that securely...	Sr. Staff Enterprise AI Architect, 2026
Governing AI	
Establish guardrails for safe, governed AI use against revenue data — including permissioning, read/write boundaries, and human-in-the-loop checkpoints.	GTM AI Operations Lead, Demand, 2026
Partner with Agent and Platform teams to define security boundaries for agent execution: sandboxing, least-privilege tool access, and runtime policy...	Lead Security Engineer, 2026
System Integrity: Ensure all AI-driven automations comply with strict IAM, security, and data protection standards.	Staff Software Development Engineer, 2026
Working with AI: directing or reviewing AI	
Evaluating and ranking code generated by AI models	Big Data Engineer for AI Training, 2024
Assessing whether a piece of Japanese text produced by an AI model is factually accurate	AI Training for Japanese sound, 2024
Evaluating and ranking code generated by AI models	Cloud DevOps Coding Mentor - Tier 3 - US, 2024
Working with AI: driving adoption	
Innovate in Hybrid and AI-Powered Learning: Reimagine how AI can support and enhance learning design, workflows, and learner experiences—from personalized...	Learning Experience Designer, 2025
Equip teams with the necessary knowledge and skills to achieve success and faster job readiness, designing new AI Services-specific processes and materials...	AI Commercial Account Management Chief, 2024
Become the Global Marketing team's expert on using Highspot and GenAI	Content Marketing Manager, 2023
New work outside AI	
Team members are required to complete mandatory Code of Conduct elearning courses upon hire and on an annual basis.	STA, 2021
Assists with UKG Pro updates, programming and updating business rules, and transitioning new divisions onto the UKG platform.	Senior Payroll Administrator, 2025
This newly-created and highly visible role - The Accounting & Measurement (A&M) Manager will be a key member of Takeda's Planet Initiative and will help...	Global Environmental Sustainability, 2022

Notes: Three lines of each kind, drawn at random from the labelled samples (Appendix Table B7): flagged lines that Sonnet 5.5 reads as new tasks, placed by the activity labels of AI work. Building and directing lines are also coded by Opus as a familiar activity with a new object and as a new activity; governing lines come from the reading of governing and other AI lines. The lines of new work outside AI include weaker cases of the kind that the second reader rejects (Section 5.1). Lines are cut at 160 characters and titles at 40.



Figure C1: Four headline numbers of Panel A of Table C4 under each variant: (a) flagged AI lines in the high-use third, (b) the change in flagged non-AI lines, (c) working with AI as a share of AI duty lines (every line counted) and (d) the exposure design. Blue: baseline.

Table C3: Duty lines of other kinds of change, drawn at random

Duty line	Job title, year
Working with AI: using AI tools in one's own work	
Embrace the future of AI-assisted technologies, enabling you to focus on the care you deliver.	Certified Veterinary Technician, 2025
Integrate AI-powered tools and workflows to unlock new levels of personalization, automation, and speed.	Head of Growth Marketing, 2026
Run the company operating cadence: all-hands meetings, board meetings, investor updates, and quarterly planning, all built with best-in-class AI tools and...	Chief of Staff, 2026
Existing task, new to the occupation	
Develop and use hand calculations, spreadsheets, programs, or other tools (Python, Excel and/or MathCAD preferred) to determine or validate miscellaneous...	Senior Pipe Stress Analysis Engineer, 2025
Data Warehouse Architecture: Oversee the structure of marketing data within Google BigQuery, ensuring a scalable single source of truth that connects...	Marketing Analytics Director, 2026
Document student progress using the HeyTutor Dashboard.	Inglewood Continuation Adult School, 2025
More of an existing AI task	
In this role, you'll lead development initiatives, contribute to our SaaS products, and tackle exciting challenges in backend architecture, frontend design,... *(2019: You'll be working on our product engineering team - building robust back-end services to help power our AI...)*	Senior Full-Stack Engineer, 2025
Contribute to product direction by identifying opportunities where AI can meaningfully improve biological design *(2019: Develop AI products for industry.)*	AI Product Manager, 2026
Partner with data scientists, AI researchers, and product engineers to deliver reliable and maintainable AI services *(2019: Collaborate with other data scientist and data engineers to design, prioritize and implement new features...)*	Senior AI/ML Engineer, 2026
Variant	
Become an expert at using the Ambra platform *(2019: Becoming an instructor on Cybrary will provide you a greater distribution advantage than any other...)*	Account Executive - Enterprise, 2021
Sign up independent dealerships as buyers to utilize ACV Auctions digital platform. *(2019: You will work alongside software and production engineers to build and ship new features that will drive...)*	Vehicle Condition Inspector, 2023
Serves as key point of contact for Government Member inquiries regarding implementation of StateRAMP *(2019: Provide FedRAMP requirements and guidance.)*	Government Procurement Specialist, 2022
Re-wording	
You will design across devices and form factors, creating a cohesive brand experience that celebrates female health and wellbeing, helping women achieve... *(2019: Help us shape our visual aesthetic and create a cohesive brand presence across all our products and platforms.)*	Product Designer, 2021
You will prepare and lead presentations of the monthly S&OP readout. *(2019: Lead the monthly S&OP process:)*	Demand Planner, Active Nutrition, 2024
Flex / rotate within various positions when needed *(2019: Rotation of position within team structure as necessary;)*	Production Team Member, 2025

Notes: Three lines of each kind, drawn at random. Using AI tools: flagged lines that Sonnet 5.5 reads as new tasks and that Opus codes as a familiar activity with a new tool. Tasks new to the occupation: tool lines that Opus confirms, 2025–26, outside computer occupations. More of an existing AI task: building or deploying lines first seen in 2025–26 with a close counterpart in 2019H1; some name a technology that is new since 2019, which the screen counts as continuing work (Appendix D measures how many new tasks the screen misses). Variants and re-wordings: the reading of flagged non-AI lines. In italics: the closest duty of 2019H1. Lines are cut at 160 characters and titles at 40.

Table C4: Robustness of the headline numbers

	Flagged AI, high	Flagged AI, low	High/low	Δ flagged non-AI	Working with AI (%)	Postings (%)	Exposure design
A. Weights, platforms, threshold and period							
Baseline	3.73	0.26	14	+2.6	25.0	5.9	1.65 (0.42)
OEWS employment weights	1.83	0.08	22	+1.8	29.9	2.7	0.79 (0.32)
CPS new-hire weights	1.69	0.08	21	+1.7	30.8	2.6	0.60 (0.26)
Without Workday and Lever	3.76	0.27	14	+3.2	25.3	6.1	1.72 (0.43)
Fixed platform mix	2.84	0.17	17	+1.5	26.7	4.6	1.60 (0.42)
Novelty threshold −0.02 (6% of 2019H2 new)	2.80	0.20	14	+1.9	25.0	5.9	
Novelty threshold +0.02 (14% of 2019H2 new)	4.66	0.32	14	+3.3	25.0	5.9	
Without 2026H2	3.71	0.27	14	+2.4	23.7	5.5	1.14 (0.29)
B. Reference, embedding model, sample and occupation codes							
Novelty threshold: 5% of 2019H2 new	2.25	0.16	14				
Novelty threshold: 20% of 2019H2 new	5.42	0.36	15				
Reference without January 2019	3.27	0.22	15				
Larger embedding model (sample)	3.10	0.26	12				
Firms posting in 2019H1 and 2025–26	3.61	0.29	13				
Without computer occupations	1.59	0.26	6				1.60 (0.54)
Major group × half-year effects							0.81 (0.42)
Confident occupation codes	4.04	0.24	17				
Codes with probability ≥ 0.75	4.23	0.20	22				
AI use over the three likeliest codes	3.60	0.16	23				
Title-based occupation codes	3.71	0.35	10				

Notes: Flagged AI, high and low: flagged AI duty lines per 100 duty lines in 2026, before reading, thirds of occupations with the most and least observed AI use; High/low: their ratio. Δ flagged non-AI: change in flagged non-AI lines per 100 duty lines, high-use occupations, 2022 to 2026 (raw, not read). Working with AI: share of AI duty lines, 2026, every duty line counted (25%; the text's 22% also allows for the keywords' catch rates, Table B4). Postings: share of postings with a working-with-AI duty, 2026. Exposure design: post (2024H1 on) minus pre (to 2022H2) coefficient for postings with a working-with-AI duty, per standard deviation of AI use (standard error), with occupations weighted by their 2019–22 postings or, in the OEWS and CPS rows, by their employment (OEWS, May 2025) and their new job starts (CPS, 2019–22, by major group, split by employment). The novelty threshold does not affect the working-with-AI share or the postings column, which count all AI lines. Panel B: each threshold makes the stated share of 2019H2 lines new; the larger embedding model (BAAI bge-base) reads 20,000 lines per half-year with its own threshold; firms: linked employer accounts posting in both periods; without computer occupations: SOC 15 dropped, thirds kept; the last three rows cut the thirds on codes with a probability of at least 0.75 (97% agreement with Opus 5.5), on each posting's expected AI use over its three likeliest codes, and on the title-based codes (Appendix A.3).

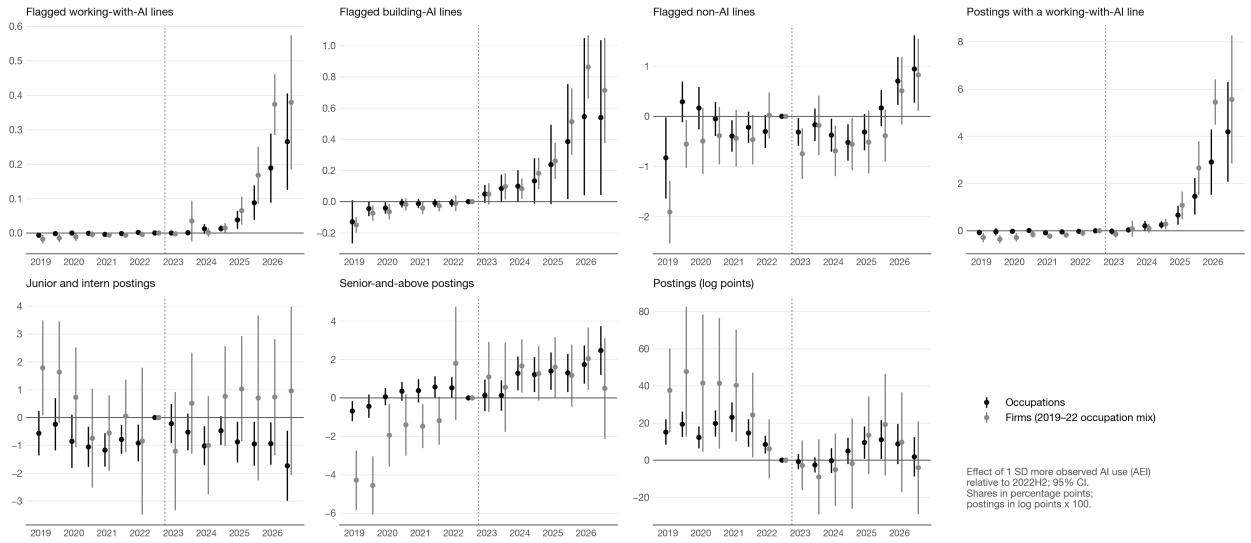


Figure C2: Occupation-level design for every outcome, and a firm-level version in which a firm's AI use is that of its 2019–22 occupation mix (230 occupations; 1,756 firms). Senior and above includes managers, directors and executives.

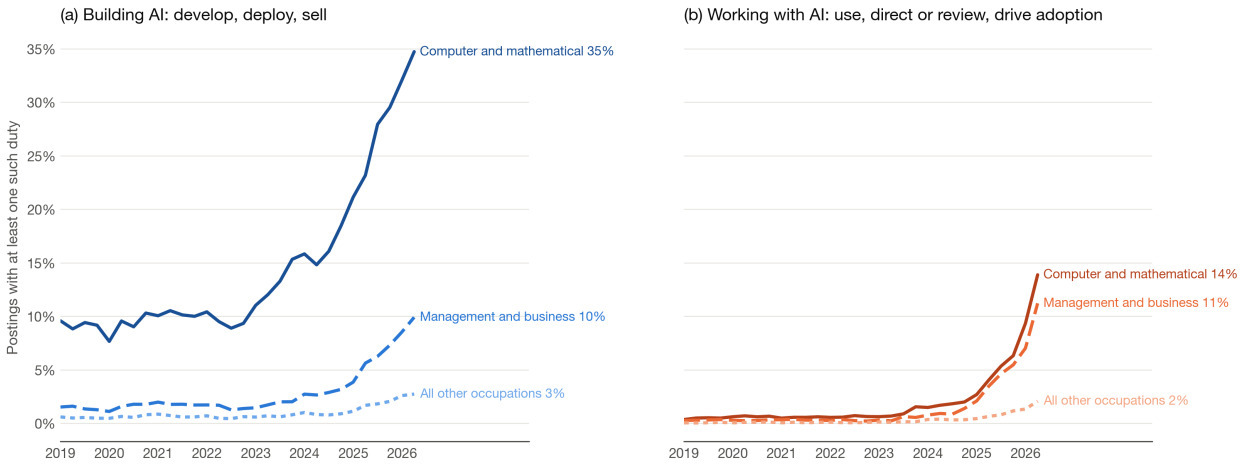


Figure C3: Share of postings with at least one duty line about building AI (a) or working with AI (b), by quarter and occupation group. Every duty line is classified: Sonnet 5.5 labels for AI-keyword sentences and a classifier trained on all labelled sentences for the rest. 2026Q3, a partial quarter, is omitted.

Table C5: The activities of AI duty lines by year (%)

Year	Build	Deploy	Sell, support	Use tools	Direct, review	Drive adoption	Govern	Lines
2019	72.5	15.7	5.5	3.4	0.4	1.8	0.6	4,186
2020	71.1	16.7	5.2	4.0	0.4	2.0	0.7	3,142
2021	63.4	22.5	7.4	3.2	0.3	1.8	1.4	6,095
2022	64.4	20.0	7.3	3.6	0.3	2.1	2.3	3,835
2023	63.8	16.6	9.7	3.5	0.9	3.1	2.4	4,195
2024	55.3	16.7	11.4	4.5	2.8	4.7	4.6	16,718
2025	46.0	17.5	12.0	8.4	0.9	10.4	4.9	44,391
2026	32.9	14.3	12.4	14.1	1.9	17.4	7.0	63,951

Notes: Sonnet 5.5 labels of every duty sentence with an AI keyword, as shares of the AI duty lines with an activity. Lines: AI duty lines with an activity. 2026: January to September.

Table C6: Posted-pay premia by occupation group

Fixed effects	Building AI	Working with AI	Postings
All postings			
Occupation × state × year	0.170 (0.012)	0.054 (0.008)	428,836
+ seniority	0.155 (0.012)	0.026 (0.007)	428,836
Firm × occupation × year	0.068 (0.004)	0.022 (0.008)	428,836
Firm × title × year	0.009 (0.007)	0.022 (0.014)	428,836
Computer and mathematical			
Occupation × state × year	0.142 (0.012)	−0.009 (0.011)	58,230
Firm × occupation × year	0.052 (0.005)	0.007 (0.009)	58,230
Management and business			
Occupation × state × year	0.203 (0.013)	0.083 (0.007)	120,551
Firm × occupation × year	0.093 (0.006)	0.029 (0.008)	120,551
All other occupations			
Occupation × state × year	0.199 (0.025)	0.074 (0.020)	250,055
Firm × occupation × year	0.078 (0.015)	0.032 (0.033)	250,055

Notes: Log annual posted pay, 2023–26, on indicators for a building-AI (build) and a working-with-AI (use) duty, with fixed effects for the cells in the first column. Pay is the midpoint of the posted range, annualised. Standard errors clustered by firm in parentheses. Postings: postings in the regression sample. Without year effects, the firm-and-title premia are 0.031 and 0.033; controlling for state laws that require pay ranges in postings changes no estimate by more than 0.01.

Table C7: The 30 largest rising themes of flagged duty lines

Theme (named by Sonnet)	Judged as	Employers 2019–22 (%)	2025–26 (%)	AI lines (%)
Integrate LLMs and AI into applications	Building AI	0.01	1.09	99
Build and operate AI platform infrastructure	Building AI	0.02	1.07	97
Leverage AI tools to work faster	Working with AI	0.01	1.00	99
Design and build AI-native workflows	Working with AI	0.01	0.96	98
Develop AI-driven product features and integrations	Building AI	0.07	0.95	96
Create and codify repeatable playbooks	Re-worded	0.22	1.06	2
Build agent platforms and tool layers	Building AI	0.03	0.87	81
Champion and drive team AI adoption	Working with AI	0.04	0.87	96
Using and adopting AI in workflows	Working with AI	0.01	0.83	99
Designing user interfaces for AI products	Building AI	0.04	0.85	97
Staying current on emerging AI capabilities	Governing/other AI	0.04	0.83	97
Building agentic AI workflows	Building AI	0.00	0.78	96
Positioning and marketing AI products	Building AI	0.03	0.80	89
Building AI agent systems and infrastructure	Building AI	0.02	0.76	100
Using AI-assisted development tools for coding	Working with AI	0.01	0.75	96
Designing and owning reliable automated workflows	Variant	0.33	1.05	5
Build and integrate LLM-based applications	Building AI	0.02	0.74	97
Ensure responsible, compliant AI governance and ethics	Governing/other AI	0.02	0.74	92
Build, launch, and sell agentic AI systems	Building AI	0.01	0.72	100
Build AI-ready data infrastructure and platforms	Building AI	0.10	0.82	93
Implement, evaluate, and improve LLM systems	Building AI	0.01	0.72	99
Lead and execute enterprise AI strategy	Governing/other AI	0.02	0.72	95
Generate and own sales pipeline	Re-worded	0.28	0.97	2
Sell AI and manage AI investments	Building AI	0.04	0.72	89
Build and operate agentic AI platforms	Building AI	0.01	0.69	98
Build backend services for AI applications	Building AI	0.03	0.70	99
Engineer AI and agentic solutions	Building AI	0.08	0.75	90
Use AI coding assistants in development	Working with AI	0.01	0.68	97
Establish governance frameworks and cadences	Re-worded	0.26	0.93	3
Use AI tools to accelerate routine work	Working with AI	0.02	0.68	96

Notes: Themes of the flagged duty lines of 2025–26 that rose at least threefold in the share of employers using them, ranked by the increase. Names and judgements by Sonnet 5.5. AI lines: share of the theme's lines labelled AI work. The themes are built on postings with confident title-based occupation codes (Appendix A.3) and dated by first capture; they are descriptive.

Table C8: The named rising themes that are not AI work

Theme (named by Sonnet)	Judged as	Employers 2019–22 (%)	2025–26 (%)	AI lines (%)
Create and codify repeatable playbooks	Re-worded	0.22	1.06	2
Designing and owning reliable automated workflows	Variant	0.33	1.05	5
Generate and own sales pipeline	Re-worded	0.28	0.97	2
Establish governance frameworks and cadences	Re-worded	0.26	0.93	3
Represent company at industry events	Re-worded	0.29	0.92	0
Develop market narrative and storytelling	Re-worded	0.28	0.87	2
Ship fast and iterate on product	Re-worded	0.13	0.69	3
Conduct customer and technical discovery	Re-worded	0.15	0.68	8
AI developer community advocacy and engagement	Variant	0.07	0.59	70
Build automated security and compliance guardrails	New task	0.06	0.58	17
Drive engineering improvements and initiatives	Re-worded	0.23	0.74	2
Own platform reliability, observability and incident practices	Re-worded	0.16	0.66	3
Build and own observability platforms	Variant	0.05	0.54	11
Make build-versus-buy and tradeoff decisions	Re-worded	0.16	0.64	2
Support go-to-market strategy and execution	Re-worded	0.15	0.63	3
Lead engineering analysis and integration	Re-worded	0.22	0.69	1
Go-to-market recruiting and GTM support	Variant	0.11	0.58	4
Executive engagement and communication	Re-worded	0.19	0.65	1
Surface insights and design across surfaces	Re-worded	0.09	0.55	4
Craft clear narratives for decision-makers	Re-worded	0.21	0.67	2
Drive readiness across operations and launches	Re-worded	0.11	0.55	2
Build trust and safety technology	Variant	0.15	0.58	4
Own systems and technical architecture	Re-worded	0.15	0.58	7
Build scalable co-selling and enterprise sales motions	Re-worded	0.05	0.47	2
End-to-end ownership of systems and outcomes	Not work	0.15	0.56	1
Unblock customers, teams, and deployments	Re-worded	0.18	0.58	1
Healthcare domain insight and clinical communication	Variant	0.13	0.53	4
Identify and reduce friction points	Re-worded	0.05	0.44	3
Identify and mitigate risks early	Re-worded	0.16	0.53	2
Partner with RevOps on tracking and reporting	Re-worded	0.07	0.44	1
Design mechanical systems for industrial robots	Variant	0.12	0.48	3
Optimize latency and throughput of real-time systems	Re-worded	0.10	0.46	13
Builders who shape and ship product	Not work	0.16	0.51	0
Define and refine ideal customer profile	Re-worded	0.08	0.43	1
Develop and maintain signals and detection tooling	Variant	0.15	0.51	6
Flag risks and blockers early	Re-worded	0.13	0.48	0
Build and support go-to-market motions	Re-worded	0.03	0.38	4
Improve win rates and balance quick wins	Re-worded	0.16	0.51	1
Drive cross-functional and systems alignment	Re-worded	0.15	0.49	2
Ensuring clean handoffs between teams	Re-worded	0.12	0.46	1

Notes: The 40 rising themes, of the 90 that Sonnet 5.5 named, that Sonnet 5.5 judges not to be AI work, ranked by the increase in the share of employers using them. Columns as in Table C7.

Table C9: The 20 largest new and rising job titles

Title	Kind	Employers, 2019–22	2024–26	AI duty (%)	Unlike neighbours (%)	Pay ($k)
Forward deployed engineer	rising	9	247	61	64	175
AI engineer	rising	10	168	95	70	176
GTM engineer	new	0	118	78	93	162
Senior AI engineer	rising	7	103	98	87	191
Staff product manager	rising	13	94	38	42	215
Applied AI engineer	rising	1	78	100	87	212
Founding account executive	rising	2	61	42	65	195
Senior product engineer	rising	11	59	31	57	192
AI/ML engineer	rising	3	49	100	61	162
Senior design engineer	rising	8	48	17	68	175
Deployment strategist	rising	4	44	33	71	160
GTM recruiter	rising	6	41	18	25	150
Member of technical staff	rising	3	39	74	75	254
Principal product marketing manager	rising	5	37	38	37	181
Senior AI/ML engineer	rising	2	35	98	64	182
Staff electrical engineer	rising	4	34	2	41	186
Founding product designer	rising	6	33	53	29	180
HVAC maintenance technician	rising	4	31	0	31	73
AI product manager	rising	1	31	100	59	155
Plumbing service technician	rising	2	30	0	40	85

Notes: Normalised job titles with at least 10 employers in 2024–26 and none in 2019–22, in any spelling (new), or at least 20 employers in 2024–26 and five times their 2019–22 count (rising), ranked by employers in 2024–26; postings for talent pools and general applications excluded. AI duty: share of the title's postings with an AI duty. Unlike neighbours: share of its 2024–26 duty lines with no close counterpart among the duties of its three nearest established titles. Pay: median posted annual pay.

C.1 Themes of the flagged lines

I cluster the 416,000 distinct flagged sentences of 2025–26 into 1,500 themes and keep those whose share of employers in 2025–26 is at least three times that of 2019–22. Sonnet 5.5 names 90 of them, the 60 that rose most and the largest whose lines are mostly not about AI, and judges each to be AI work by side or, if not, a new task, a variant, a re-wording or not work; Opus 5.5 re-reads them and judges the AI themes on the same four kinds (Appendix B.6). Of the 60 fastest-rising themes, 49 concern AI: integrating language models into applications, building AI platforms and agents, using AI tools to speed up work, designing AI-native workflows and leading AI adoption (Figure C4a). Among the rising non-AI themes, Sonnet classifies all but one—building automated security and compliance guardrails—as re-wordings or templates, variants of familiar work or non-duty text (Table C8), whereas applying the same clustering to 2020–21 cleanly recovers COVID-19 duties. Genuine new work outside AI is too dispersed across topics to form large rising clusters, so panel (b) groups the individual non-AI new-work lines of 2023–26 by topic: about a third involve non-AI technologies such as blockchain software and humanoid robotics, another third involve new products and markets such as crypto assets and electric vehicles, and most of the rest involve new regulations, compliance regimes and clinical practices, alongside a small number of AI duties that the AI labels missed.

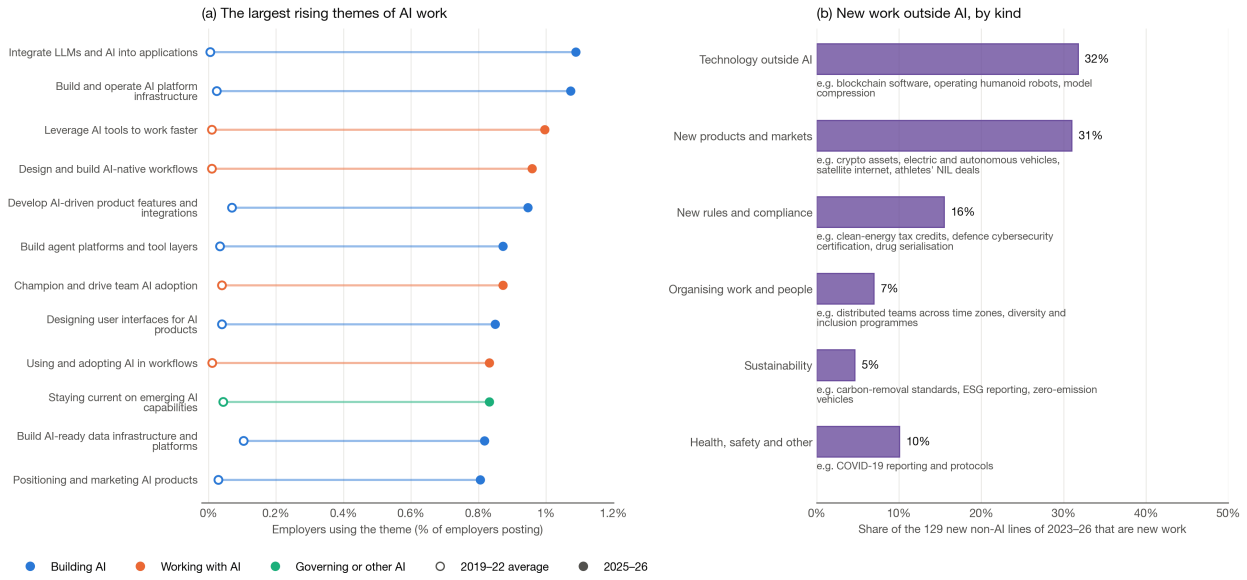


Figure C4: What the new work is. (a) The largest rising themes of AI work among the flagged duty lines: the share of employers using each theme in 2019–22 (open) and 2025–26 (filled). Themes: clusters of the distinct flagged sentences of 2025–26; rising: used by at least three times the share of employers. Names and sides by Sonnet 5.5 from recent and older example sentences; the themes not about AI are in Table C8. (b) The flagged non-AI lines of 2023–26 that are new work (Section 4.3; Appendix Figure B2), by Sonnet's judgement of what makes them new; examples paraphrased from the lines; shares rounded.

C.2 The occupation-level design: further outcomes

Figure C5 adds major group × half-year effects to the design of Section 5.5, so that occupations are compared only with others in their major group. Compared that way, occupations that use AI more gain no more new work about AI (−0.16, standard error 0.14) or outside it (−0.05, 0.08),

while duties that only ask workers to use AI tools rise somewhat (0.04, 0.02): the gradient in new work runs between broad groups of occupations.

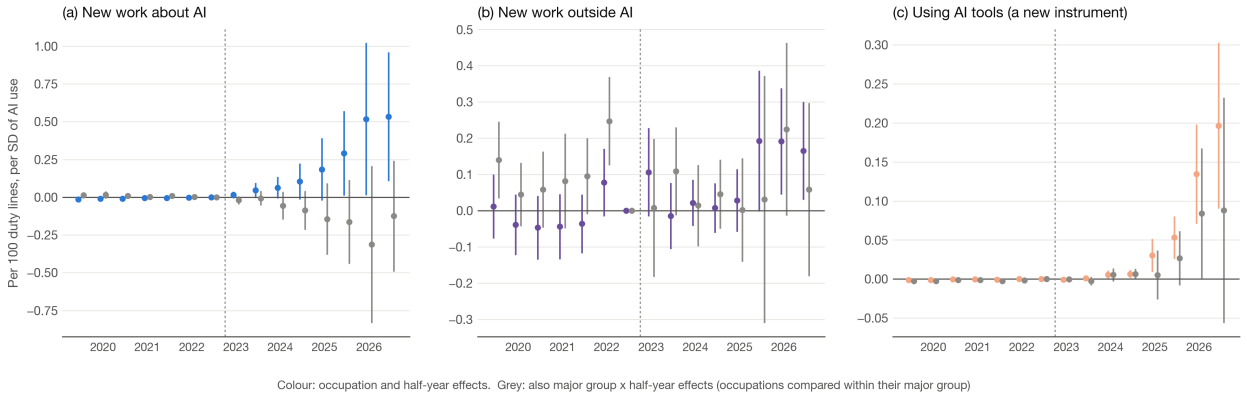


Figure C5: The design of Figure 6 with major group × half-year effects added (grey) beside the baseline (colour): coefficients on standardised observed AI use by half-year, relative to 2022H2, with 95% intervals.

Figure C2 gives the same design for AI duties and flagged lines, which every duty line measures, and a firm-level version in which a firm's AI use is that of its 2019–22 occupation mix. Flagged non-AI lines, mostly turnover of wording, rise with AI use only in 2026 and vary with it before 2022 as well. Occupations with higher observed AI use already differed before 2022 in their senior share, posting volumes and non-AI wording churn, and observed AI use correlates closely both with pre-existing task-exposure ratings (correlation 0.64; Eloundou et al., 2024) and with an occupation's 2019–21 share of remote postings. Entered alongside AI use, the pre-existing remote-work share absorbs the post-2022 gradient in postings with a working-with-AI duty (+2.94 points per standard deviation, against −0.16 for AI use; Appendix D), reflecting the close overlap between teleworkable, screen-based work and the tasks where language models apply. With occupations weighted by their employment rather than their postings, the effect on postings with a working-with-AI duty (1.7 points per standard deviation, standard error 0.4) is about half the baseline, and without the partial last half-year it is 1.1 (Appendix D).

The postings are also becoming more senior: between 2022 and 2026 the share of junior titles fell from 24% to 20% of postings, and the share of senior and staff titles rose by about half, a shift that began in 2024. It was larger in occupations that use AI more, but those occupations were already becoming more senior before 2022 (Figure C2), so the occupation design cannot attribute it to AI.

D Checks

This appendix collects the checks of the headline measure, new work per 100 duty lines (Figure 1), under other choices of frame, dates, readings and screens (Table D1), and the checks of the other results. The paper's basis is postings dated by their stated date and readings in mixed batches (Sections 3 and 4.3); every block of the table is on that basis, and new work follows the rule of Section 2.

Intervals and what counts. The 95% intervals of the headline, which cover the sampling of employers and of the read lines, are 0.36–0.69 for the high-use third in 2022 and 2.28–2.89 in 2026, or 1.92–2.89 allowing also for the second reader (Readings, below). In the high-use third the figure

Table D1: New work per 100 duty lines under other choices

	Most AI use, 2022	Most AI use, 2026	Least AI use, 2026	All occupations, 2026
Paper (stated dates, mixed-batch readings)				
Paper	0.53	2.58	0.22	1.24
Readings in batches of one class and cell	0.53	2.02	0.22	1.01
Opus 5.5's readings of a fifth of the batches	0.37	2.40	0.34	1.15
AI shares measured for 2019–22, the low third and governing AI	0.43	2.56	0.20	1.22
Composition (paper's basis)				
Firms posting in every year 2022–26	0.55	2.79	0.22	1.32
Inside firm × occupation cells present in both years	0.57	2.97		1.23
Without the 1% of firms with most postings	0.53	2.55		1.32
Occupations (paper's basis)				
Thirds on title-based occupation codes	0.55	2.46	0.26	1.23
Thirds of Eloundou et al. exposure	0.52	2.32	0.24	1.23
Screen (paper's basis; threshold rebuilt)				
Threshold: 5% of 2019H2 new ($\tau = 0.627$)	0.36	1.60	0.14	0.78
Threshold: 7.5% ($\tau = 0.646$)	0.45	2.11	0.19	1.02
Threshold: 12.5% ($\tau = 0.672$)	0.64	2.98	0.25	1.43
Threshold: 15% ($\tau = 0.682$)	0.75	3.30	0.28	1.60
Threshold: 20% ($\tau = 0.700$)	1.00	3.89	0.33	1.89
Reference without the January 2019 crawl ($\tau = 0.652$)	0.53	2.41	0.22	1.17
Frame and dates (paper's basis)				
Without postings with an unresolved location	0.52	2.58	0.22	1.22
Dated by first capture	0.52	2.56	0.21	1.22
Each posting in every half-year it stayed up	0.53	2.48	0.19	1.16
Dated by the stated posting date (within ten years)	0.54	2.64	0.22	1.28
Bulk-poster cutoff prorated for 2026	0.53	2.58	0.22	1.24
Without Workday (flat text)	0.51	2.59	0.22	1.25
Without Workday and Dayforce	0.52	2.64	0.23	1.28

Notes: New work per 100 duty lines (Figure 1) in the thirds of occupations with the most and least observed AI use, and in all occupations. Each block states its basis: the Paper row is the basis of the Composition, Occupations and Screen blocks, and the first row of the last block is the basis of the frame variants. Blank: not computed for the least-AI-use third.

was 1.6 in 2025. In 2026, 18% of the postings of the high-use third have at least one duty read as a new task, against 5% in 2022 and 2% of the postings of the low-use third in 2026. Sonnet reads 61% of the flagged building-AI lines of 2026 as new tasks and 90% of the working-with-AI lines; if every flagged AI line counted as new work, the 2026 figure for the high-use third would be 4.7 rather than 2.6. Of the AI lines read as new tasks in 2026, 8% (building AI) and 16% (working with AI) describe a new activity and 76% and 79% a familiar activity done with or on AI (Appendix B.6).

Frame and dates. Dropping the 28,835 postings whose location could not be resolved, dating every posting by its first capture, counting each advert in every half-year between its first and last capture (so that an advert that stays up, or is reposted unchanged, counts in later years too), dating postings by their stated date within ten years of capture rather than one, and scaling the bulk-poster cutoff to the nine months of 2026 each move the headline by little more than a tenth of a point, as does dropping Workday, whose pages carry text only from 2022, or Workday and Dayforce, the platforms with flat text (Table D1). Of the postings of 2019–26, 87% carry a stated date within the year before their first capture and are dated by it; the first capture comes a median of 37 to 67 days after the stated date, with a 90th percentile of 114 to 534 days and a 95th percentile of up to 731, depending on the year.

Composition. In the 2,520 firms that posted in every year 2022–26, which hold 40% of 2026 postings, new work in the high-use third goes from 0.55 to 2.79 per 100 duty lines (all postings: 0.53 to 2.58), and working with AI is 25% of AI lines in 2026 (every line counted; all postings: 25%). Decomposing the change over firm-by-occupation cells, cells present in both years hold 22% of 2026 lines, since the archive's thin sample makes cells come and go, and inside them the rate goes from

0.57 to 2.97; cells entering in 2026 are at 2.47. The rise happens inside firms and occupations at the rate of the whole. The top 1% of firms by 2026 postings (145 firms) hold 24% of 2026 postings and 15% of 2026 AI lines, and without them the headline is 0.53 to 2.55. Counting each distinct sentence once per firm and year, AI lines per 100 duty lines are 4.1 in 2026 against 3.3 when every line counts.

Occupations. On thirds cut by the title-based occupation codes (Appendix A.3), which rest on the title rather than the duties, the high-use third goes from 0.55 to 2.46 and the low-use third is at 0.26 in 2026; on thirds of Eloundou et al.'s exposure, rated before the period, the high-exposure third goes from 0.52 to 2.32 and the low-exposure third is at 0.24. The high–low gap in new work widened by 0.06 points a year over 2019H2–2022; extended to 2026 that trend gives a gap of 0.5, against the 2.4 observed, 1.8 points above trend (1.6 on exposure thirds). In the occupation-level design, the exposure measure alone gives 1.93 points per standard deviation; the occupation's 2019–21 share of remote postings, entered alongside observed use, takes the gradient (+2.94 against −0.16), and its posting growth from 2019 to 2021 does not (1.60 for use). By the second half of 2026 (to September), an occupation one standard deviation higher in AI use has 4.2 points more postings with a working-with-AI duty. The average effect over 2024–26 of 1.7 points is 0.8 (standard error 0.4) with major-group-by-period effects, 0.79 (0.32) with occupations weighted by OEWS employment and 0.60 (0.26) by new job starts in the CPS, and 1.1 without the partial last half-year.

Readings. Read in batches of one class and cell, new work in the high-use third is 0.53 in 2022 and 2.02 in 2026, against 0.53 and 2.58 on the mixed-batch readings; the difference is the AI lines, which the one-class batches read as new tasks less often (Appendix B.6). On Opus's readings of a fifth of the same batches, the figure is 0.37 in 2022 (0.05–0.69) and 2.40 in 2026 (1.92–2.88), against 2.58 (2.28–2.89) on Sonnet's, so the interval that covers either reader is 1.92–2.89 in 2026 and 0.05–0.69 in 2022; for all occupations in 2026 it is 0.95–1.38. The main estimate applies the new-task shares of 2023–24 to the flagged AI lines of 2019–22, pools the AI shares across thirds and gives governing and other AI lines the pooled share of the two sides. Read directly (in one-class batches), flagged AI lines of 2019–22 are new tasks 5% (building) and 14% (working with AI) of the time, governing and other AI lines 44%, and AI lines of the low-use third in 2026 31% and 69%; with those shares in place of the borrowed ones the headline is 0.43 to 2.56 in the high-use third and 0.20 in the low-use third. The screen misses new tasks of both kinds, and the second reader calls fewer non-AI lines new tasks: scaled by the recall rates of Table B1, Appendix B and the reading without the screen (below), and across the two readers and prompts, new work outside AI in the high-use third in 2026 lies between 0.4 and 1.7 per 100 duty lines and new work about AI between 1.9 and 3.9; holding each recall rate fixed between 2022 and 2026, their rise since 2022 lies between 0.1 and 0.5 points outside AI against 1.8–3.7 about AI. Recall need not be fixed: the reading without the screen, below, measures it in both years and finds a larger rise outside AI.

Without the screen. Reading random duty lines of 2022 and 2026 in the high- and low-use thirds, flagged or not and in every band of similarity (3,000 lines per cell, Opus on 600), gives the rates that the screen and the reading together would find if the screen missed nothing; new work is counted by the rule of Section 2, with the three-question reading for the non-AI lines read as new tasks. In the high-use third in 2026, new work about AI is 3.0 per 100 duty lines (standard error 0.29) and new work outside AI 1.5 (0.23) by Sonnet, and 2.8 and 1.2 by Opus; in 2022, they are

0.22 and 0.42 (Sonnet). In the low-use third in 2026 they are 0.14 and 0.13. The rates come from a reading of their own, so the screen's recall is measured within it: in the high-use third in 2026 the flagged lines hold 75% of the new work about AI and 58% of that outside AI by Sonnet (56% and 65% by Opus), and the bounds above use these rates. The new work about AI that the screen misses is 24% (Sonnet) to 41% (Opus) of the lines about building, governing, directing or adopting AI that the screen counts as continuing (Table 4). Part of the unscreened count outside AI is AI work that the AI labels miss, such as building MCP servers or training frontier models. Without the screen, new work about AI in the high-use third rose by 2.8 points between 2022 and 2026 and new work outside AI by 1.1 points according to Sonnet (2.8 and 0.6 according to Opus), so the ordering of growth holds without the screen as well. The rise outside AI exceeds the recall-adjusted bounds above because those bounds hold each recall rate fixed, while in Sonnet's reading the screen catches less of the new work outside AI in 2026 than in 2022.

Screen. Table D1 gives the headline with the threshold set so that 5, 7.5, 12.5, 15 or 20% of 2019H2 duty lines count as new, each with the posting panel, reading samples and estimate rebuilt at that threshold, and with the January 2019 crawl dropped from the reference (Appendix B). The levels move with the threshold and the contrast between the thirds does not.

Drafted adverts and boilerplate. Benefits and legal text, which no reading counts as work, is flagged 10% of the time in 2019H2 and 34% in 2026 (duty lines: 10% and 17.4%), the drafting change that the screen picks up; read with the paper's prompt, 99% of 800 such flagged lines of 2022 and 2026 are not duties and none is a new task, by either model. Appendix C gives the recurrence of AI lines across firms.

Sectors on each basis. The balance between building AI and working with it differs by sector: in information firms, 27.1% of 2026 postings have a building-AI duty against 15.6% with a working-with-AI duty, whereas in finance, health care and retail the two are much closer (8.9% against 9.1% in finance), every duty line counted with the classifier's labels. On that basis, in 2026, information firms have 6.3 building-AI and 2.3 working-with-AI lines per 100 duty lines, finance 2.1 and 1.3, health care 0.4 and 0.3, and retail 0.4 and 0.3; corrected for the keywords' catch rates, the corresponding rates are 8.9 and 2.6, 3.2 and 1.5, 0.6 and 0.3, and 0.5 and 0.4. The correction raises the building side everywhere and changes no ordering. Across the other exhibits, Figure 1 and the occupation design count every duty line with the classifier's labels and read the new ones; Figure C3, the sector comparison and the pay regressions count every duty line with the classifier's labels; Figure 3b counts keyword lines; and the overall share of working with AI applies the catch-rate correction.

Pay. Table D2 gives the pay regressions with their identifying samples. Within firm, title and year, 1,751 postings in 500 cells identify the building-AI premium and 880 in 270 the working-with-AI premium, of 428,836 postings with pay. The AI indicators miss AI duties that have no keyword and that the classifier does not find, so some comparison postings carry the duty; under classical misclassification with precision 0.98 and sensitivity 0.60 (building) and 0.88 (working with AI), the estimates understate the premium by a factor that depends on the share of postings with the indicator in the identifying sample, and dividing by it gives 0.104 (0.006) and 0.024 (0.008) within firm, occupation and year and 0.015 (0.012) and 0.024 (0.015) within firm, title and year. In

the firms that posted in every year 2022–26 (185,340 postings with pay), the premia within firm, occupation and year are 0.065 (0.006) and 0.017 (0.010), and within firm, title and year −0.008 (0.009) and 0.047 (0.027). Adding seniority and state to the firm, occupation and year cell gives 0.045 (0.004) and 0.007 (0.010). Split by what the worker does with AI (Figure 7), working-with-AI duties that are new work (driving adoption, directing or reviewing AI) and duties that only use AI tools carry 0.068 (0.012) and 0.017 (0.008) within occupation, state and year; 0.025 (0.011) and 0.011 (0.006) adding seniority; 0.049 (0.010) and −0.020 (0.008) within firm, occupation and year; 0.026 (0.015) and −0.019 (0.006) adding seniority and state; and 0.051 (0.026) and 0.006 (0.012) within firm, title and year. The differences between the two, with standard errors from the clustered covariance of the two estimates, are 0.050 (0.015), 0.014 (0.013), 0.068 (0.014), 0.045 (0.017) and 0.045 (0.029).

Table D2: Posted-pay premia: specifications, identifying samples and variants

Fixed effects	Building AI	Working with AI	Postings	Build: switching postings / cells	Work: switching postings / cells
All postings with pay, 2023–26					
Occupation × state × year	0.170 (0.012)	0.054 (0.008)	428,836	171,739 / 2,566	159,919 / 2,114
Occupation × state × year × seniority	0.148 (0.011)	0.023 (0.007)	428,836	137,913 / 4,223	123,137 / 3,328
Firm × occupation × year	0.068 (0.004)	0.022 (0.008)	428,836	42,665 / 6,247	26,335 / 3,958
Firm × occupation × seniority × state × year	0.045 (0.004)	0.007 (0.010)	428,836	20,839 / 4,632	10,665 / 2,494
Firm × title × year	0.009 (0.007)	0.022 (0.014)	428,836	1,751 / 500	880 / 270
AI indicators from keyword lines only					
Firm × occupation × year	0.065 (0.005)	0.024 (0.008)	428,836	34,458 / 4,986	23,787 / 3,596
Firm × title × year	0.003 (0.010)	0.021 (0.015)	428,836	1,308 / 358	813 / 246
States and dates with a pay-range law					
Firm × occupation × year	0.062 (0.005)	0.011 (0.010)	192,822	19,050 / 3,097	11,564 / 1,892
Firm × title × year	0.009 (0.013)	0.056 (0.035)	192,822	677 / 209	289 / 91
Bottom of the posted range					
Firm × occupation × year	0.066 (0.007)	0.017 (0.010)	428,836	42,665 / 6,247	26,335 / 3,958
Firm × title × year	0.025 (0.016)	−0.013 (0.028)	428,836	1,751 / 500	880 / 270
Top of the posted range					
Firm × occupation × year	0.069 (0.004)	0.024 (0.008)	428,836	42,665 / 6,247	26,335 / 3,958
Firm × title × year	0.007 (0.007)	0.027 (0.015)	428,836	1,751 / 500	880 / 270
Firms posting in every year 2022–26					
Occupation × state × year	0.174 (0.022)	0.062 (0.011)	185,340	65,319 / 1,516	58,316 / 1,216
Firm × occupation × year	0.065 (0.006)	0.017 (0.010)	185,340	24,651 / 2,953	16,274 / 2,031
Firm × title × year	−0.008 (0.009)	0.047 (0.027)	185,340	1,064 / 297	429 / 128
Corrected for the recall of the AI labels (last two columns: the misclassification factors)					
Firm × occupation × year	0.104 (0.006)	0.024 (0.008)	428,836	factor 0.66	factor 0.94
Firm × title × year	0.015 (0.012)	0.024 (0.015)	428,836	factor 0.58	factor 0.90

Notes: Log annual posted pay, 2023–26, on indicators for a building-AI and a working-with-AI duty, with the fixed effects in the first column; standard errors clustered by firm in parentheses. Switching postings / cells: postings, and fixed-effect cells, in which the indicator varies. Keyword lines only: indicators from Sonnet's census of AI-keyword sentences, without the classifier. Corrected: the estimate and standard error divided by the misclassification factor in the last columns.

Work new to the job, by activity. Appendix B.2 explains why matching lines to O*NET's work activities does not give a reliable task for each line. A looser version is still informative about change over time. Opus 5.5 labelled 14,664 duty lines of 2019–26, drawn by year with AI lines oversampled, with one of O*NET's 313 intermediate work activities, or none; a classifier on the sentence embeddings learns the labels (held out, it gives Opus's activity for 40% of lines and has it among its three likeliest for 64%) and scores every duty sentence. Each occupation's activity boundary is the set of activities that make up at least half a per cent of its duty lines in a random half of its 2019 postings, and the crossover is the share of its duty lines in a year that fall outside that boundary; the other half of 2019 is the placebo, what sampling and classifier noise alone produce. Over the 69 occupations with at least 2,000 such lines, the crossover is 14.0% in the 2019 placebo half, 15.4% in 2022 and 15.2% in 2026, 14.4% in the high-use third and 16.3% in the low-use third in 2026 (13.8% and 13.4% in the placebo), and the same without Workday and at boundaries of a quarter or one per cent. Opus's rewordings of 1,000 2019 lines keep the classifier's activity for 65%

of lines and cross their occupation's boundary 27.9% of the time, against 27.1% for the originals, so wording alone moves the measure by less than a point. AI lines are 2.8% of the crossing lines of 2025–26 against 4.1% of all lines: the new AI duties mostly sit inside activities the occupation already had. On this measure, jobs have not moved into activities outside their 2019 mix since 2022.

Tools in every O*NET family. Extending Section 5.4 to all 48 O*NET 27.1 software categories with hot technologies, grouped into 16 families with 179 terms, with Opus 5.5 checking 80 matched lines in each cell as before, the share of postings asking for a family their occupation did not use in 2019–20 rose by 0.88 points between 2022 and 2026 for the five families of Section 5.4 (0.04 over 2019–22), by 0.60 for the eleven other families (0.31 over 2019–22, mostly collaboration tools in 2020–22) and by 1.38 for all sixteen. Figure D1 shows each family's change with its interval. Business intelligence (0.42) and workflow automation (0.31) move most; business systems (0.13), office software (0.13) and cloud and development tools (0.11) rise by about a tenth of a point; operating systems, web development, spreadsheets and computer-aided design do not move. Most families rise a little, by a median of 0.05 points: a general increase in the naming of tools as adverts lengthen, which the two data families of Section 5.4 exceed several times over, and which the placebo of Appendix B.2 (non-use defined on 2019, new tools counted in 2020–21) does not show before 2022. The precision of the matches is low where an occupation did not use cloud, social-media or web tools, so those families lean on the correction.

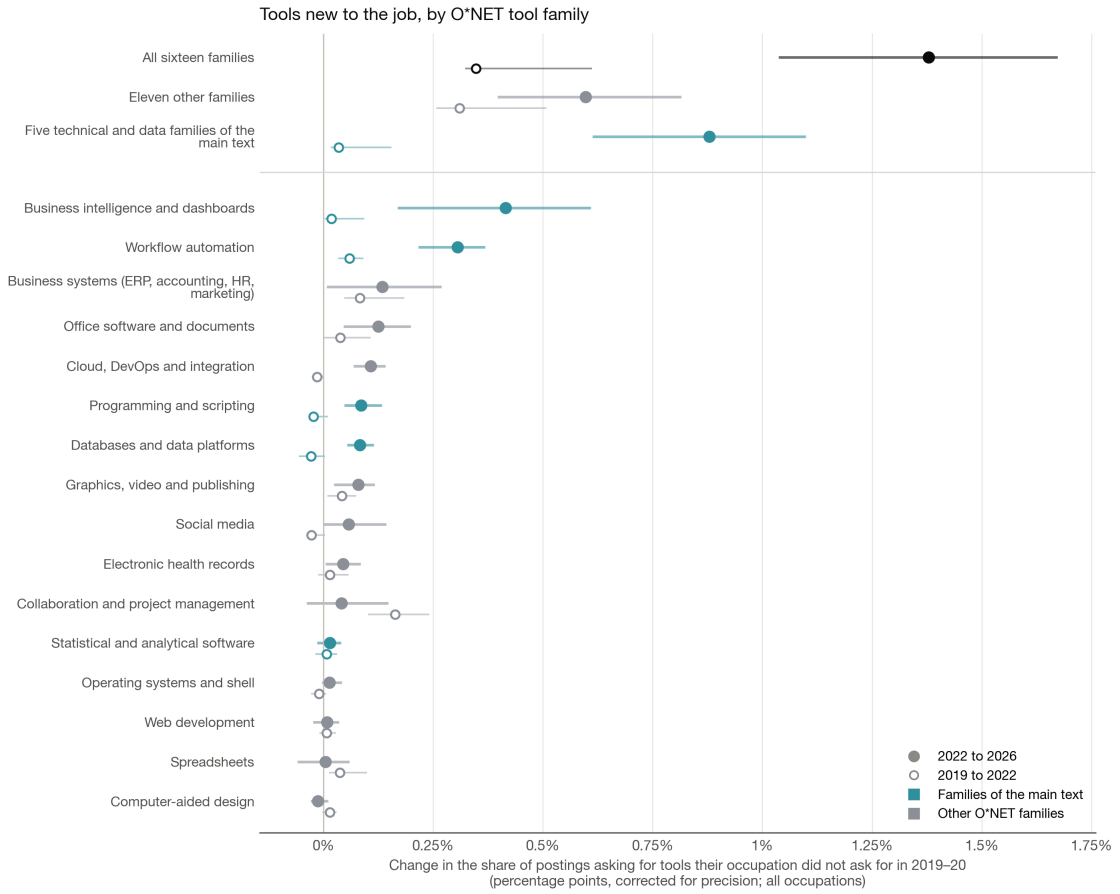


Figure D1: Tools new to the job, by O*NET tool family: change in the share of postings asking for a family their occupation did not use in 2019–20, corrected for the precision of the matches, 2022 to 2026 (filled) and 2019 to 2022 (open), with 95% intervals from resampling employers and the validation sample; all occupations. Teal: the five families of Section 5.4; grey: the eleven other families of O*NET 27.1's hot technologies; black: all sixteen.